



机械工程基础I

投影

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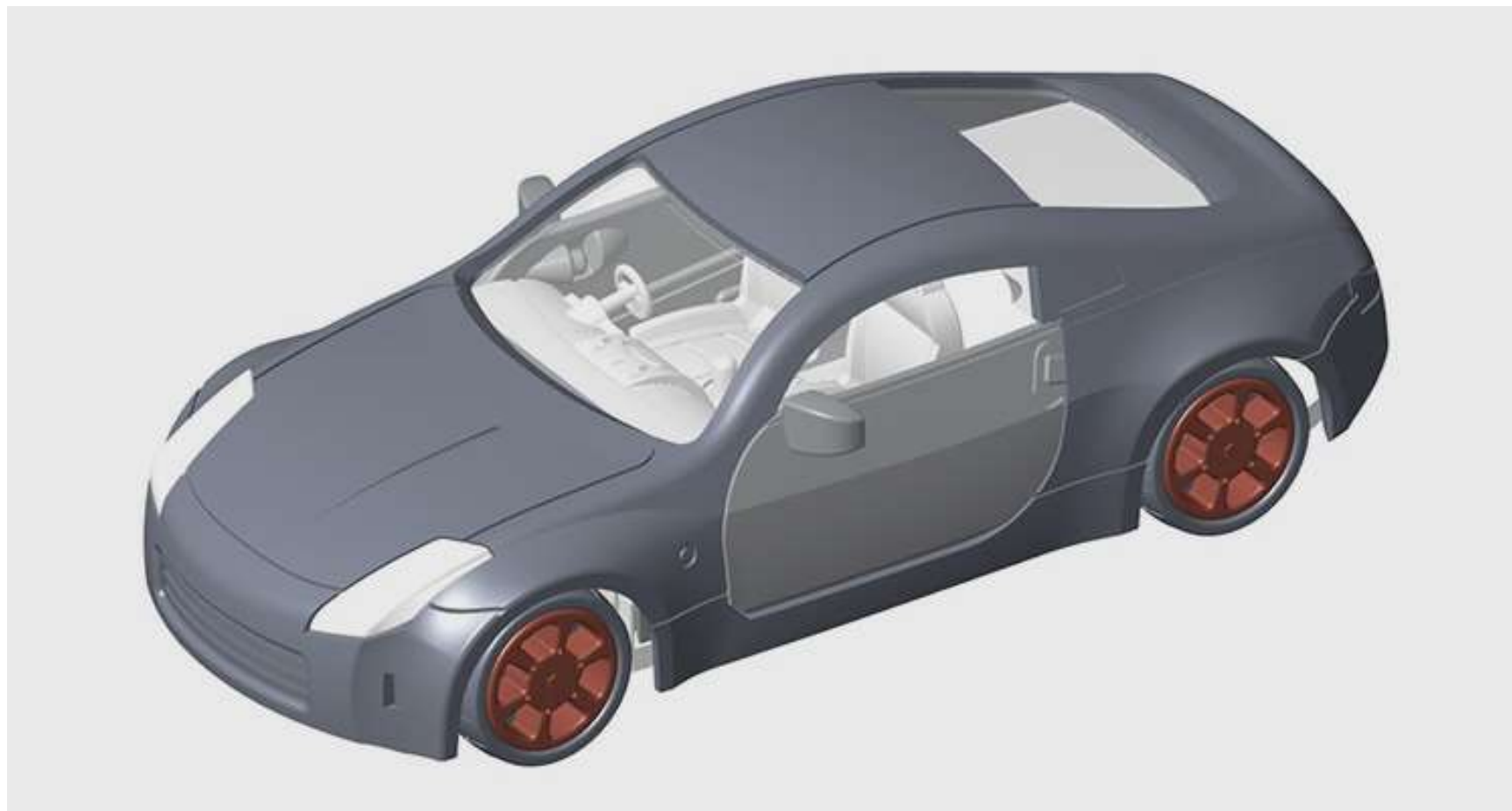


- 几何体
- 投影法
- 点的投影
- 直线的投影
- 平面的投影
- 体的投影

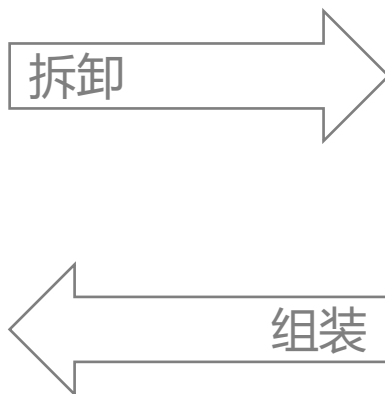








■ 部件：功能单元



■ 零件：最小制造单元



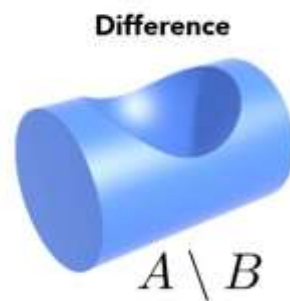
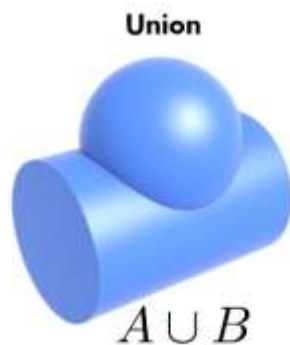
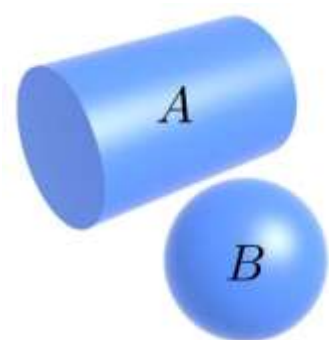
■ 零件：最小制造单元

■ 组合体



■ 区别：工艺结构、标准化结构

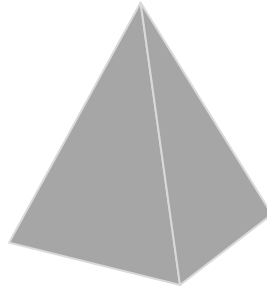
■ 组合体



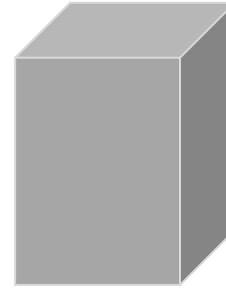
■ 基本几何体



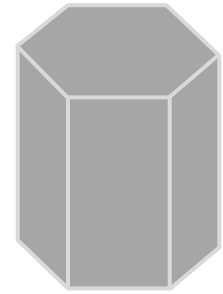
■ 平面基本体



三棱锥



四棱柱



六棱柱

■ 曲面基本体



圆柱



圆锥



球

■ 对称体



对称面

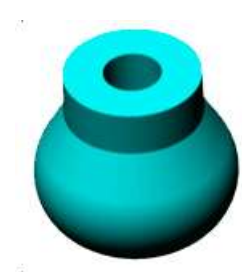
■ 非对称体



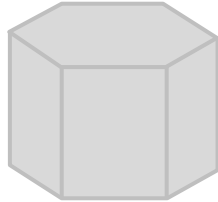
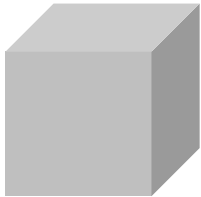
■ 回转体



对称轴



■ 拉伸体



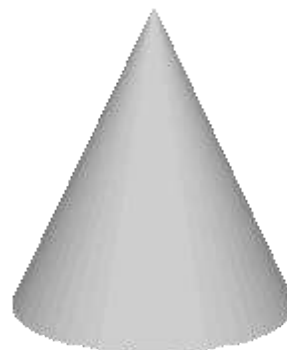
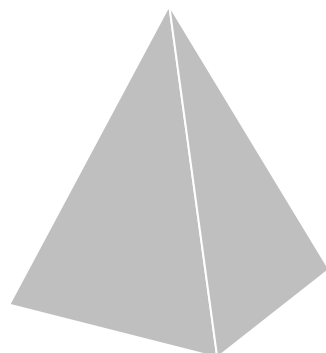
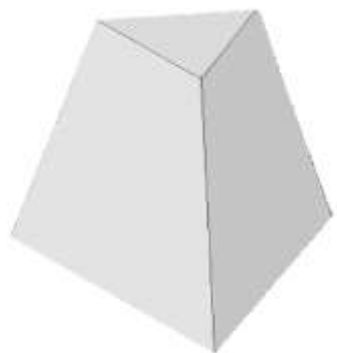
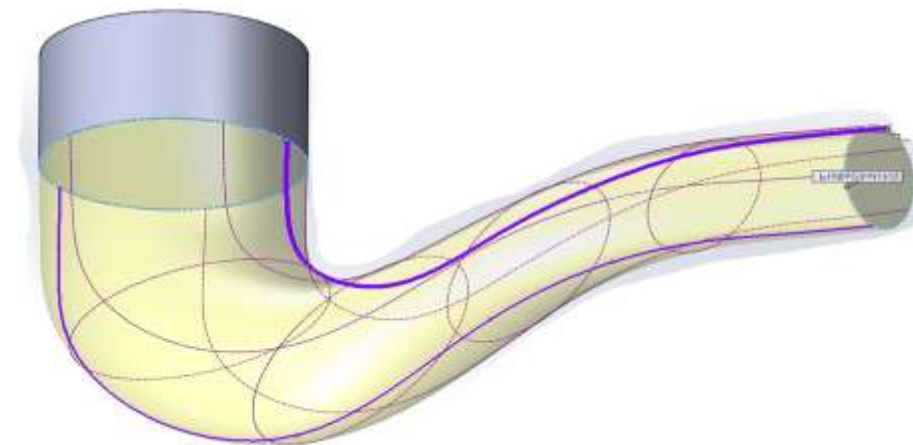
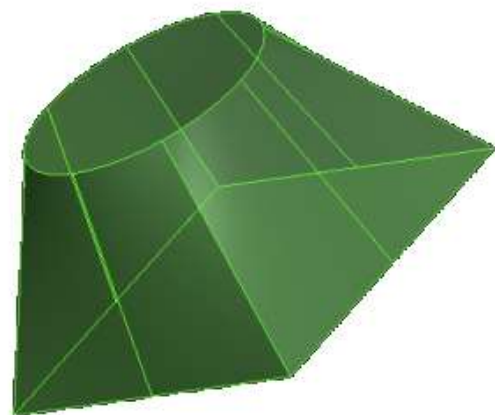
沿着垂直该面的方向扩展

■ 扫掠体 sweep



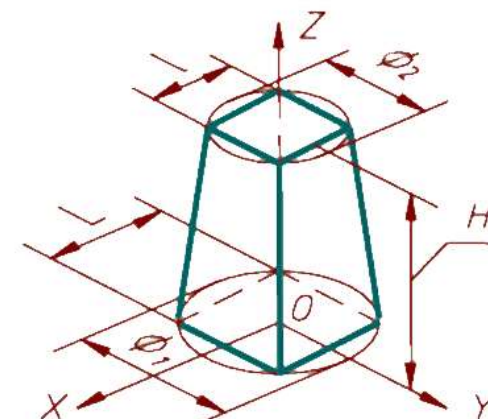
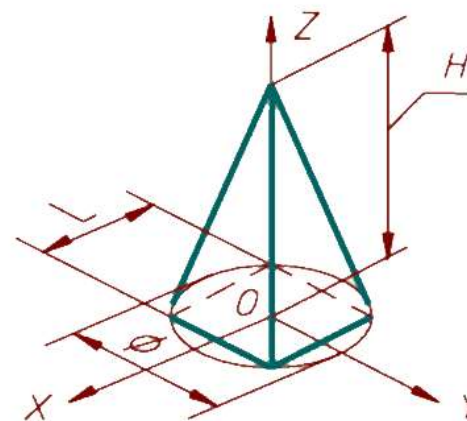
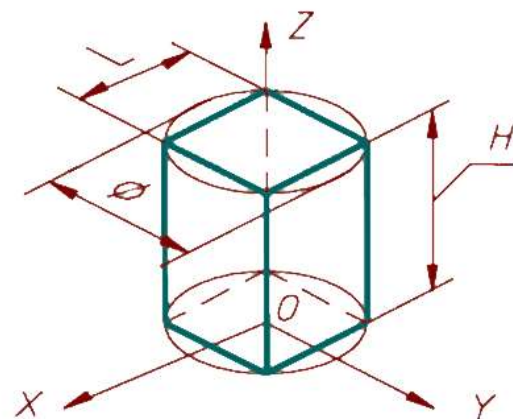
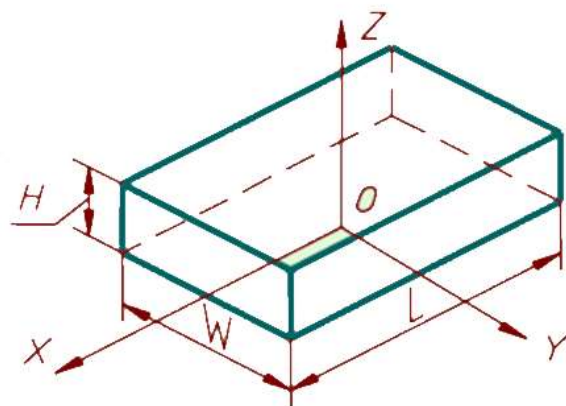
沿着垂直给定路径扩展

■ 放样体

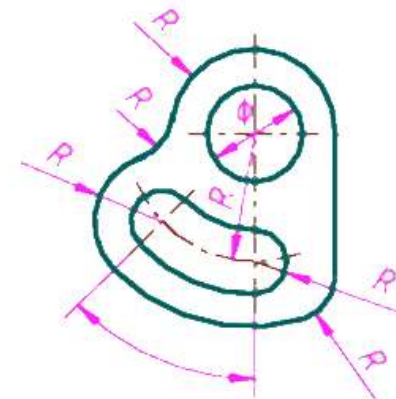
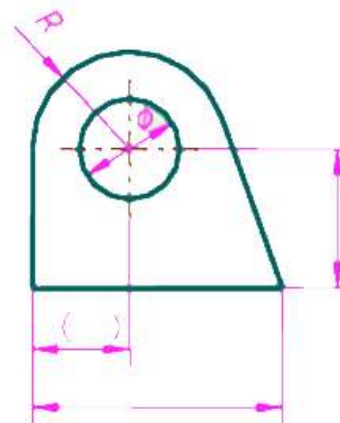
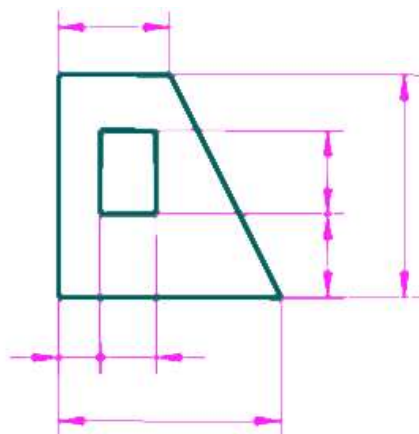
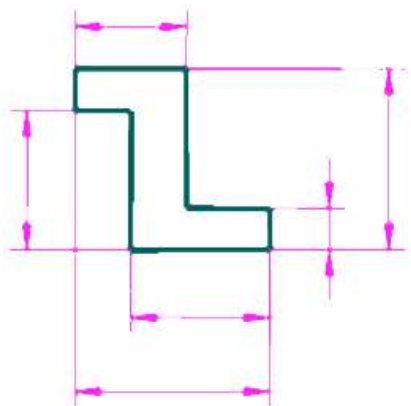


1. 沿着垂直给定路径扩展
2. 初始轮廓渐变到目标轮廓

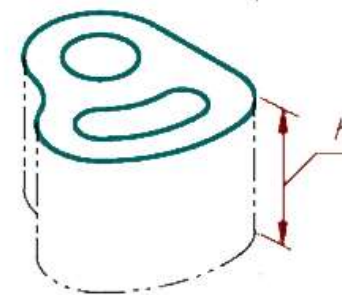
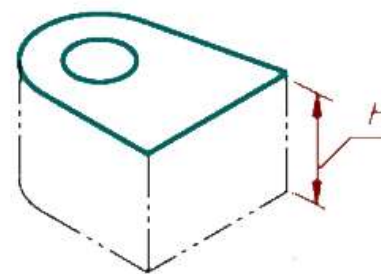
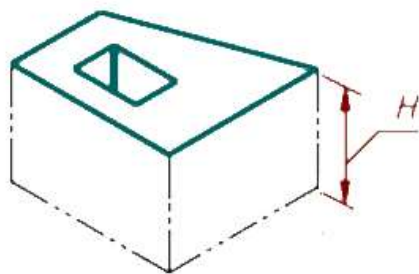
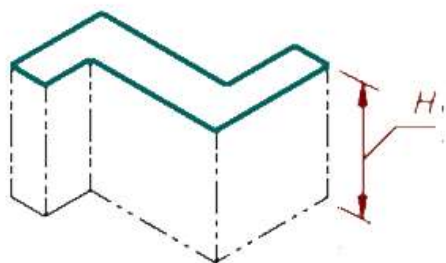
■ 拉伸体



■ 拉伸体

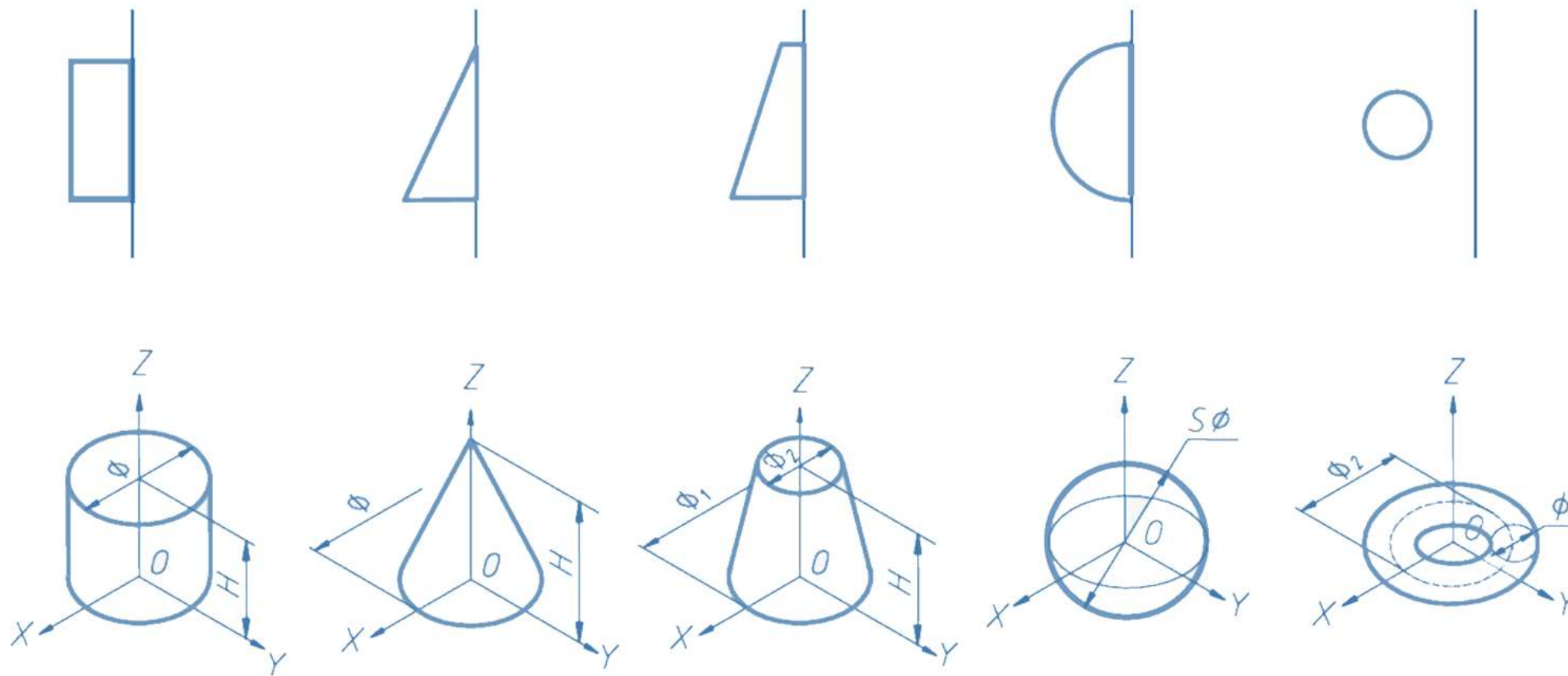


截面尺寸

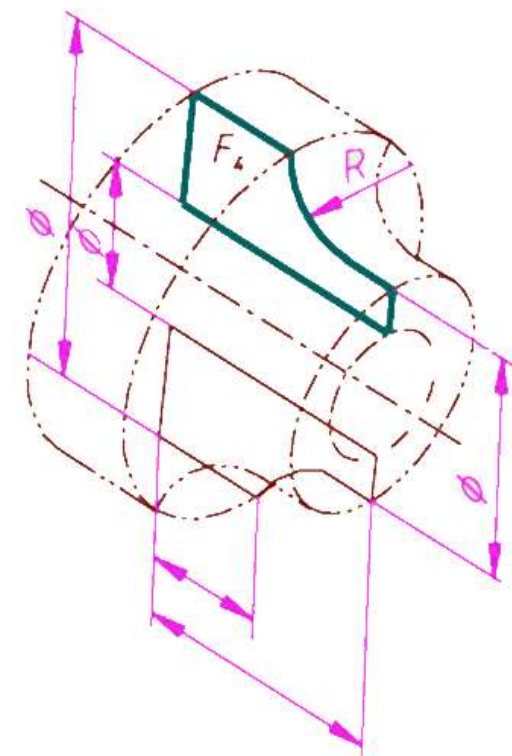
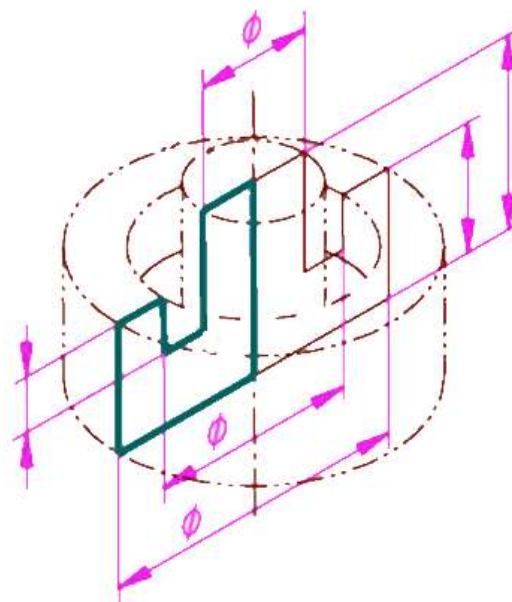
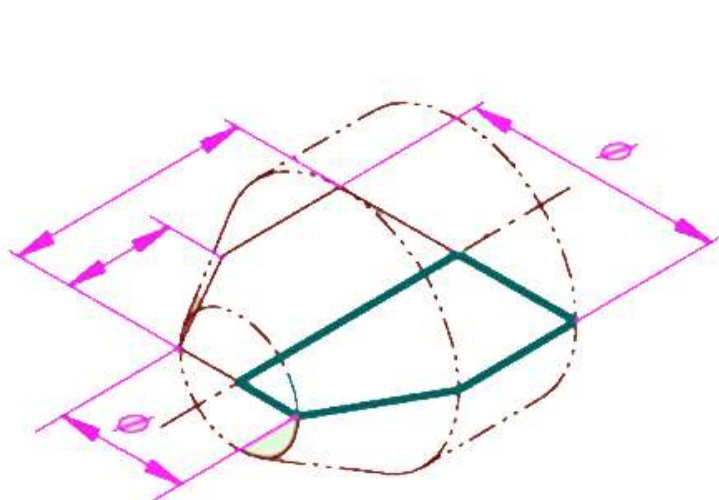


拉伸长度!

■ 回转体

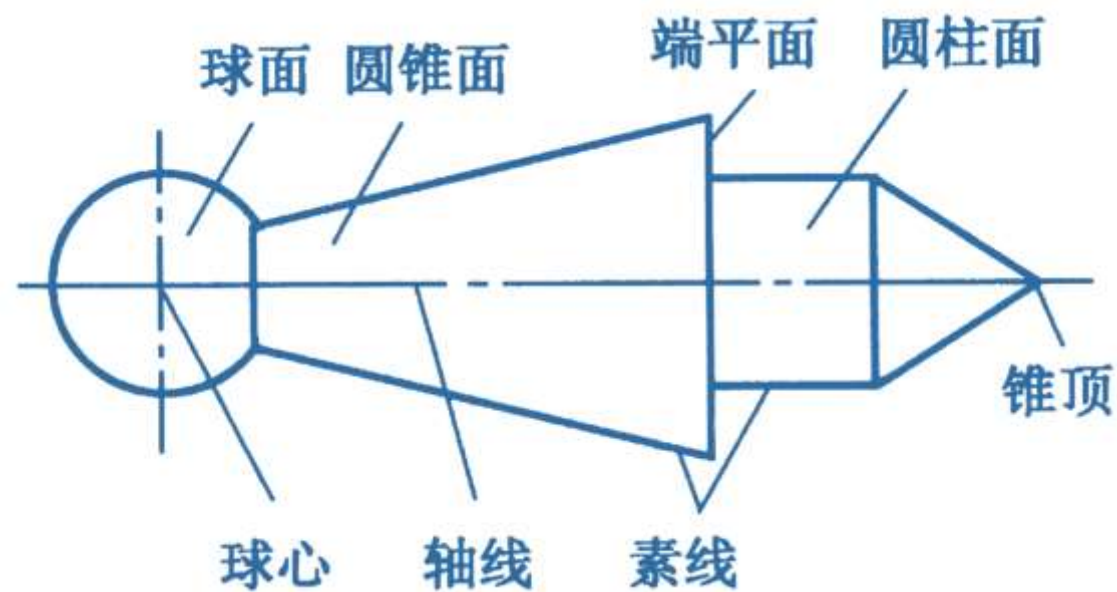


■ 回转体

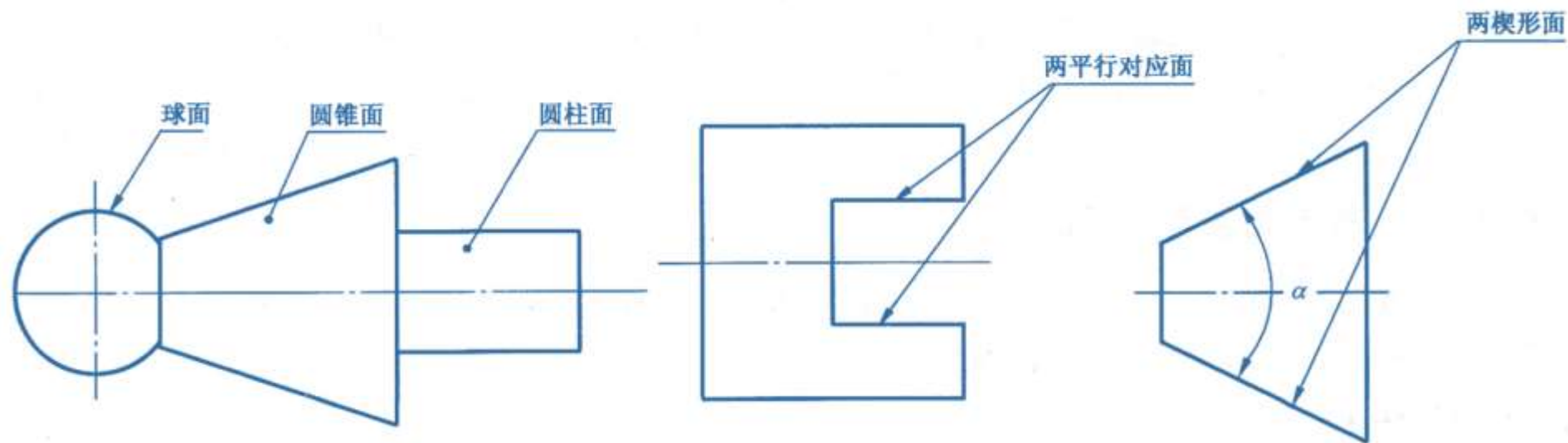


1. 注意截面形状
2. 注意旋转轴位置

■ 点、线、面、体

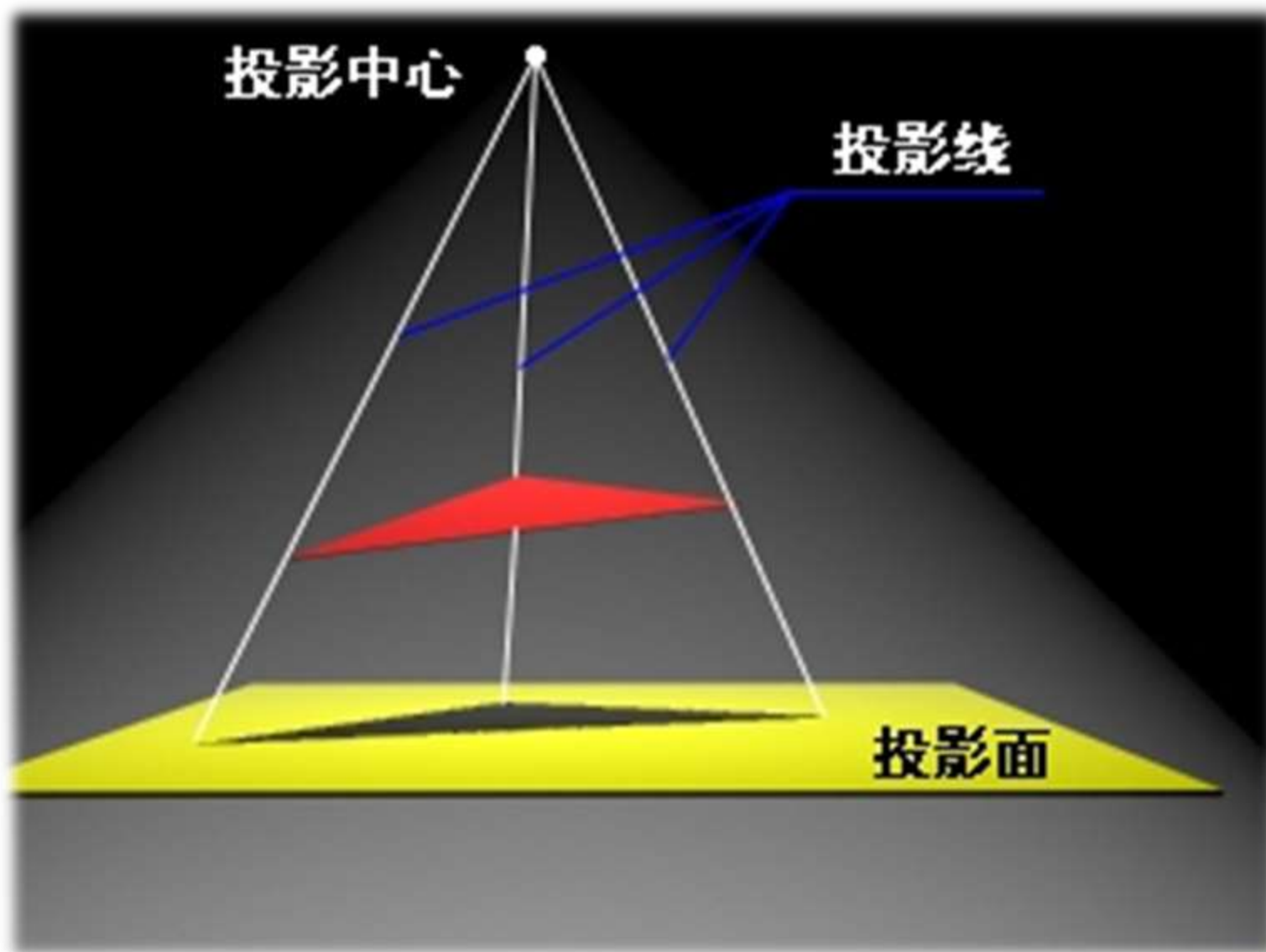


- **组成要素** (轮廓要素, 指几何体(零件) 的面或面上的线, 如圆柱面、圆锥面、平面、素线、曲线、曲面等)
- **导出要素** (中心要素, 由一个或几个公称组成要素导出, 如轴线、球心、圆心、两平行平面的中心平面等)



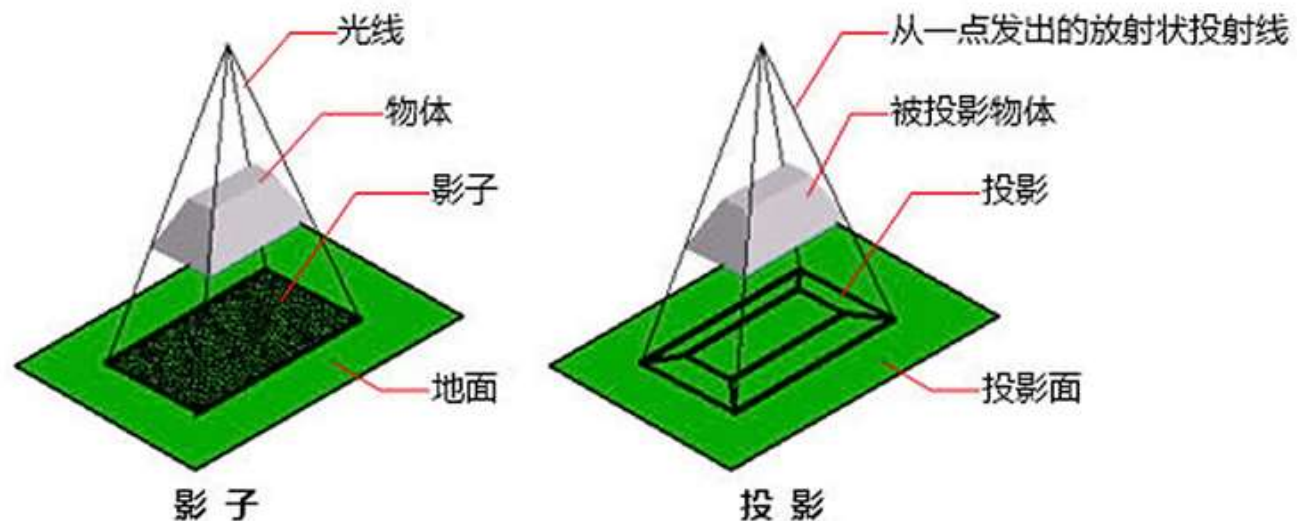
任何机械零件都是由各种要素有机地组合在一起的。

- 投影线通过物体向选定的面投射，并在该面上得到图形的方法称为**投影法**。
- 根据投影所得到的图形称为投影图，简称**投影**。
- 投影法中得到投影的面称为**投影面**。

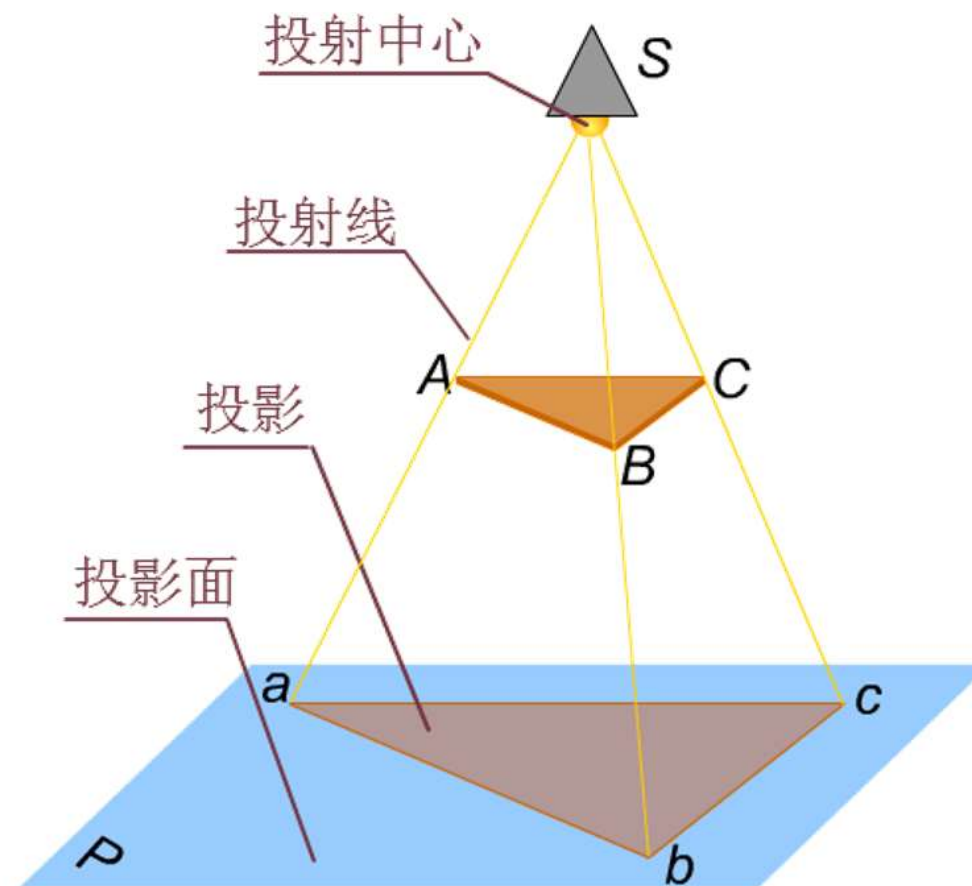


■ 中心投影法

➤ 投射线汇交于一点的投影法称为中心投影法

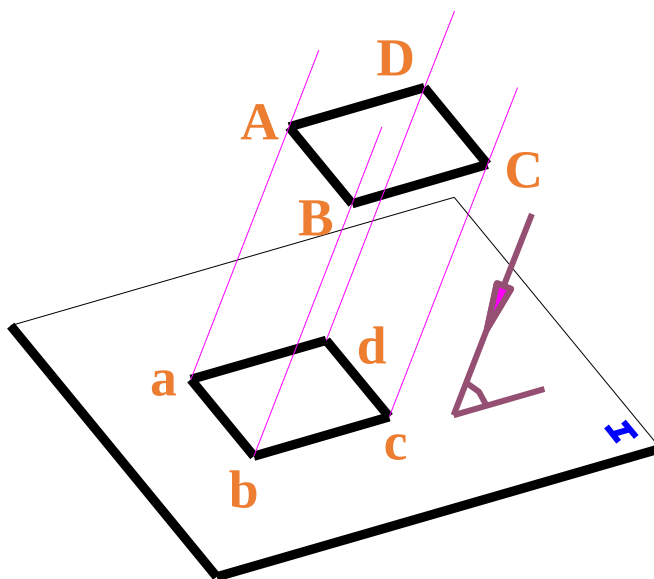


影子和投影的区别



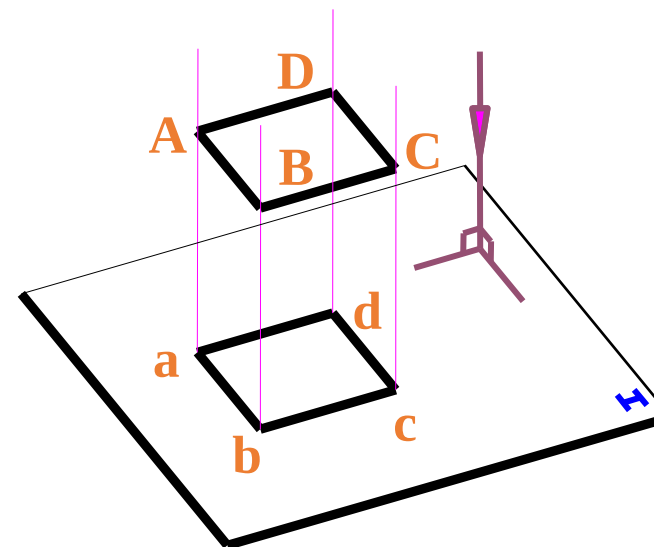
■ 平行投影法

✓ 斜投影法



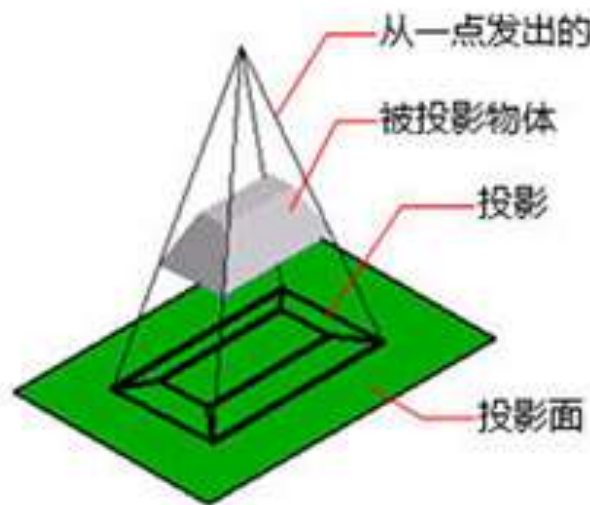
➤ 斜投影法得到的图形称为斜投影。

✓ 正投影法

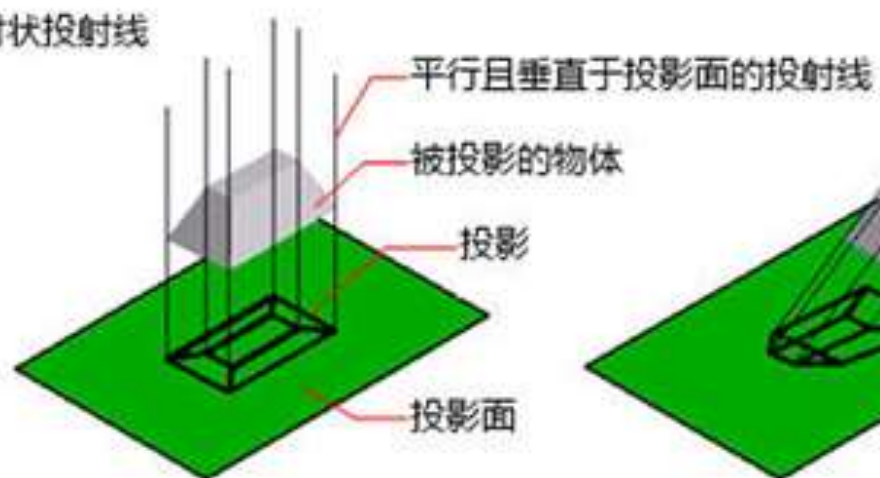


- 投影线彼此平行，投影线垂直于投影面；
- 正投影法所得到的图形称为正投影。

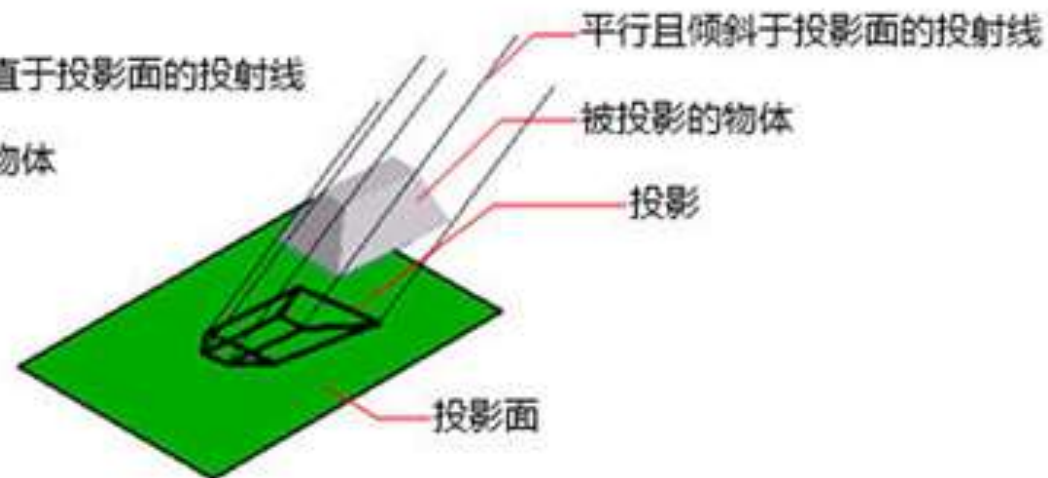
■ 投影法



中心投影

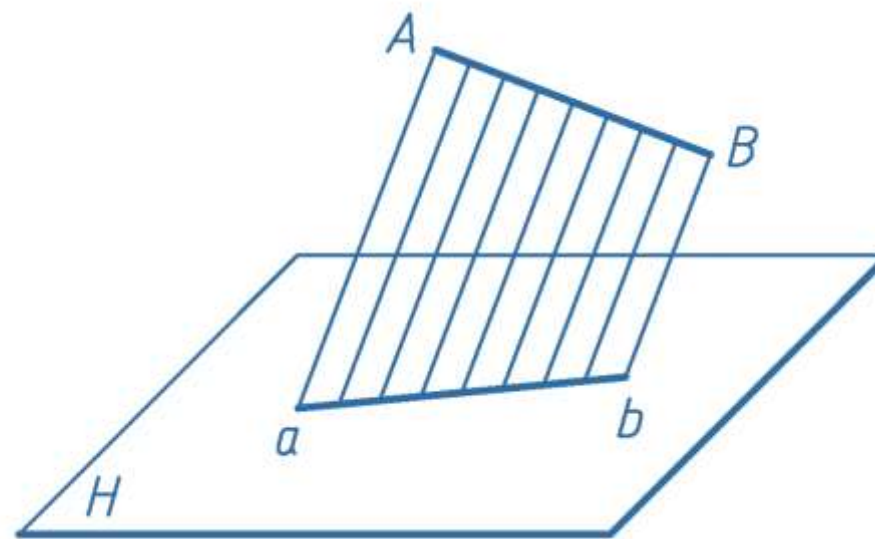


正投影



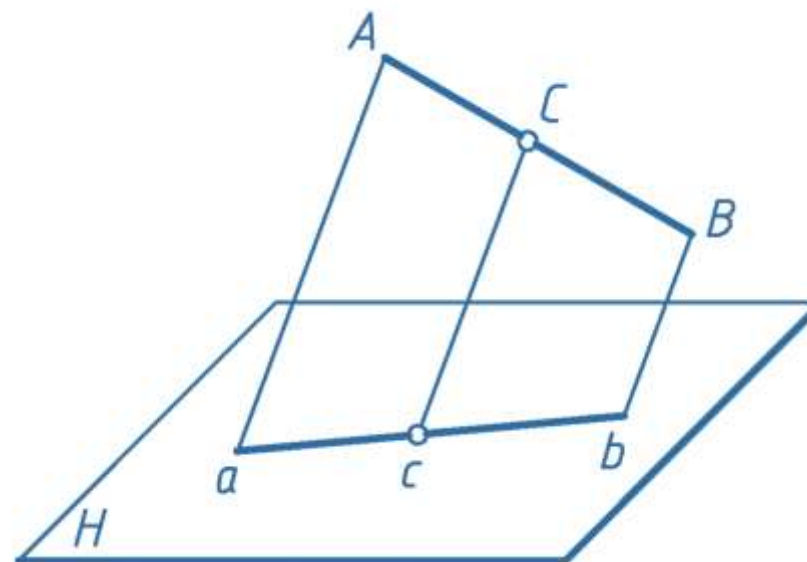
斜投影

- 同素性
- 从属性不变
- 平行性不变
- 简单比不变
- 相仿性
- 积聚性
- 全等性



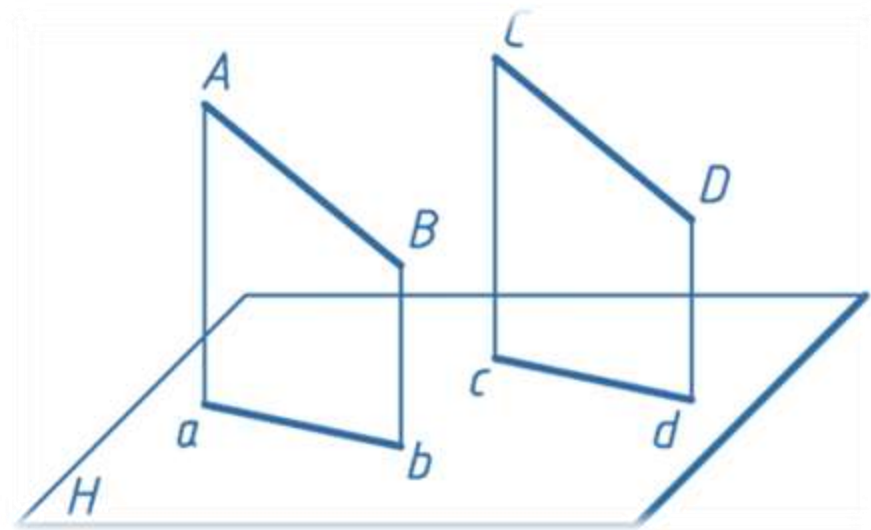
- ✓ 点的投影是点
- ✓ 直线的投影一般仍是直线

- 同素性
- 从属性不变
- 平行性不变
- 简单比不变
- 相仿性
- 积聚性
- 全等性



- ✓ 若点在直线上，则该点的投影一定在该直线的投影上
即 C 在 AB 上，则 c 在 ab 上

- 同素性
- 从属性不变
- 平行性不变
- 简单比不变
- 相仿性
- 积聚性
- 全等性

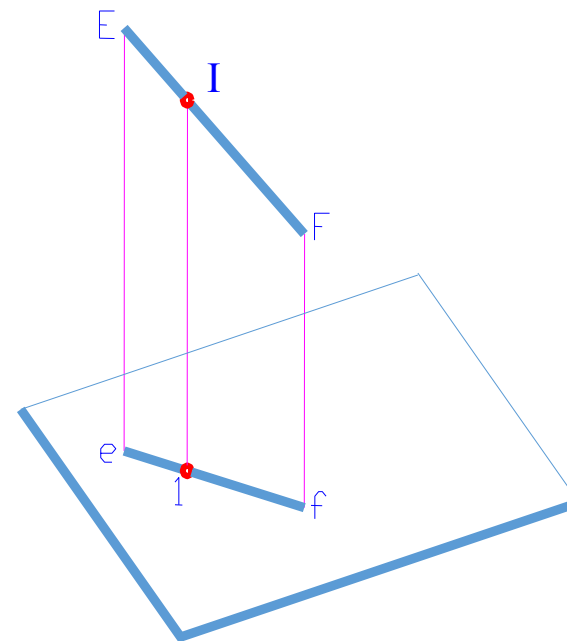


✓ 两平行直线的投影一般仍平行

$$\because AB \parallel CD$$

$$\therefore ab \parallel cd$$

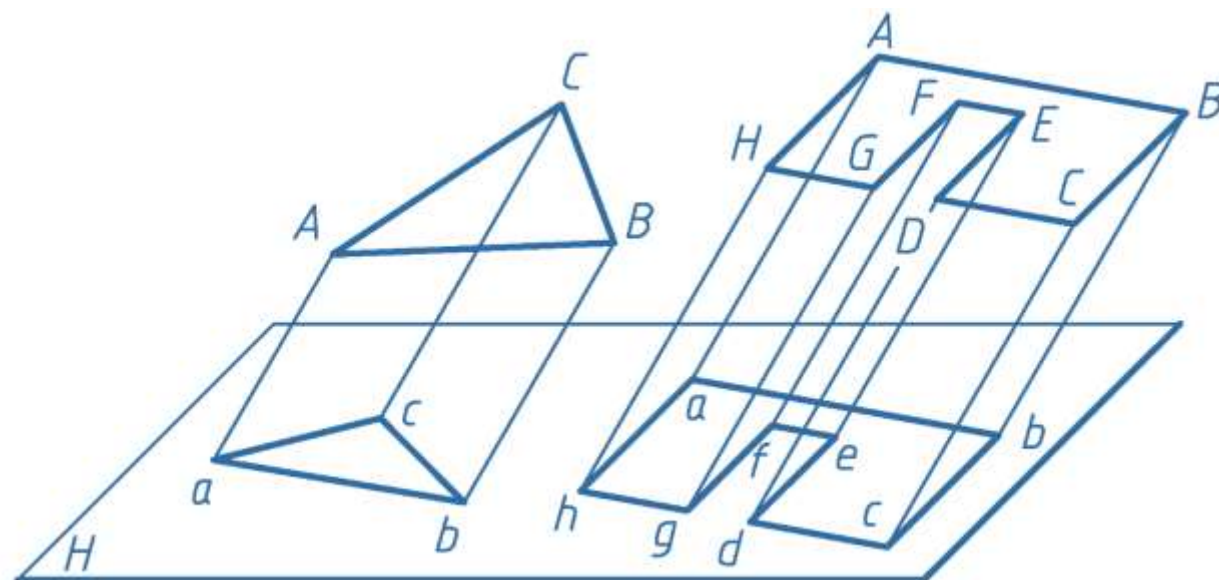
- 同素性
- 从属性不变
- 平行性不变
- 简单比不变
- 相仿性
- 积聚性
- 全等性



✓ 一条直线上任意三个点的简单比是平行投影的不变量

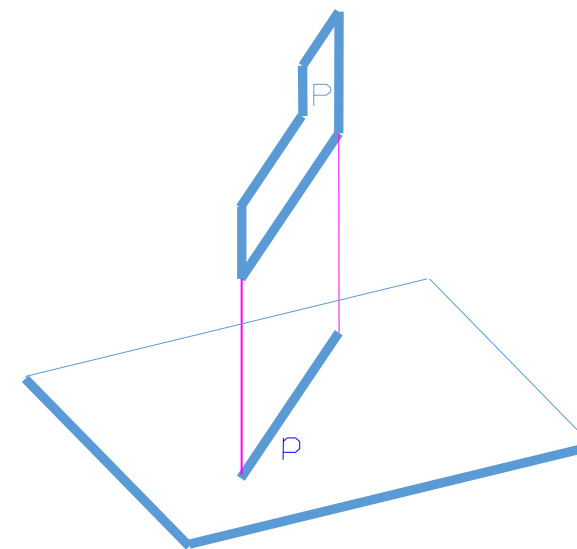
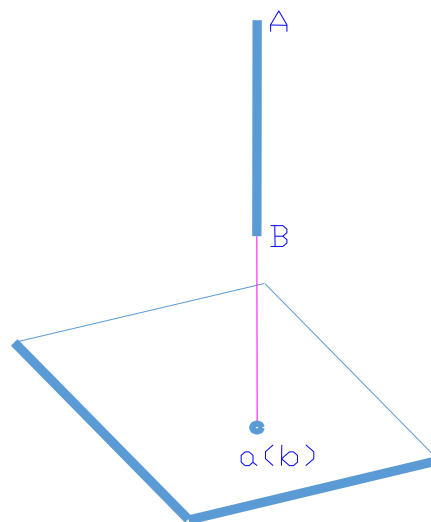
$$EI / IF = ei / if$$

- 同素性
- 从属性不变
- 平行性不变
- 简单比不变
- 相仿性
- 积聚性
- 全等性



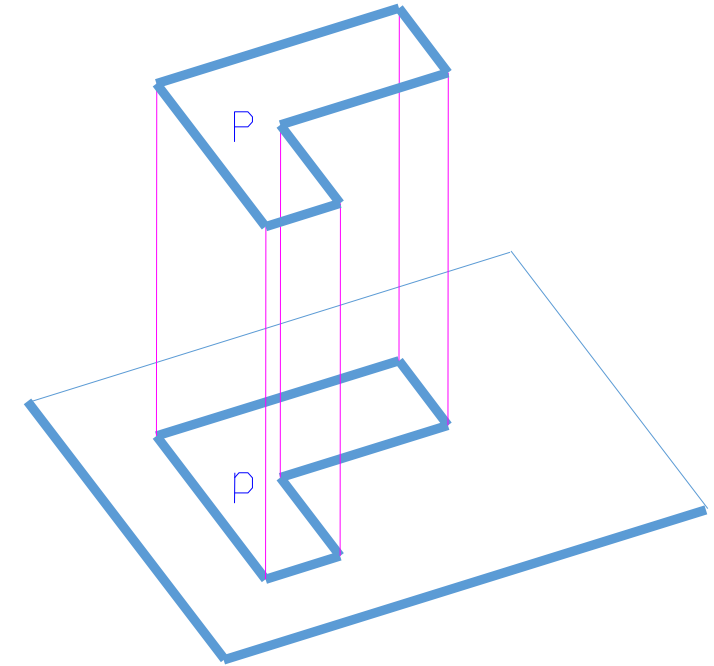
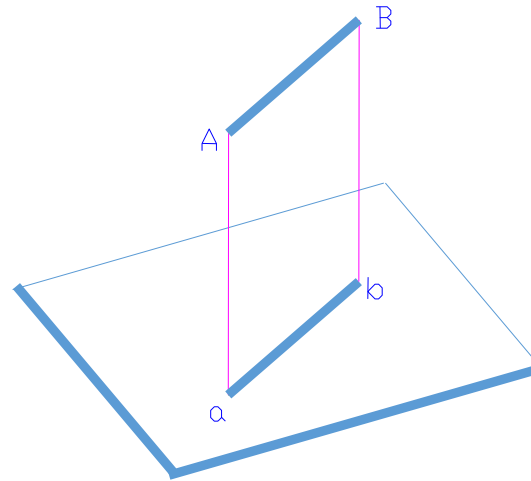
✓ 平面投影后，与原形的对应线段保持定比性，表现为投影形状与原形的边数相同、平行性相同、凸凹性相同及边的直线或曲线性质不变

- 同素性
- 从属性不变
- 平行性不变
- 简单比不变
- 相仿性
- 积聚性
- 全等性



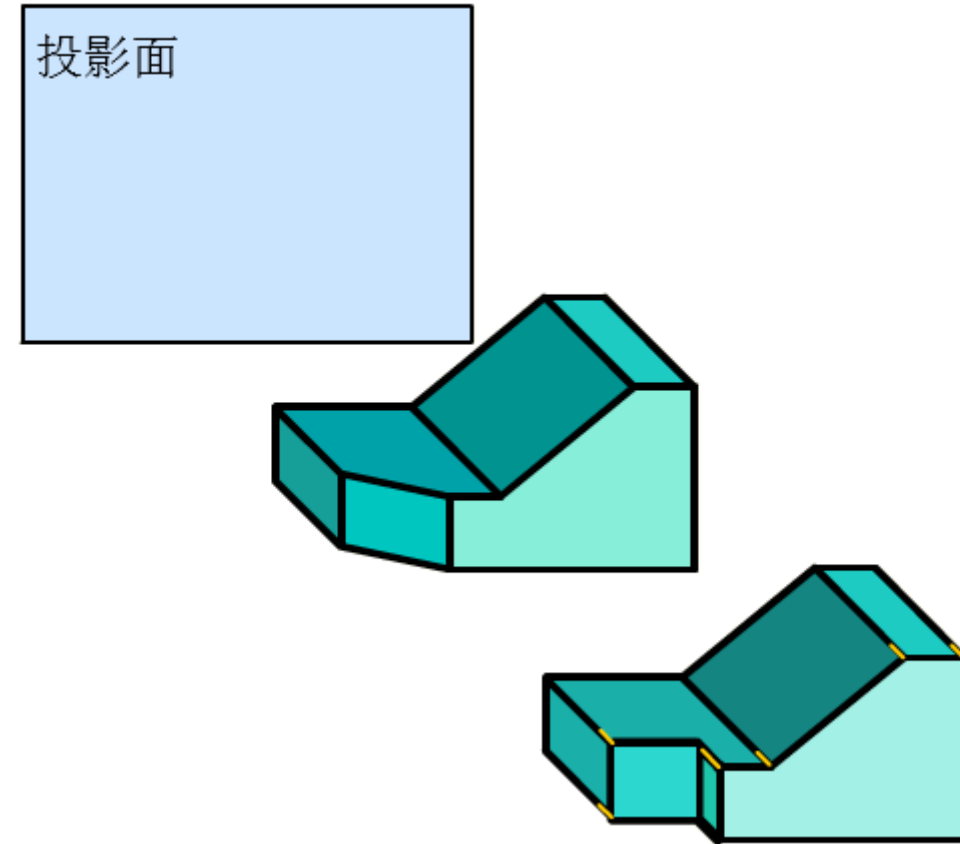
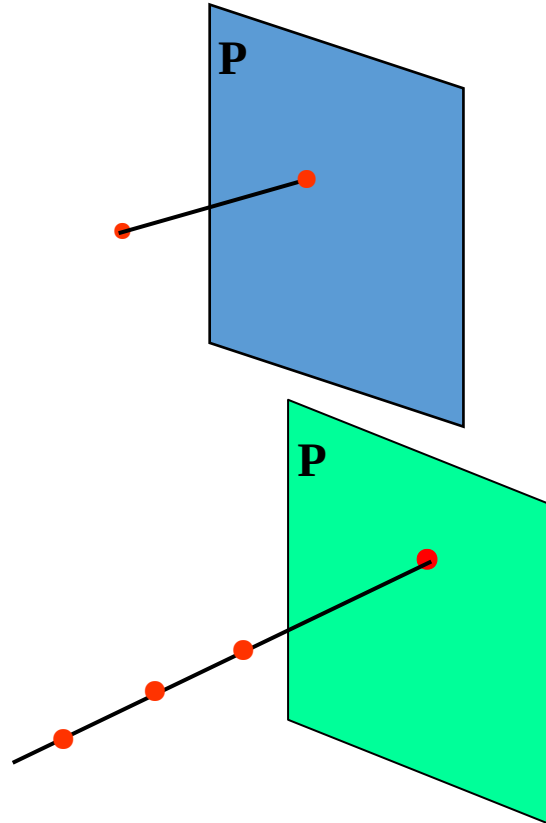
- ✓ 当直线平行于投射方向时，直线的投影为点
- ✓ 当平面图形平行于投射方向时，其投影为直线

- 同素性
- 从属性不变
- 平行性不变
- 简单比不变
- 相仿性
- 积聚性
- 全等性



- ✓ 当线段平行于投影面时，其投射长度反映线段的实长
- ✓ 当平面图形平行于投影面 时，其投影与原平面图形全等

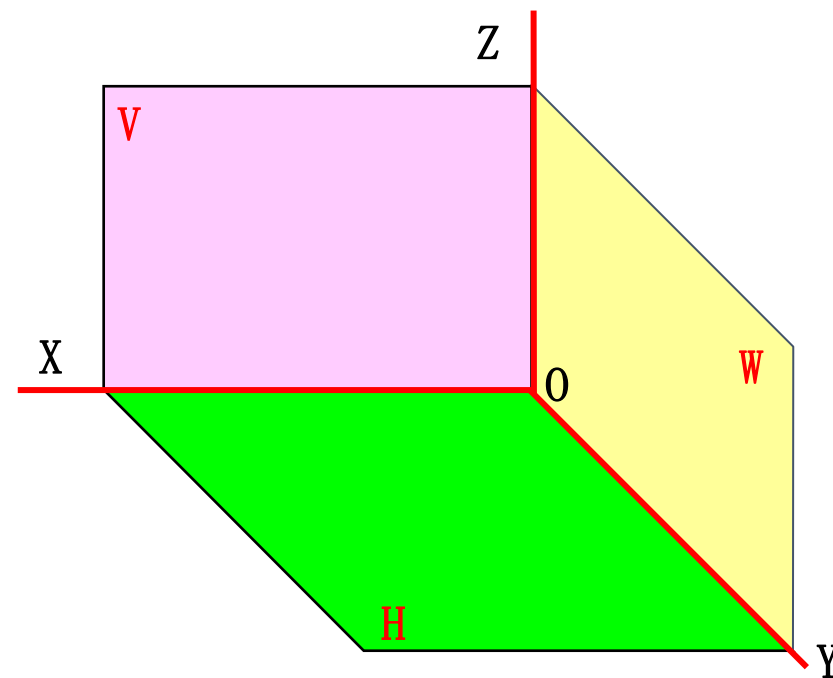
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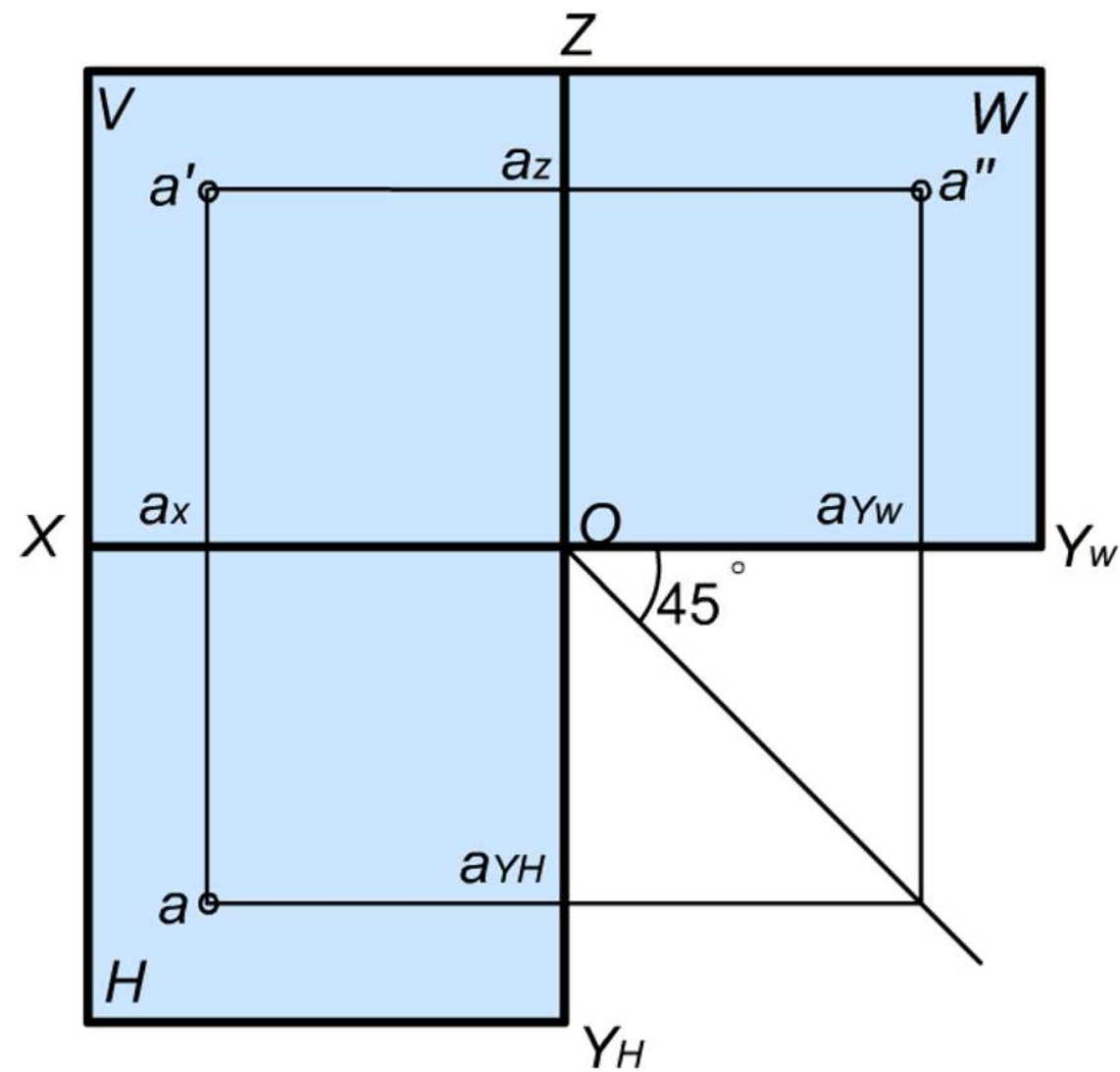
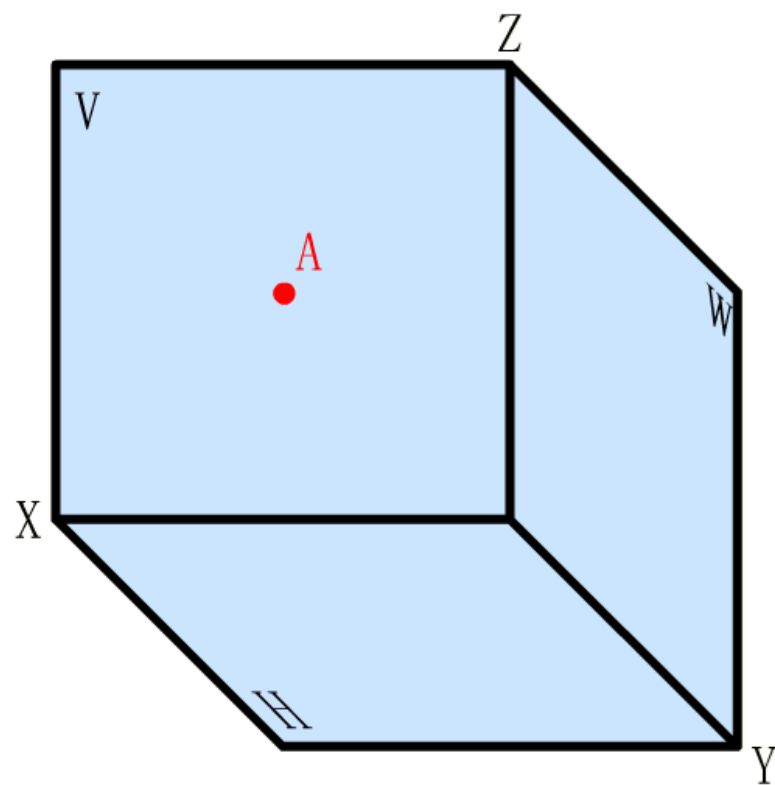


- 物体在一个投影面上所得到的正投影不能唯一地确定其空间形状

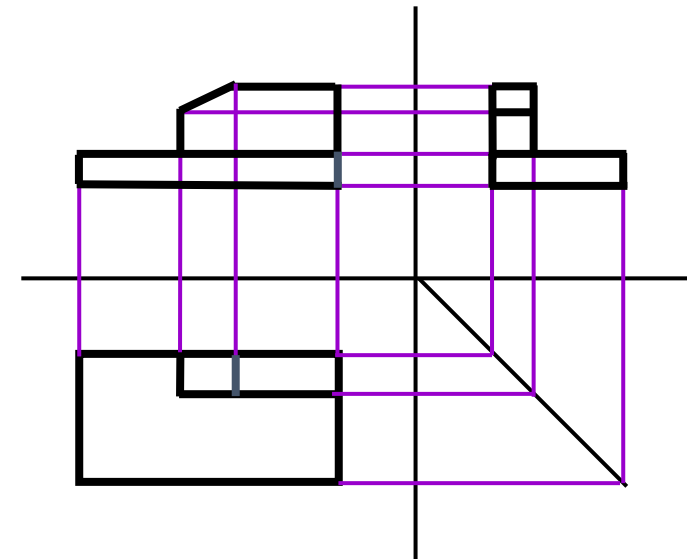
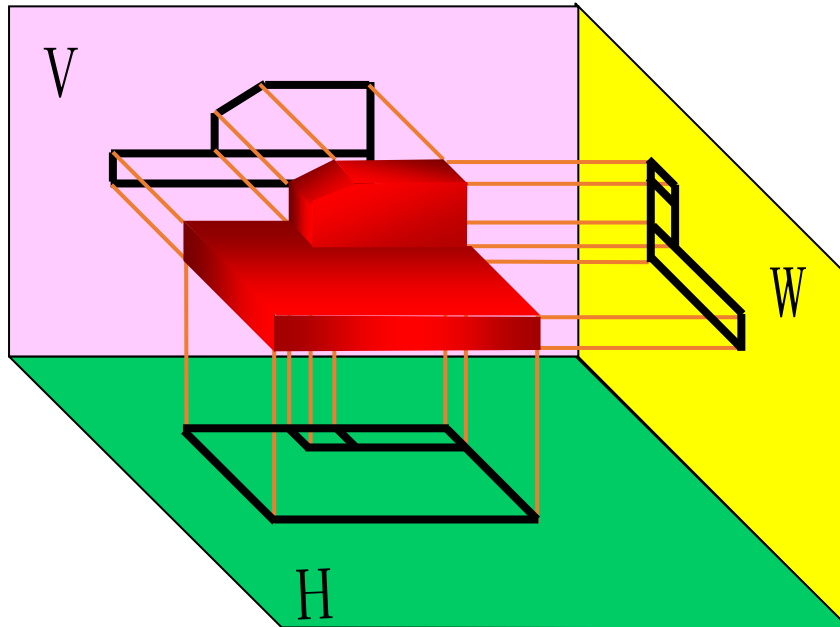
定义常用投影面：

- 正面投影面（简称**正面**或 **V** 面）
 - 水平投影面（简称**水平面**或 **H** 面）
 - 侧面投影面（简称**侧面**或 **W** 面）
- OX轴 V面与H面的交线
 - OY轴 H面与W面的交线
 - OZ轴 V面与W面的交线





将物体放在三个投影面之间，用正投影法分别向三个投影面投射，就得到了三个视图，称为三面视图，简称为三视图

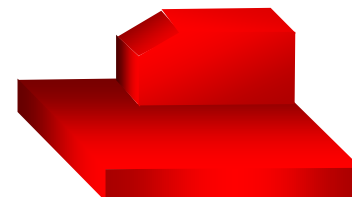
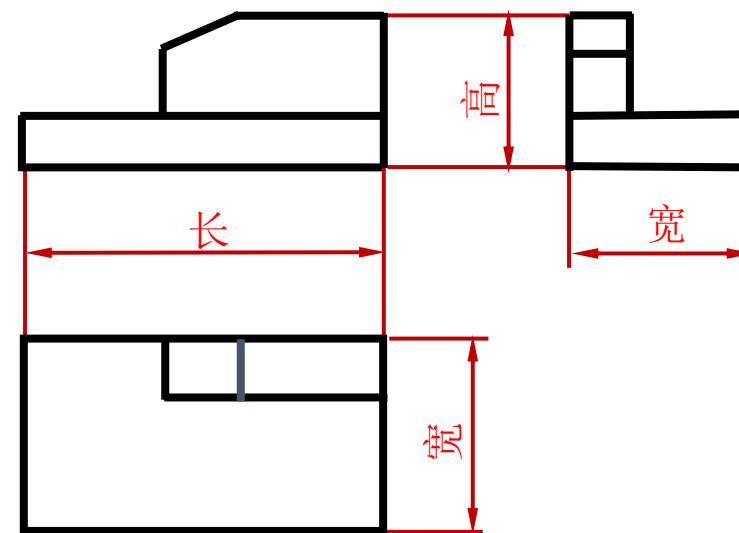


- 主视图 正面投影
- 俯视图 水平投影
- 左视图 侧面投影

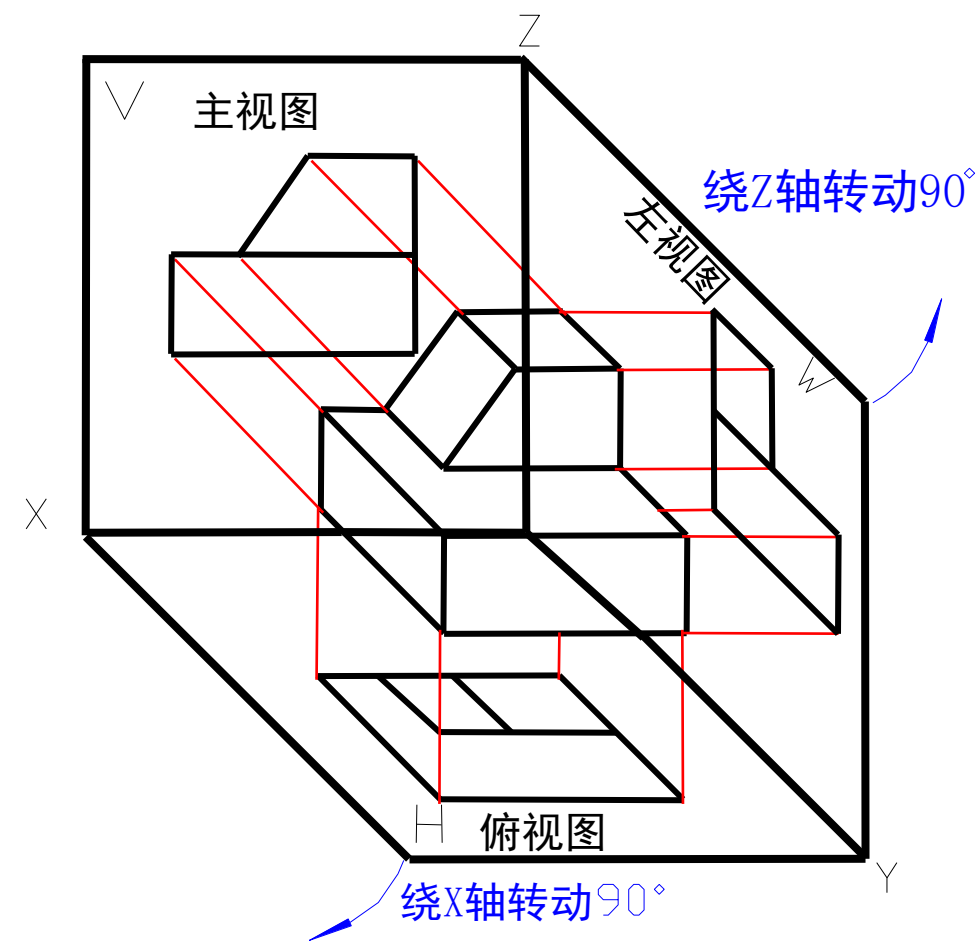
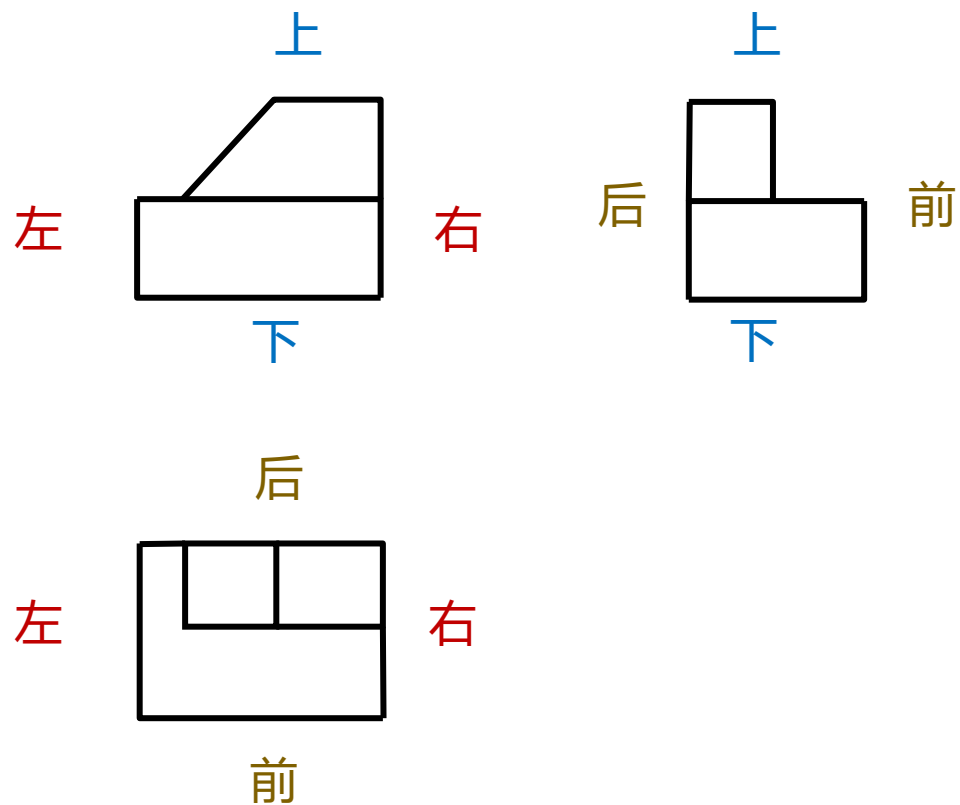
■ 长对正

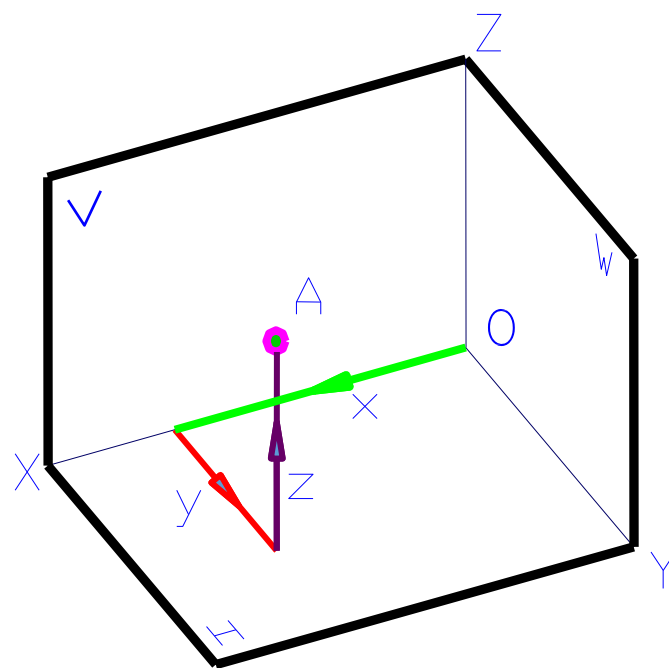
■ 高平齐

■ 宽相等

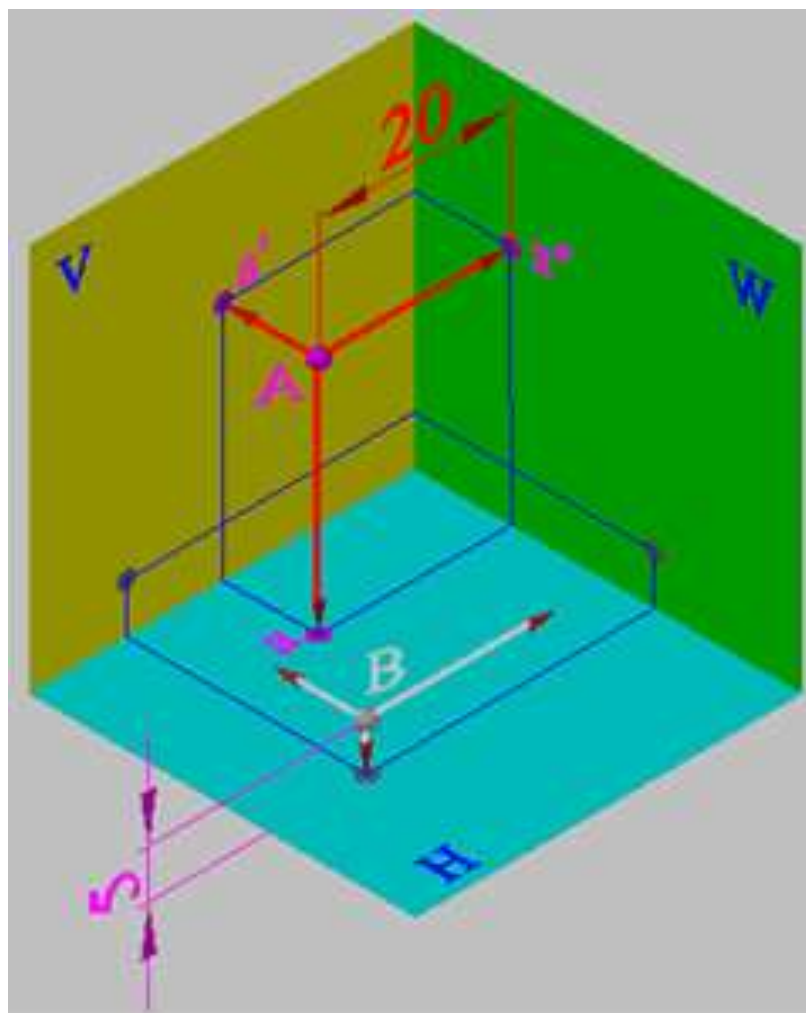


三视图反映物体的位置关系：左右-上下-前后





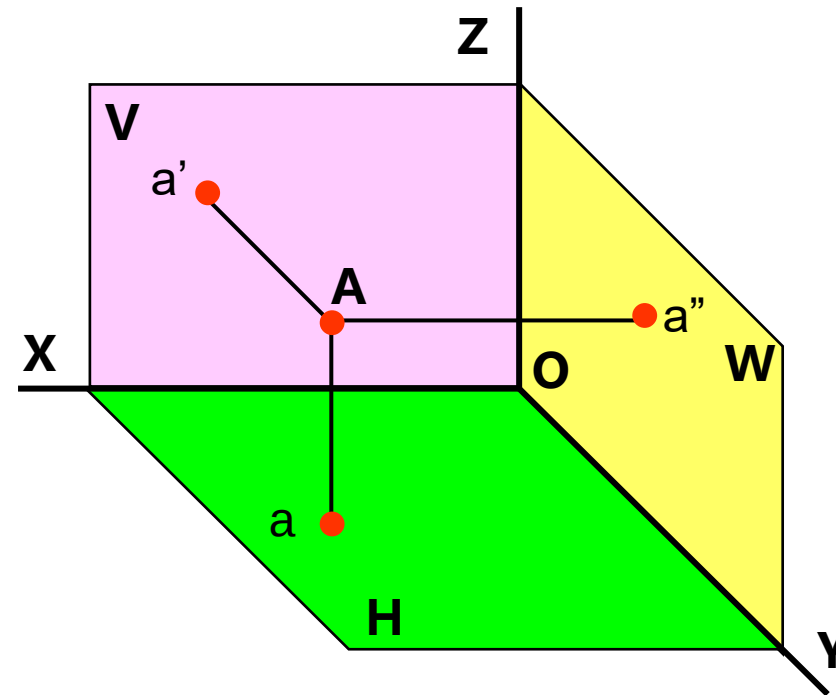
点A和B

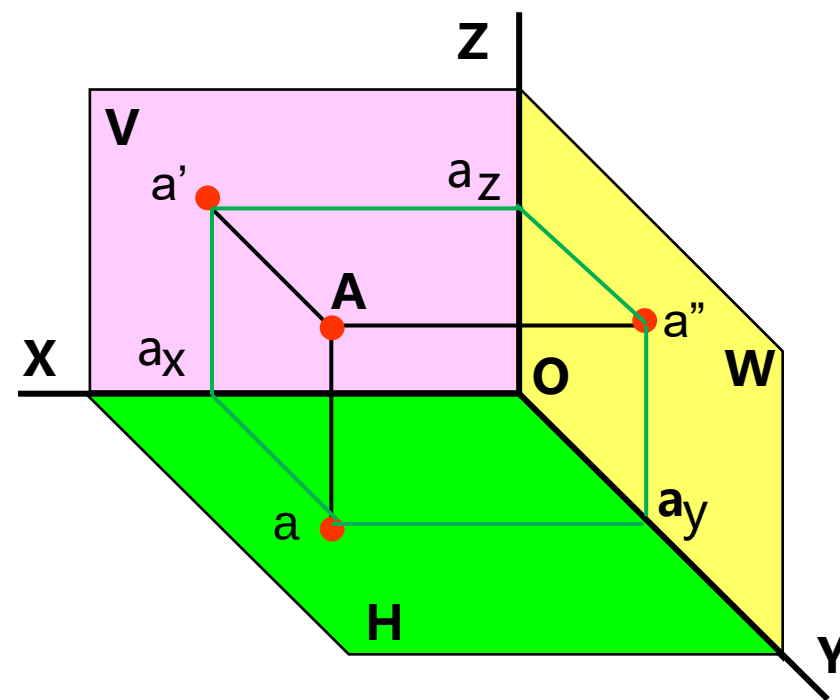
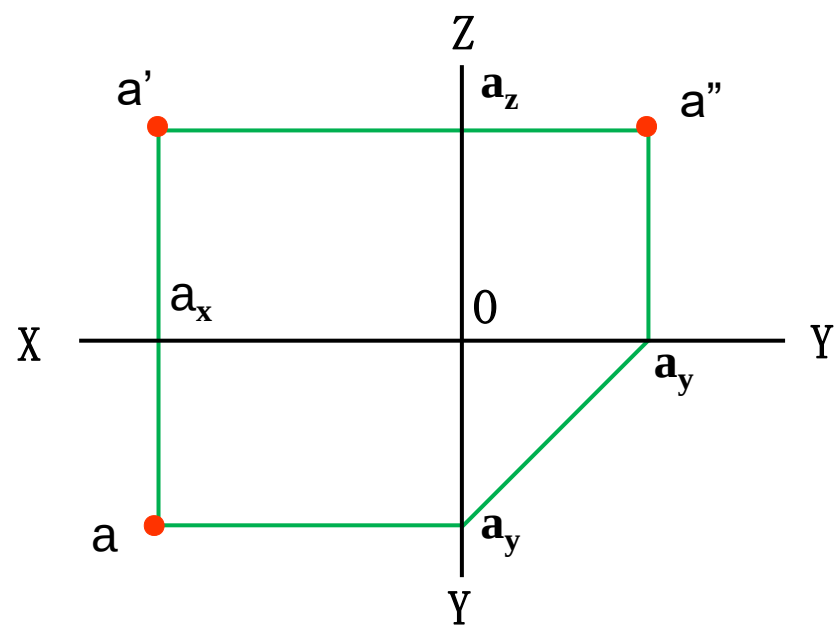


a' 点A的正面投影

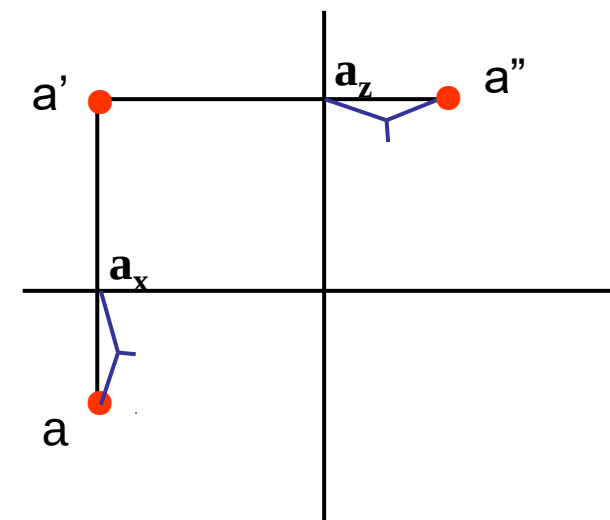
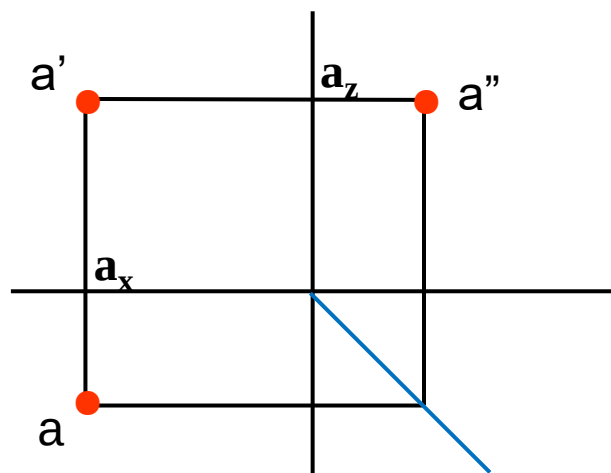
a 点A的水平投影

a'' 点A的侧面投影





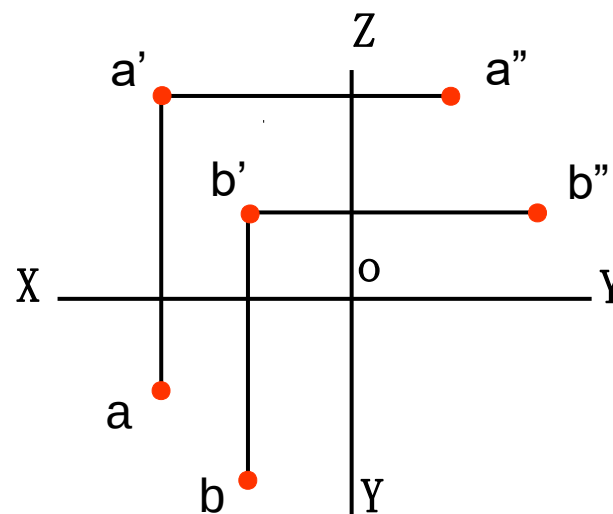
已知点的两个投影，求第三投影



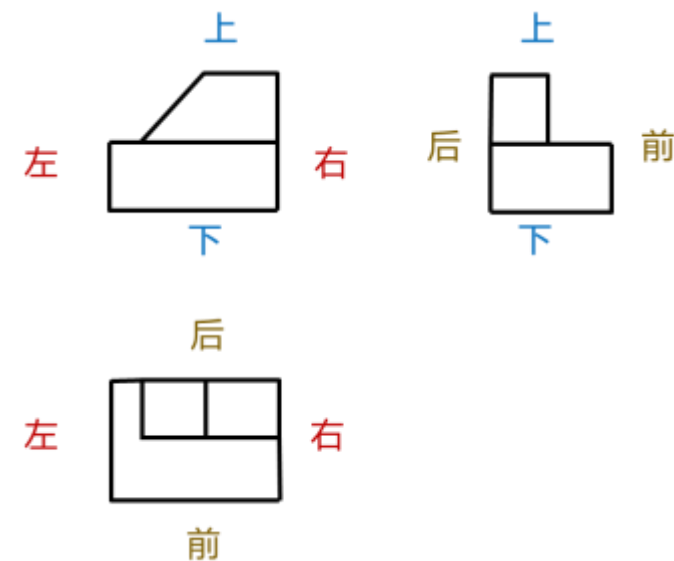
两点的相对位置指两点在空间的上下、前后、左右位置关系

判断方法：

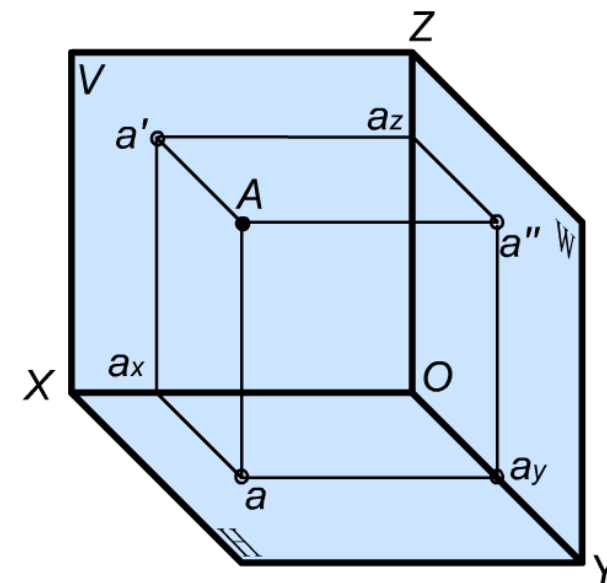
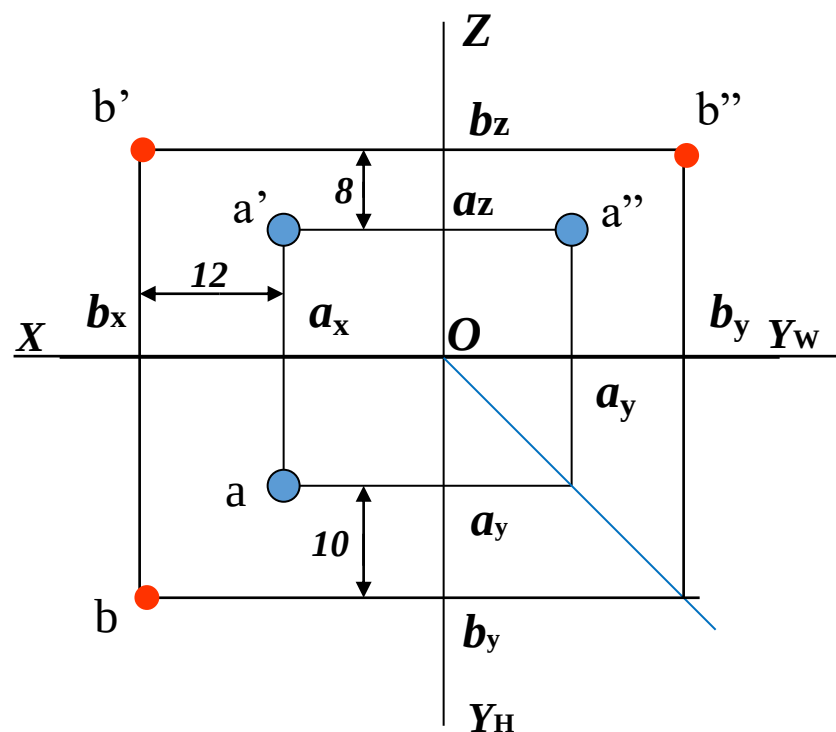
- x 坐标大的在 左
- y 坐标大的在 前
- z 坐标大的在 上



B 点在 A 点之右、之前、之下



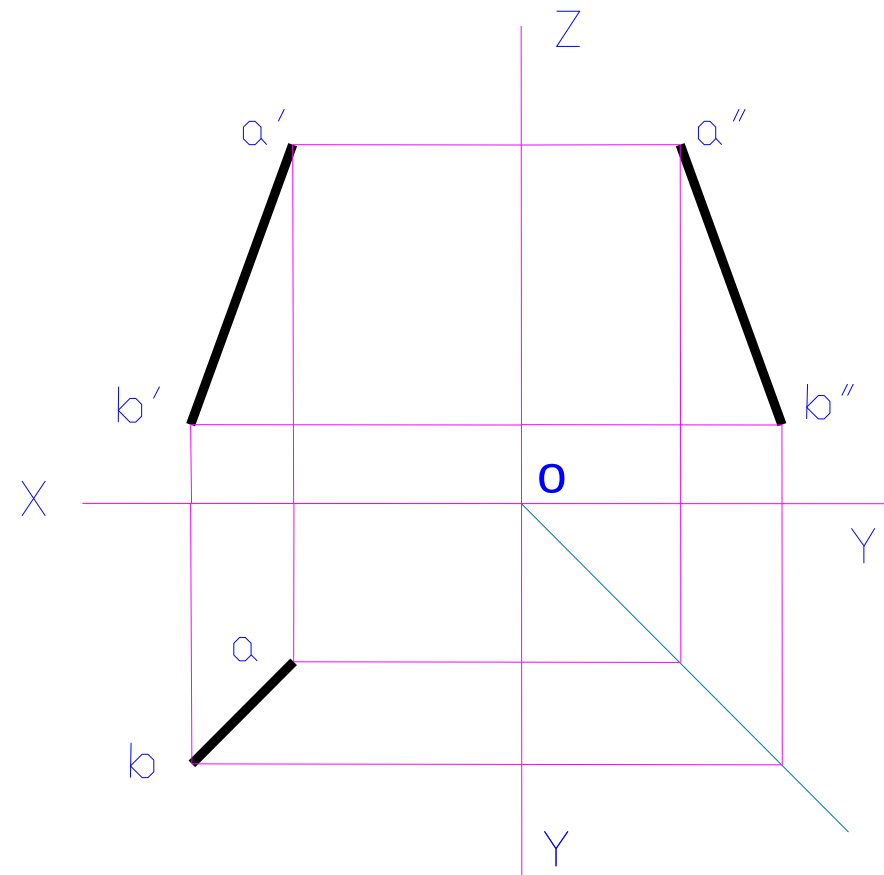
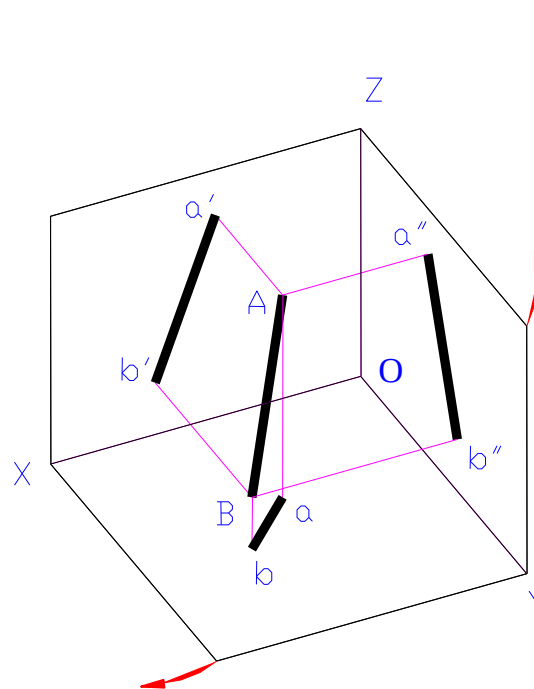
已知点 A 的三投影，另一点 B 在点 A 上方8 mm、左方12 mm、前方10 mm处，求：点 B 的三个投影。

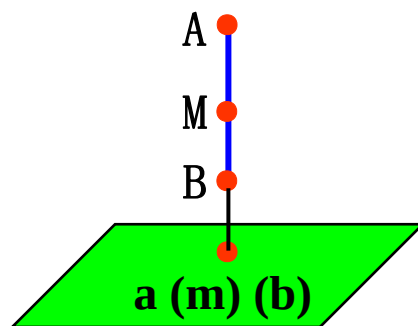




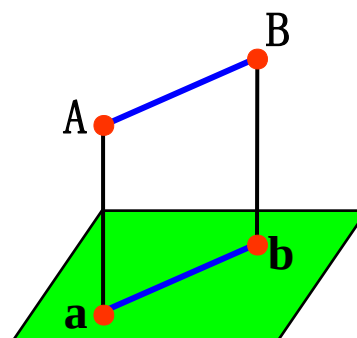
连点成线

- ◆ 一般情况下，直线的投影仍为直线
- ◆ 两点确定一条直线，将两点的同面投影用直线连接，就得到直线的同面投影

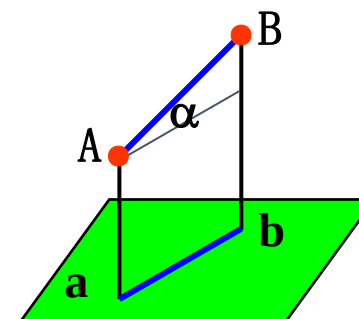




直线垂直于投影面
投影重合为一点

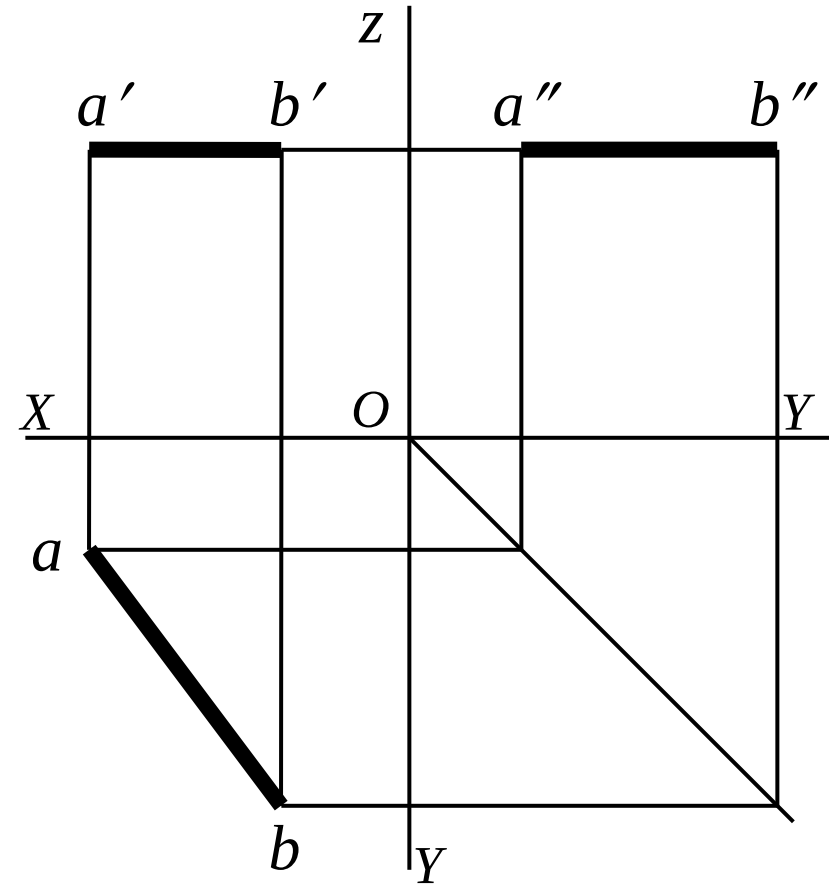
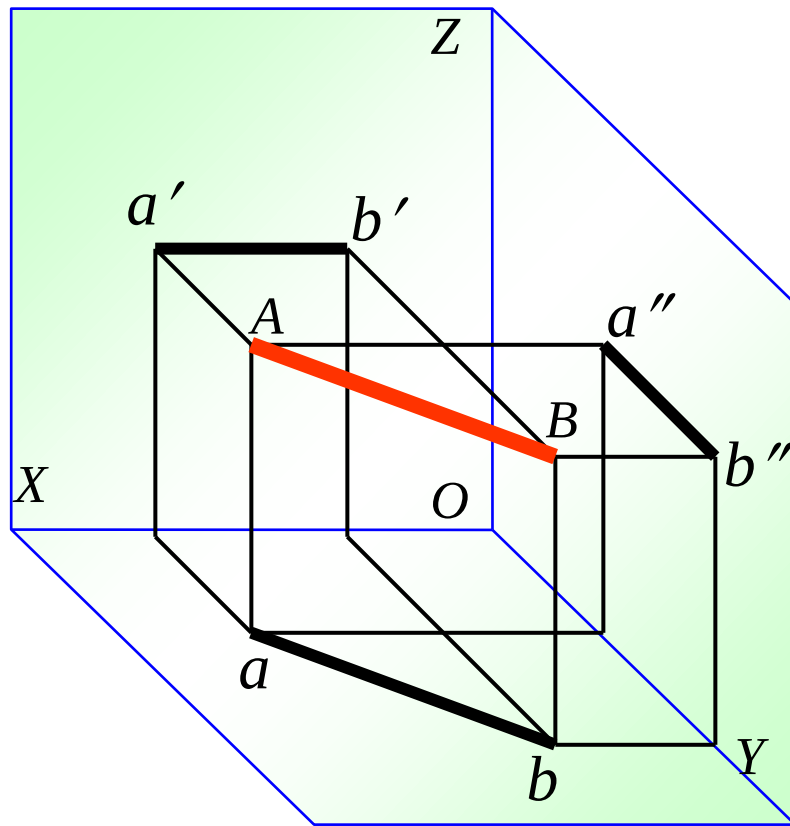


直线平行于投影面
投影反映线段实长
 $ab=AB$

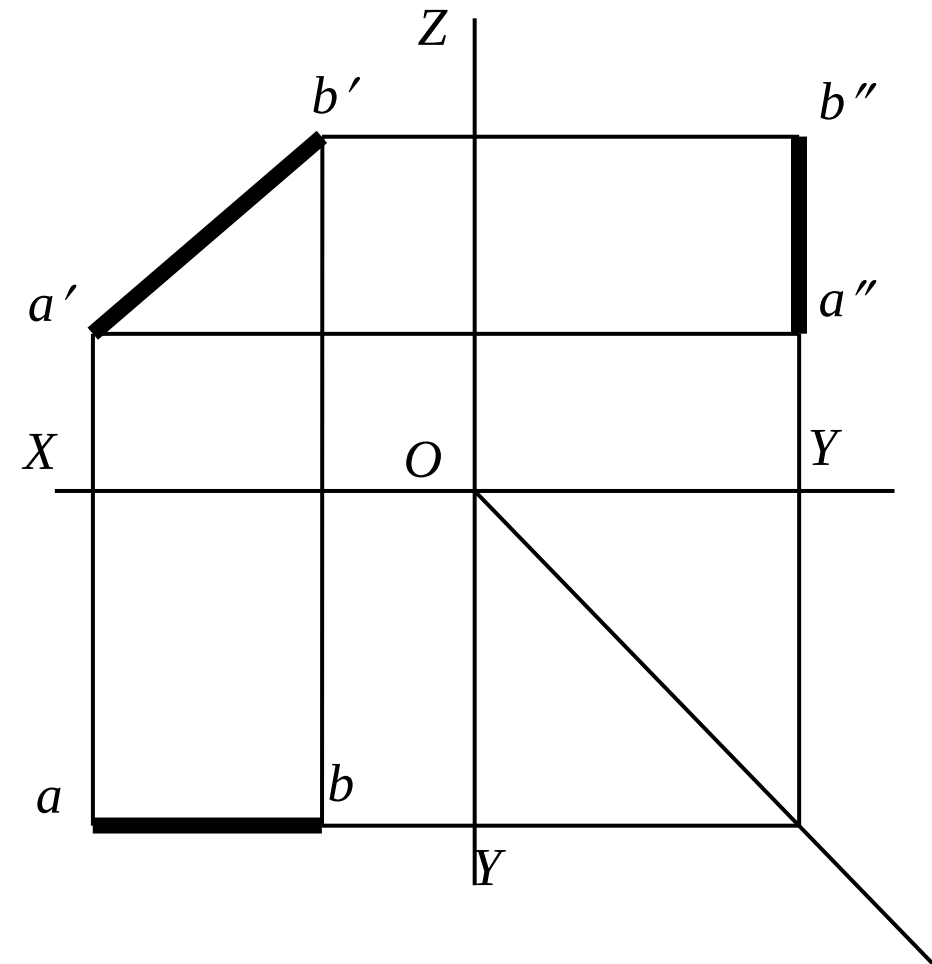
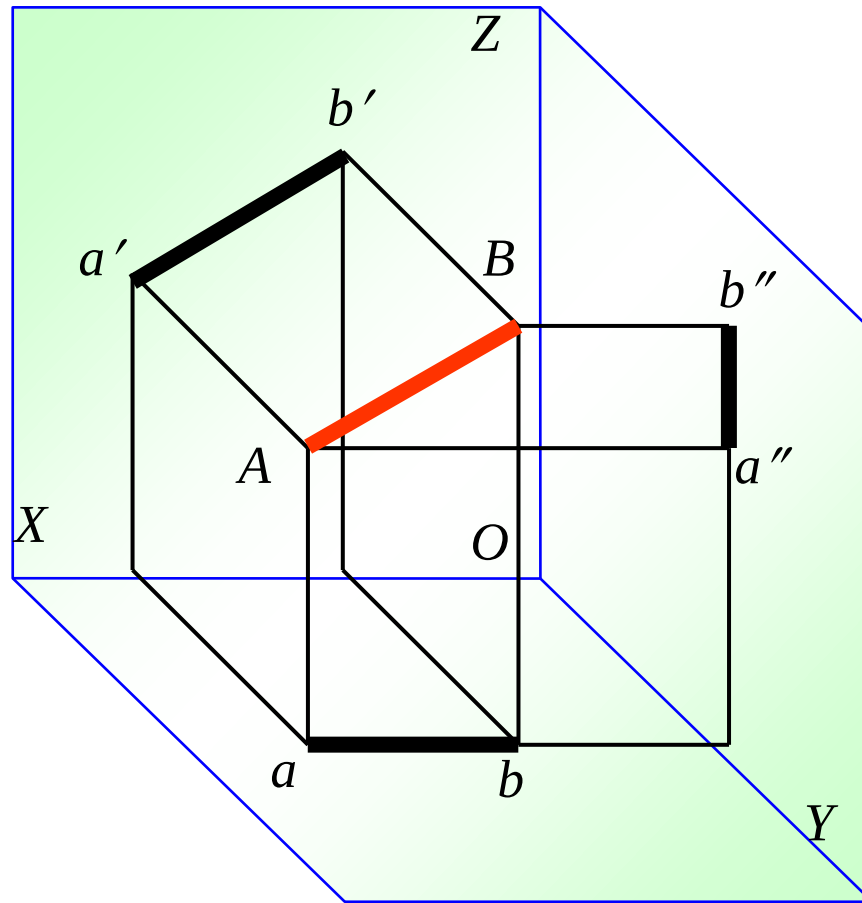


直线倾斜于投影面
投影比空间线段短
 $ab=AB \cdot \cos \alpha$

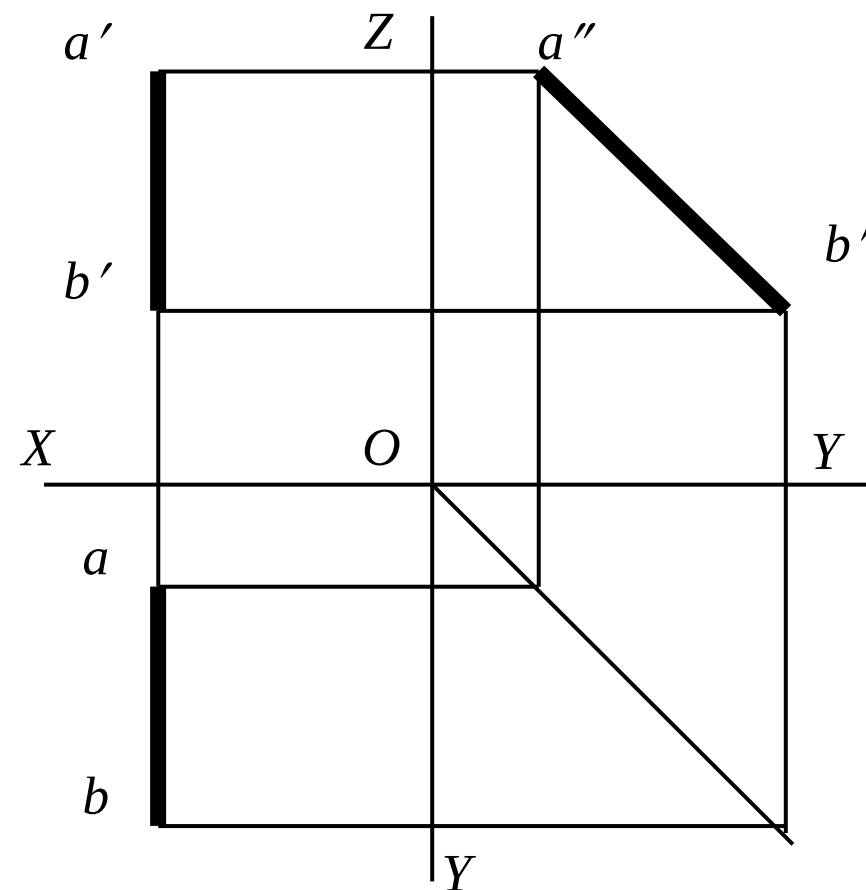
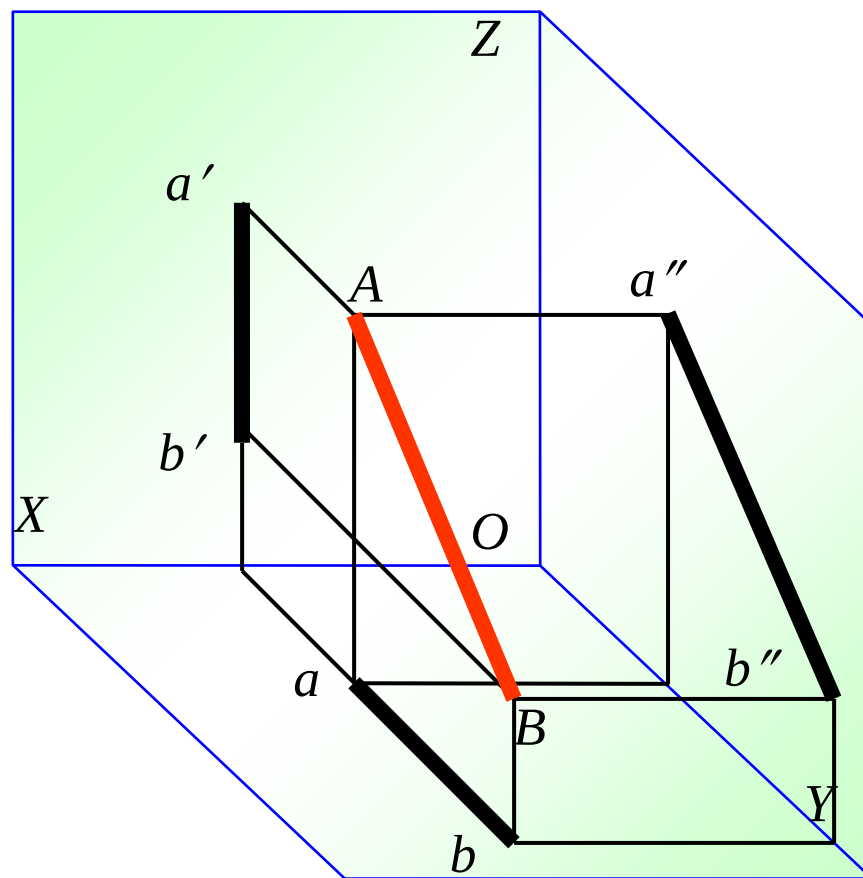
■ 水平线的投影

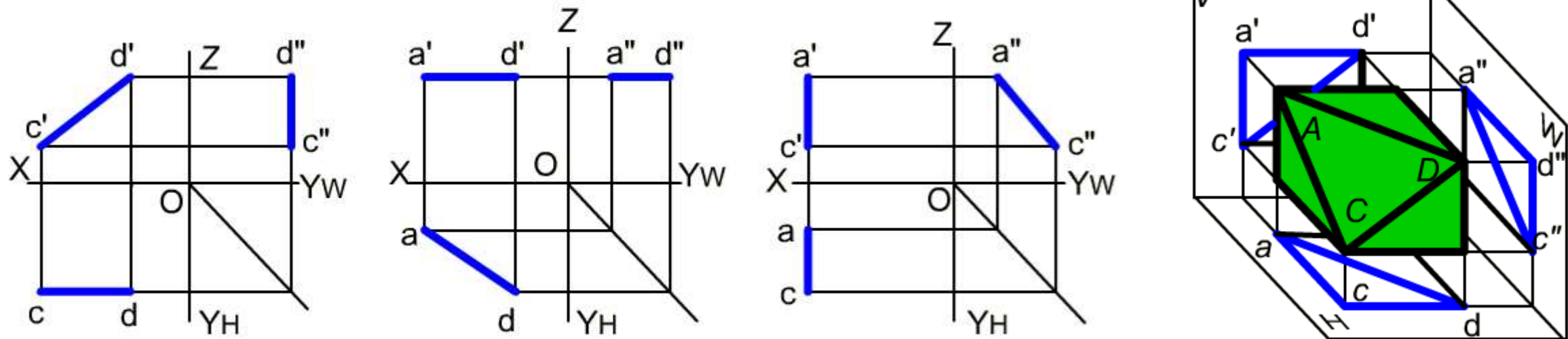


■ 正平线的投影



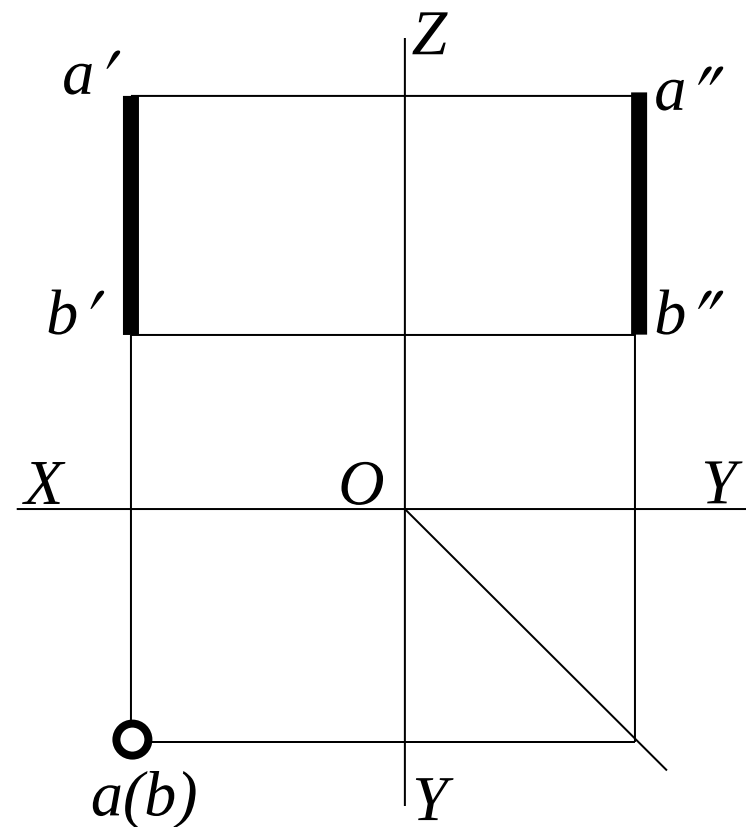
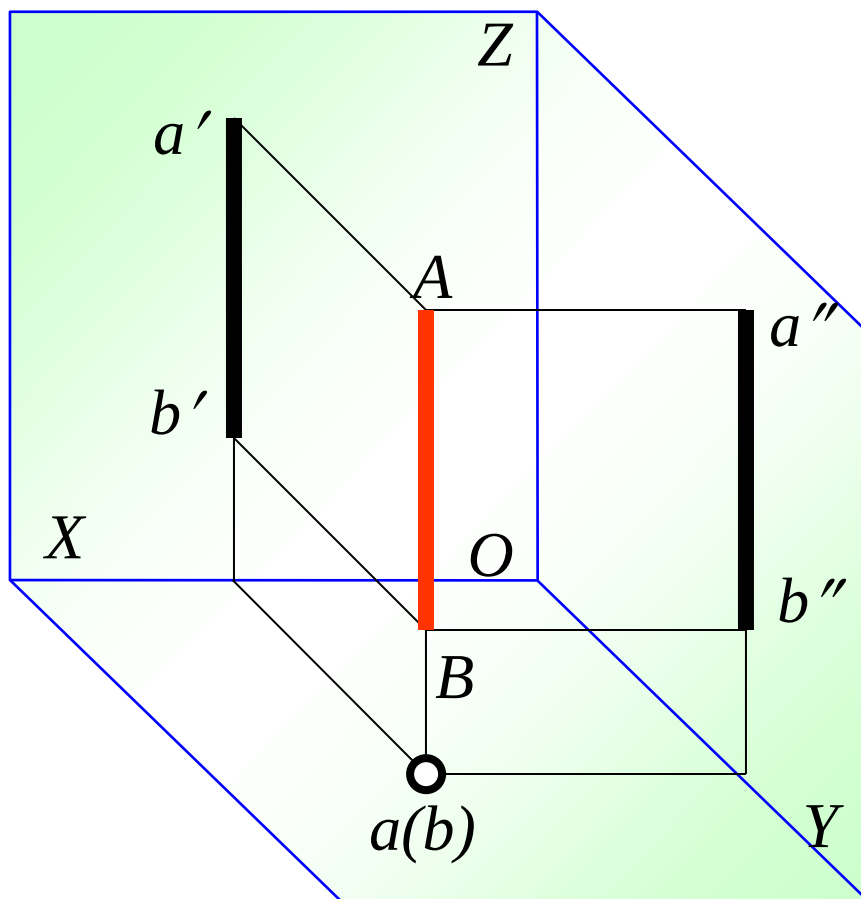
■ 侧平线的投影



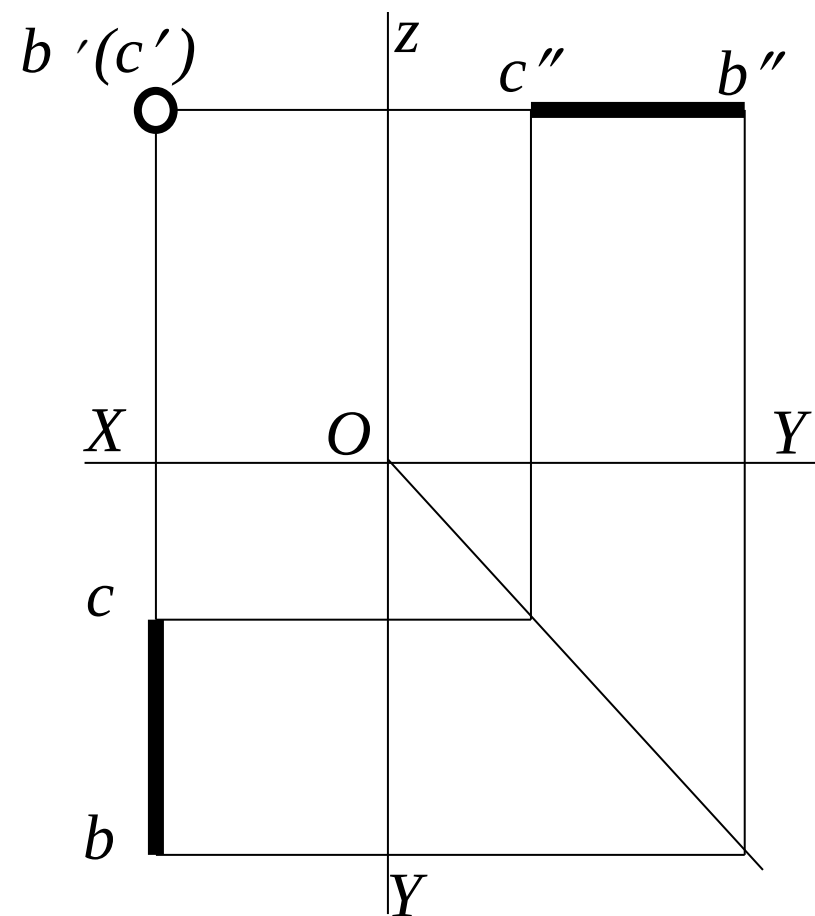
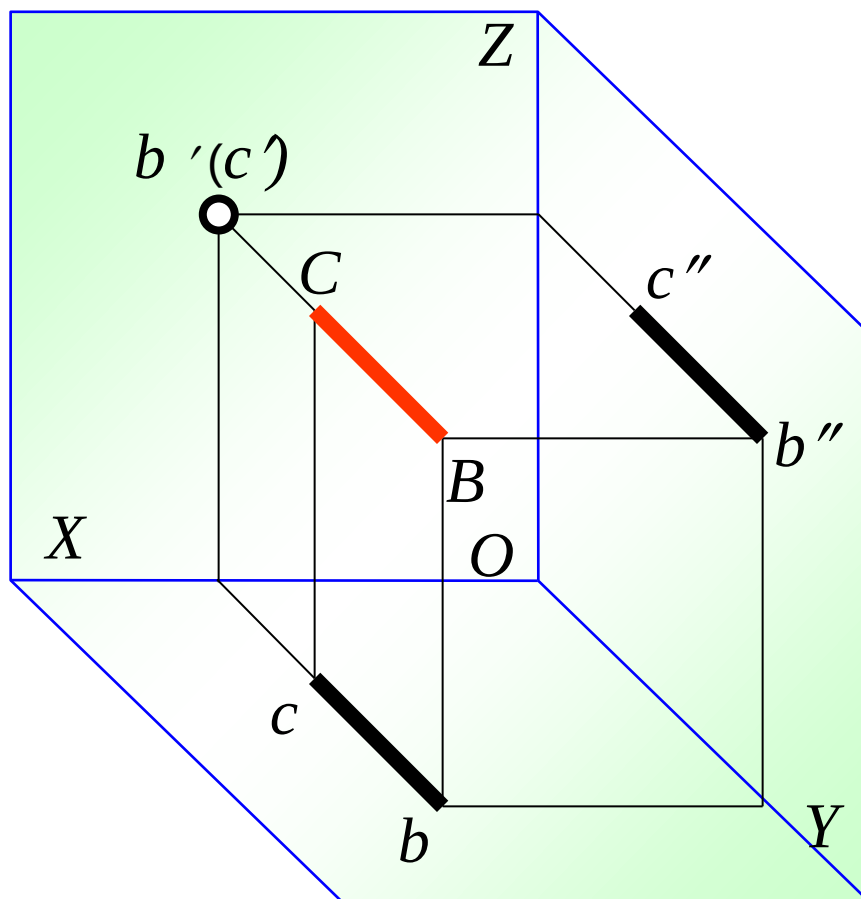


- ✓ 在与线段平行的投影面上，该线段的投影为倾斜的线段，反映实长。
- ✓ 其余两个投影分别平行于相应的投影轴，且都小于实长。

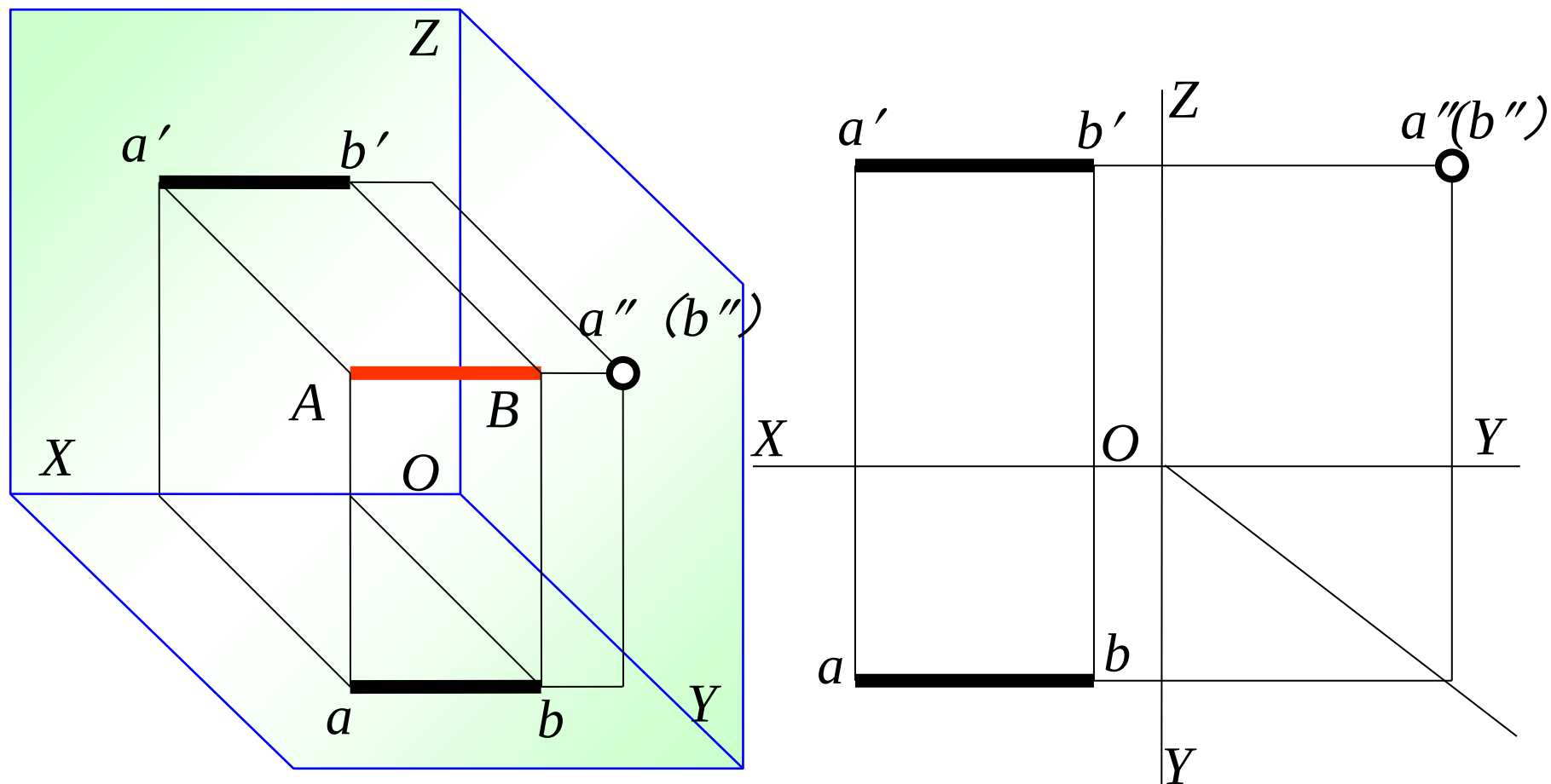
■ 铅垂线的投影

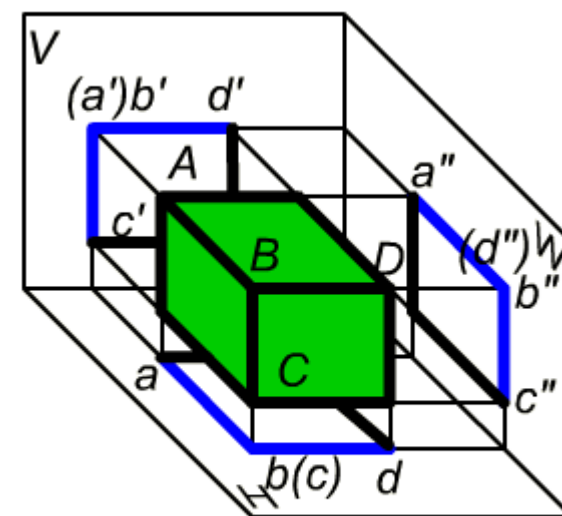
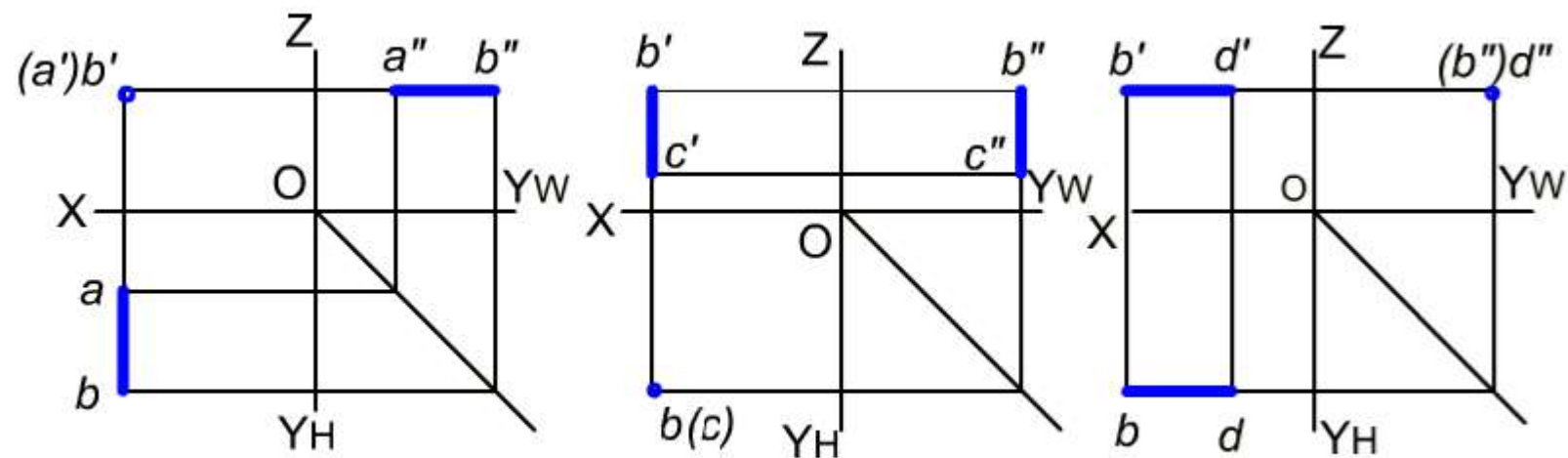


■ 正垂线的投影



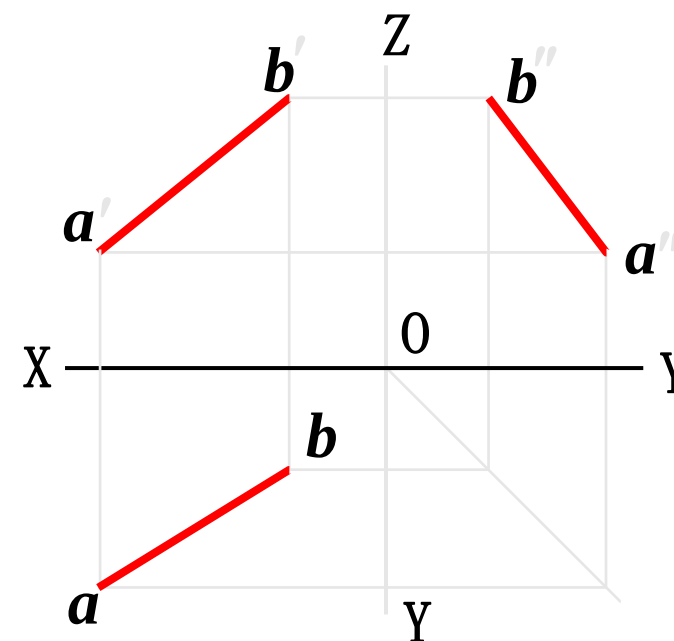
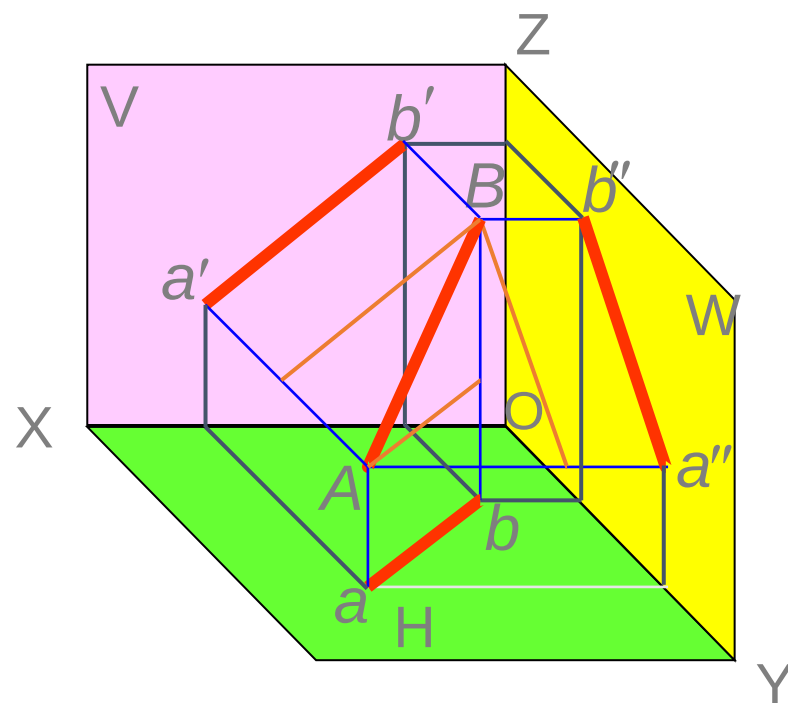
■ 侧垂线的投影



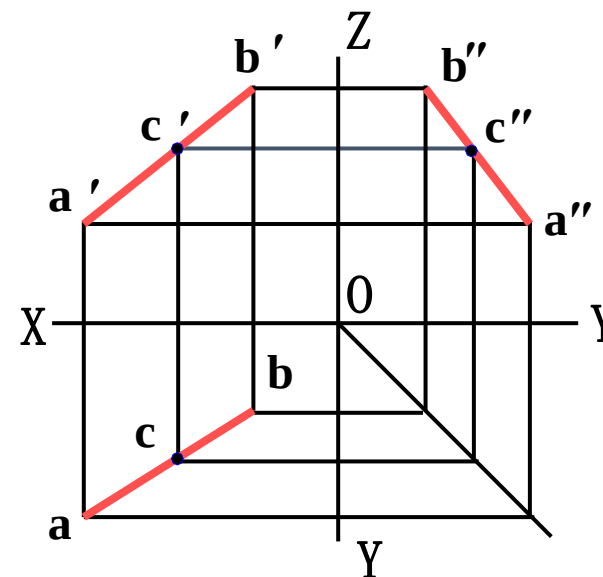
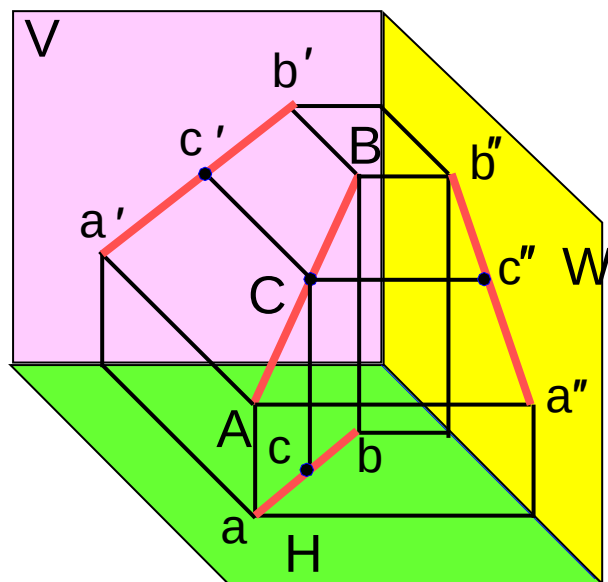


- ✓ 在与线段垂直的投影面上，该线段的投影积聚成一点。
- ✓ 其余两个投影垂直与相应的投影轴，且反映该线段的实长。

■ 一般位置直线的投影

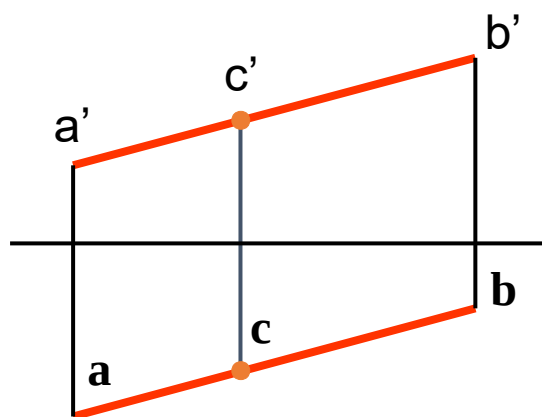


- ✓ 由于一般位置直线对三个投影面都倾斜，因此，其三个投影都是倾斜线段，且都小于该直线段的实长

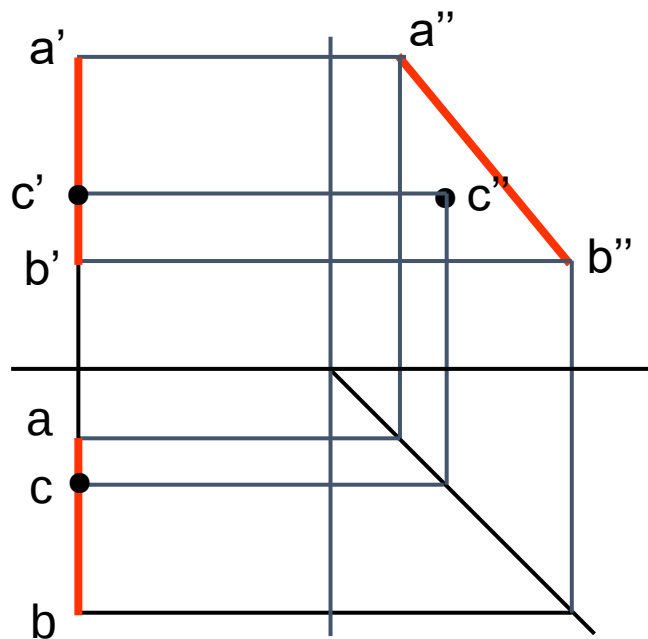


- ✓ 若点在直线上, 则点的投影必在直线的同名投影上
- ✓ 点的投影将线段的同名投影分割成与空间线段相同的比例

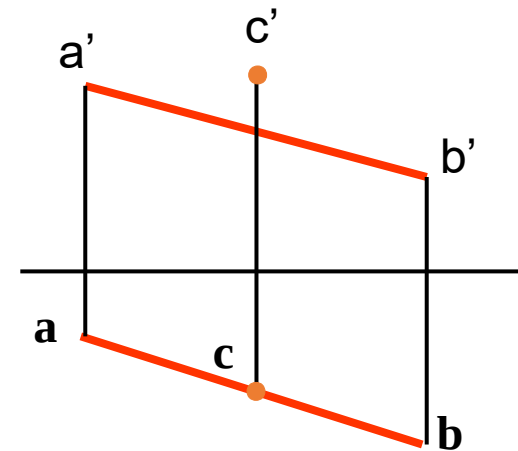
判断点C是否在线段AB上。



在



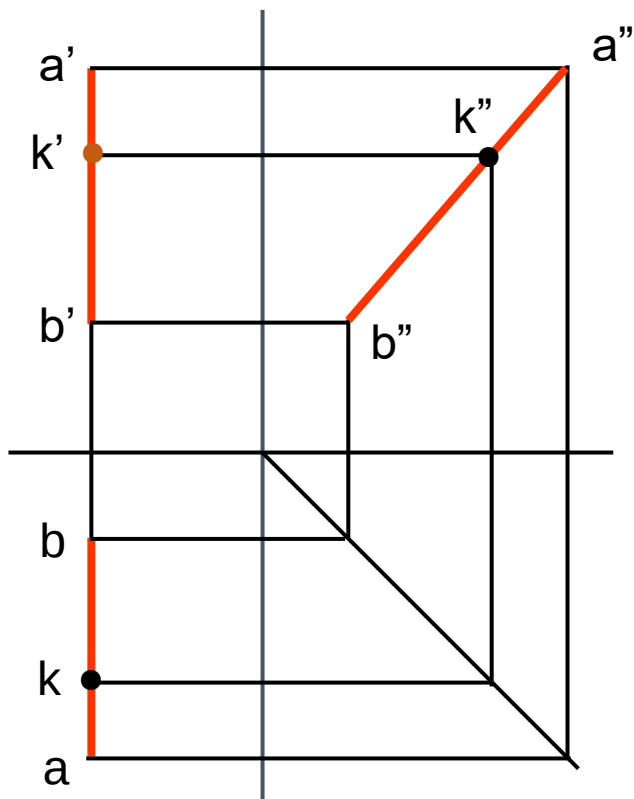
不在



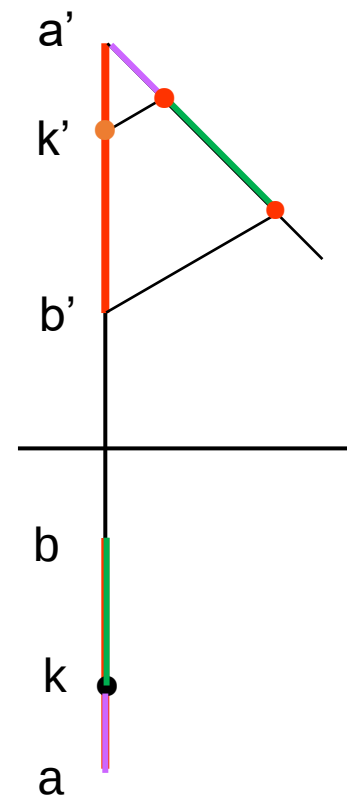
不在

已知点K在线段AB上，求点K正面投影。

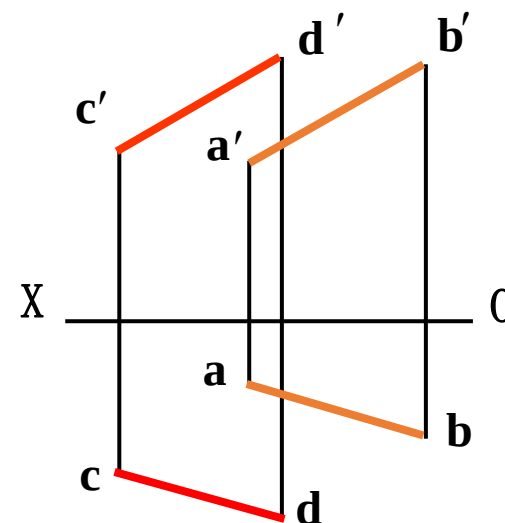
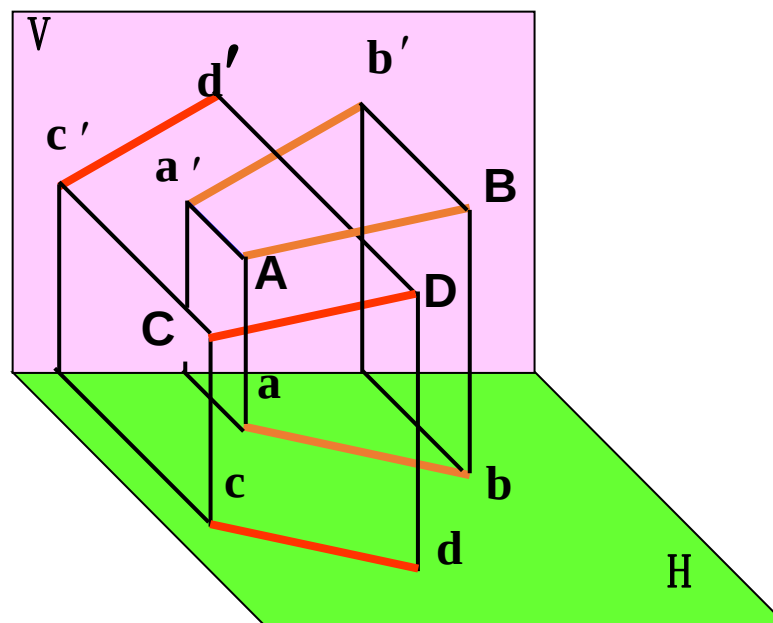
解法一：应用第三投影



解法二：应用定比定理

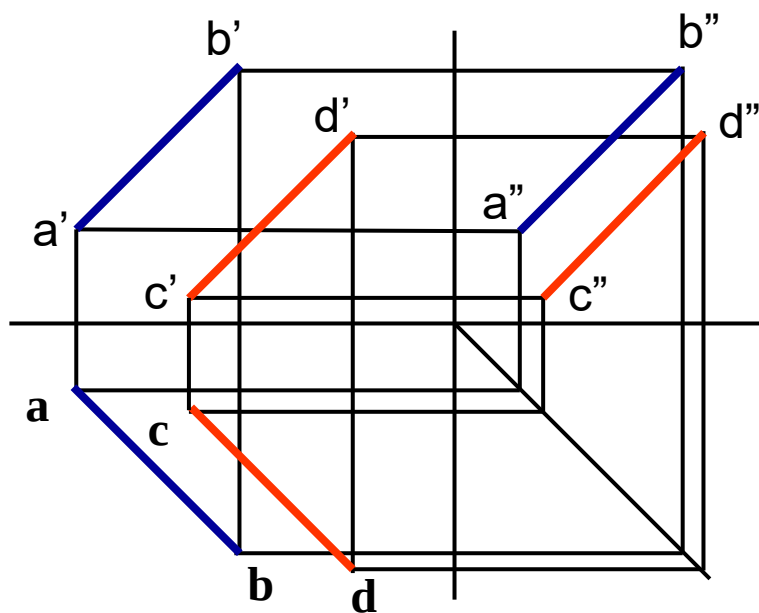


■ 平行



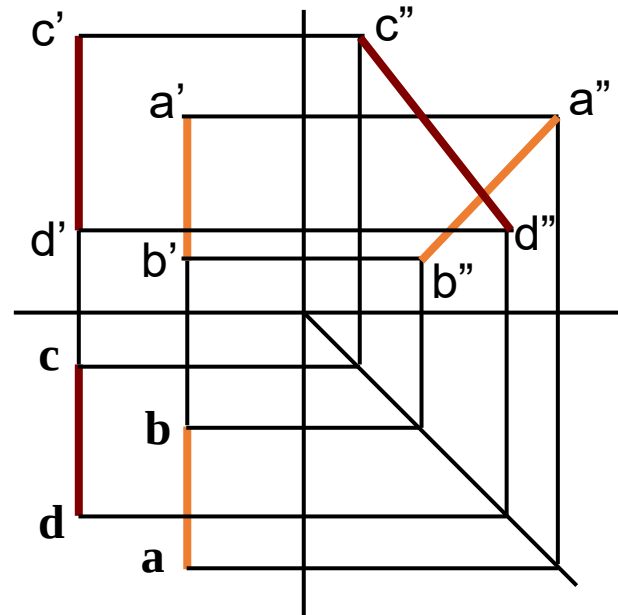
✓ 空间两直线平行，则其各同名投影必相互平行，反之亦然

判断两直线是否平行。



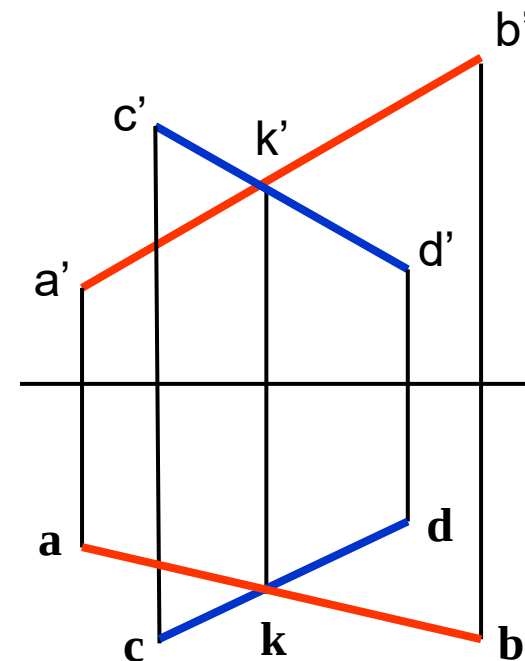
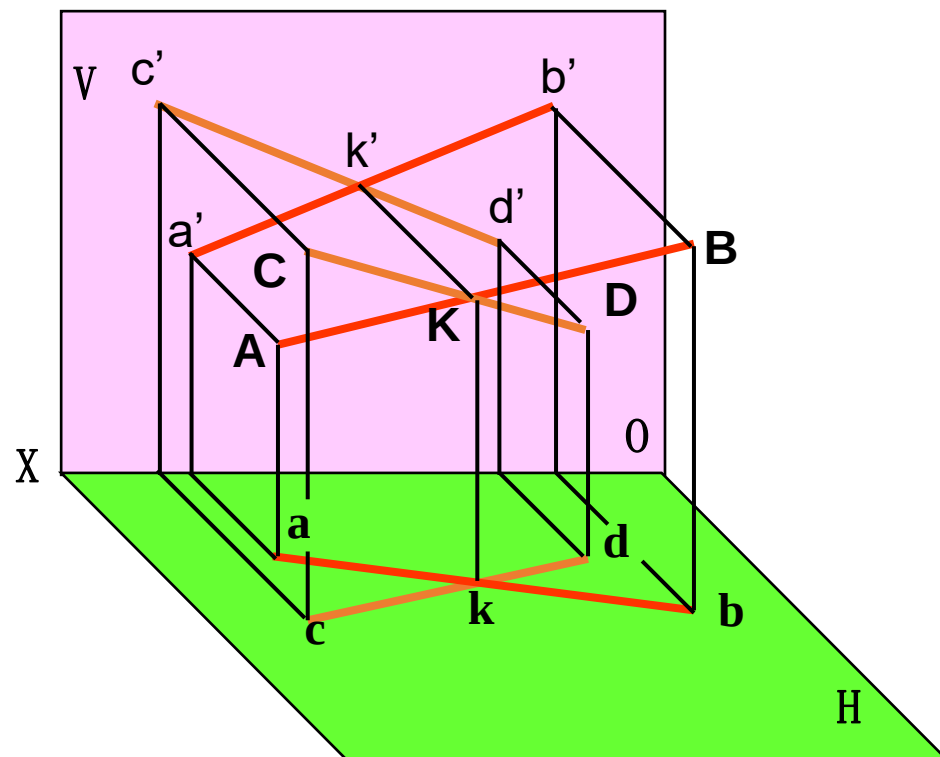
- ✓ 对于一般位置直线，只要有两组同名投影互相平行，空间两直线就平行

判断两直线是否平行。



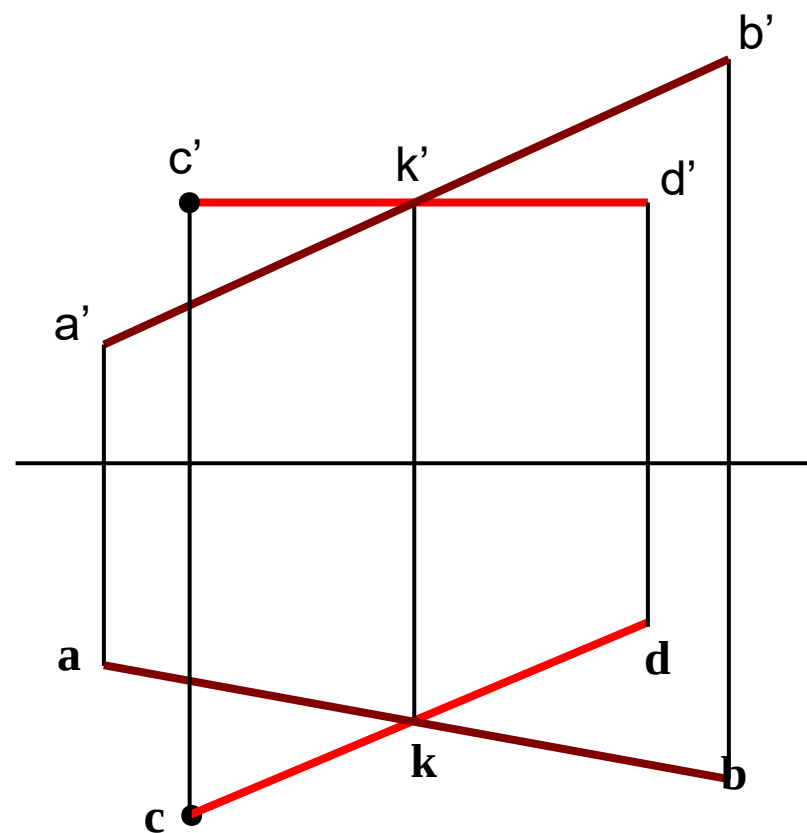
- ✓ 对于特殊位置直线，只有两组同名投影互相平行，空间直线不一定平行

■ 相交

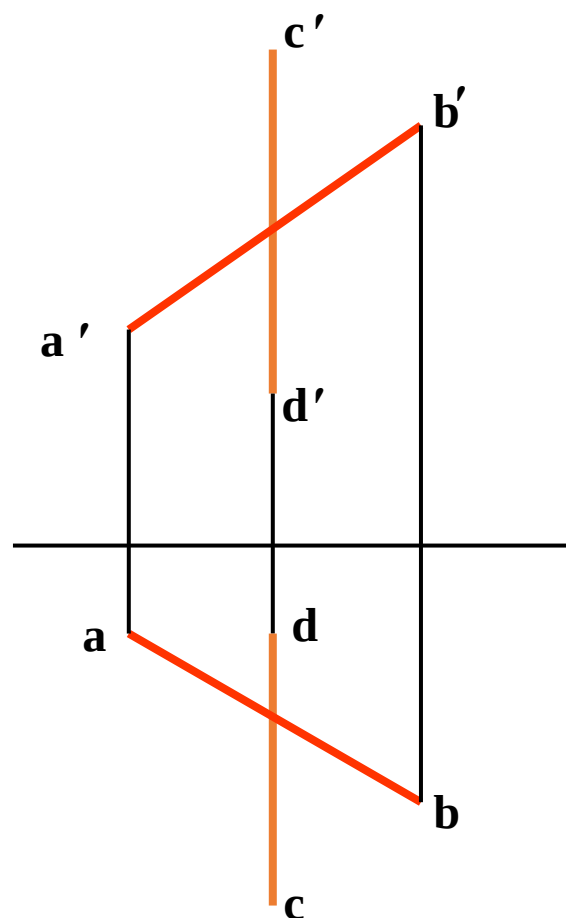


✓ 若空间两直线相交，则其同名投影必相交，且交点的投影必符合空间一点的投影特性

过C点作水平线CD与AB相交。



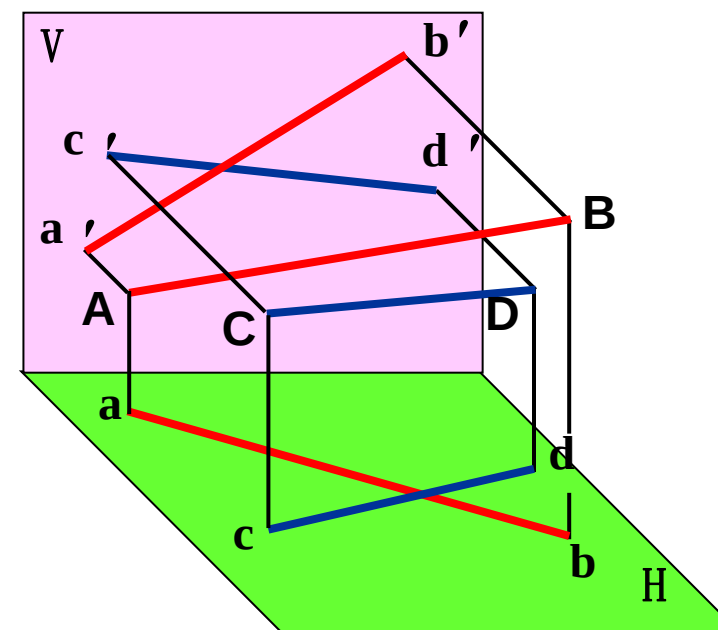
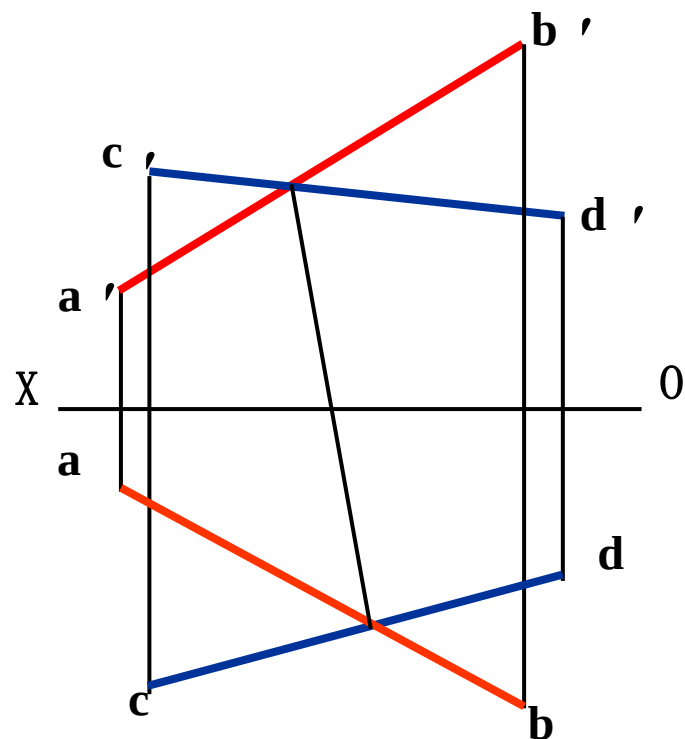
判断直线AB、CD是否相交。

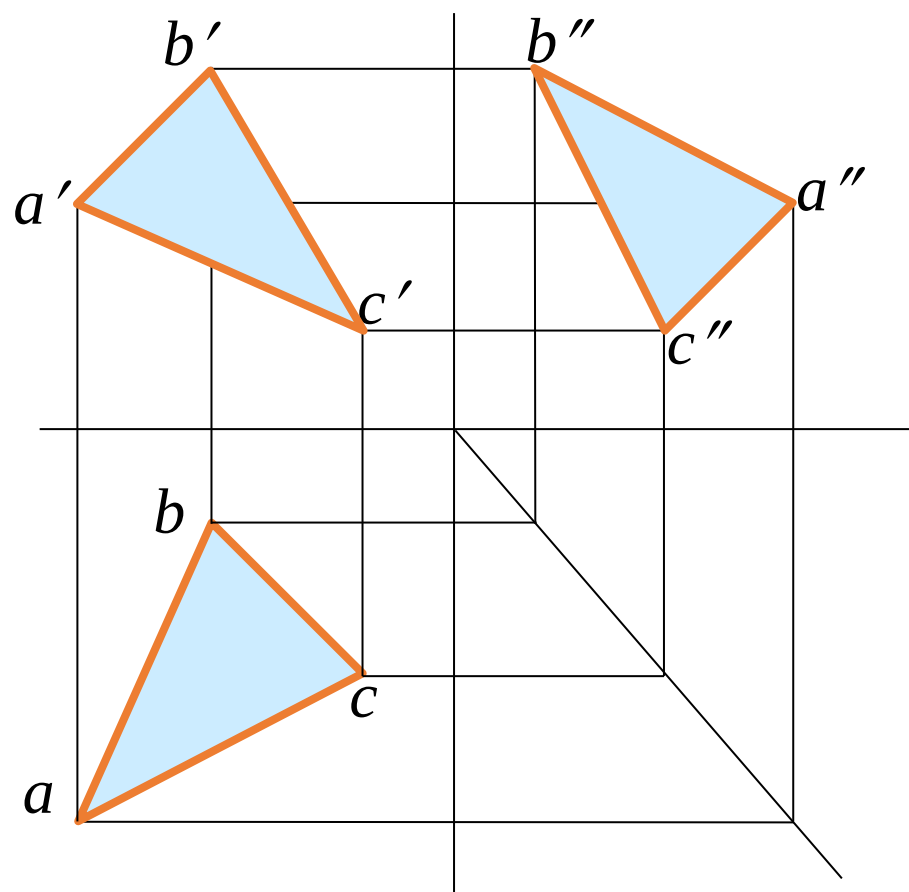


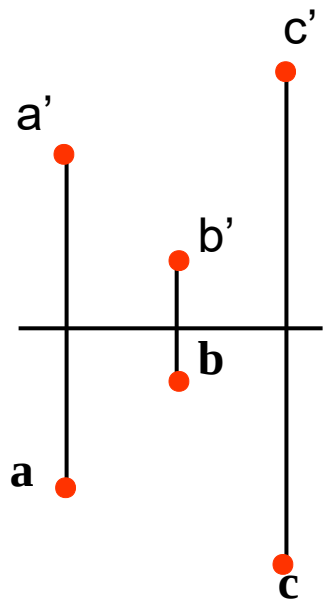
判断方法？

- ◆ 应用定比定理
- ◆ 利用侧面投影

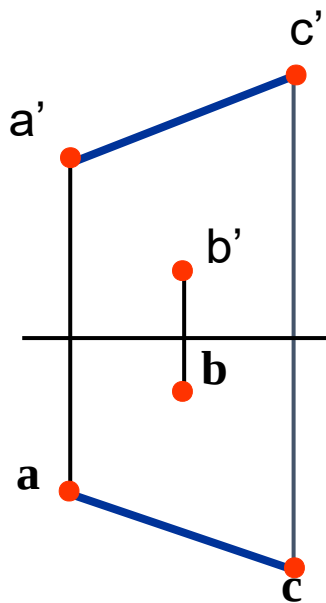
■ 相错



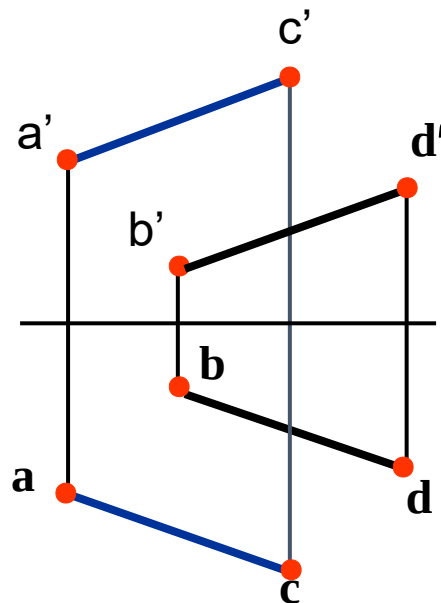




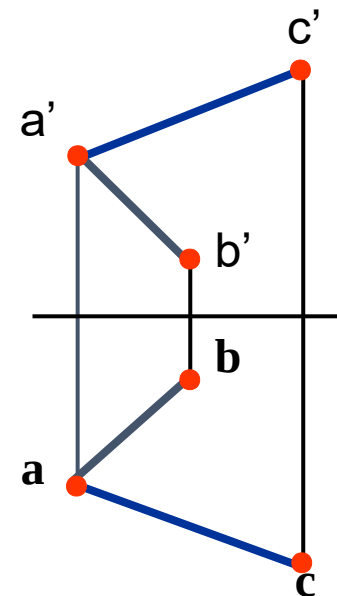
✓ 不在同一直线上的三个点



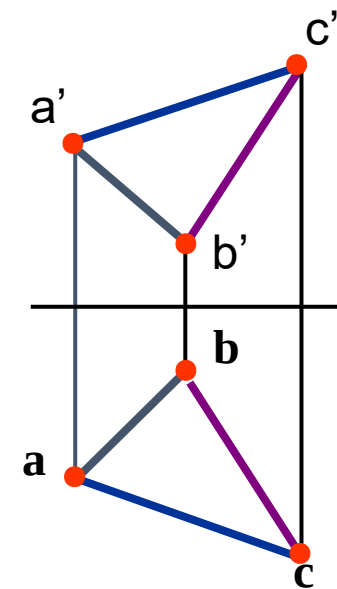
✓ 直线及线外一点



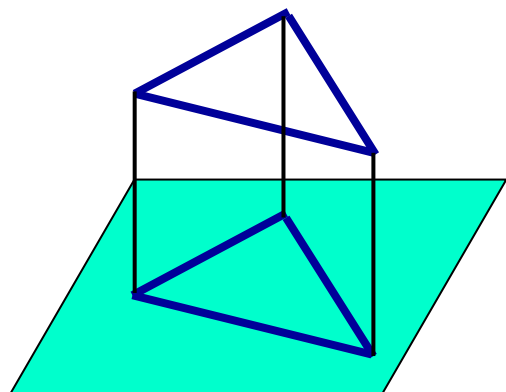
✓ 两平行直线



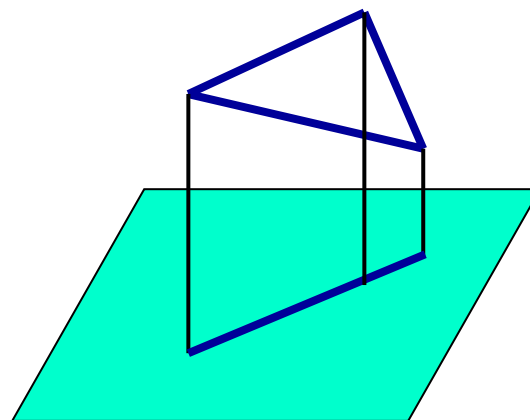
✓ 两相交直线



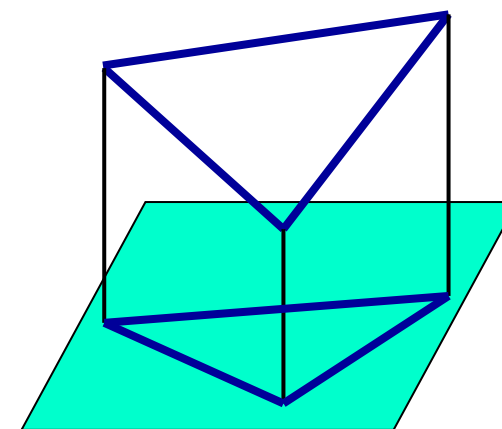
✓ 平面图形



平行

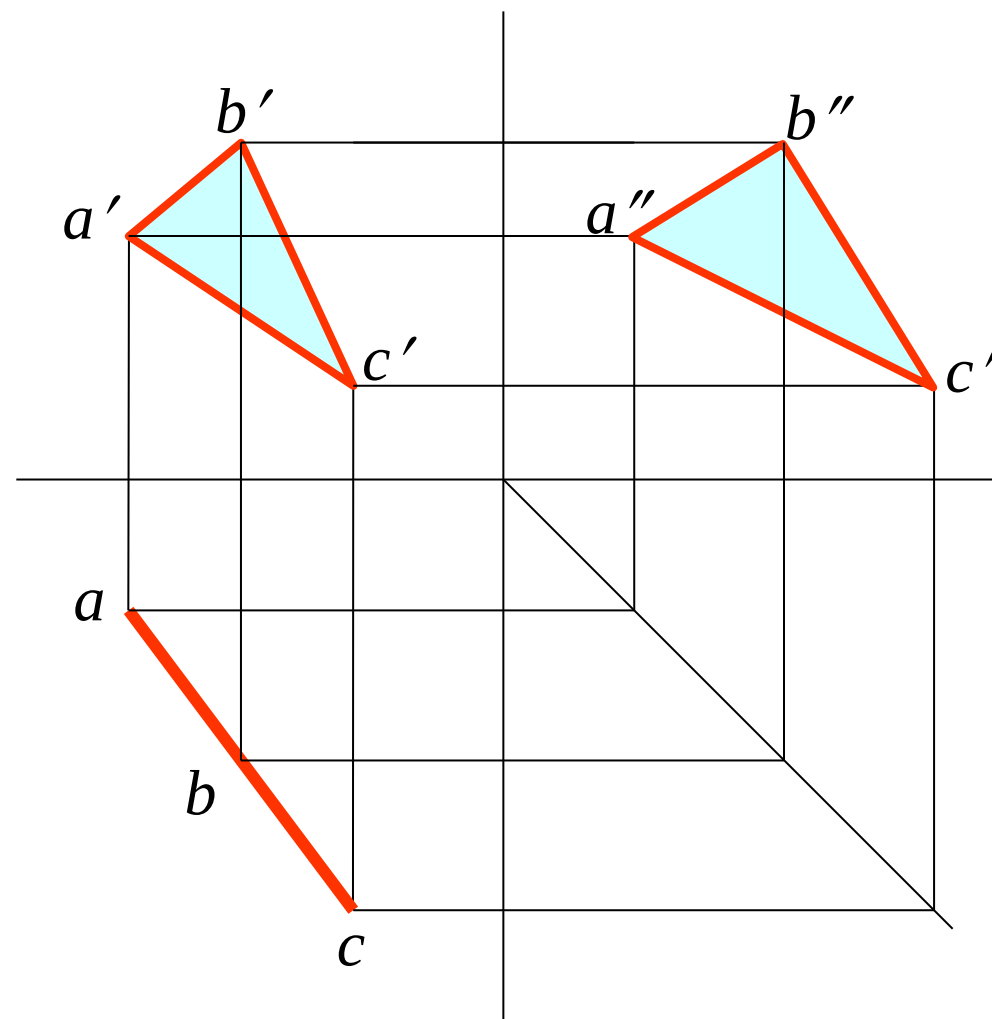
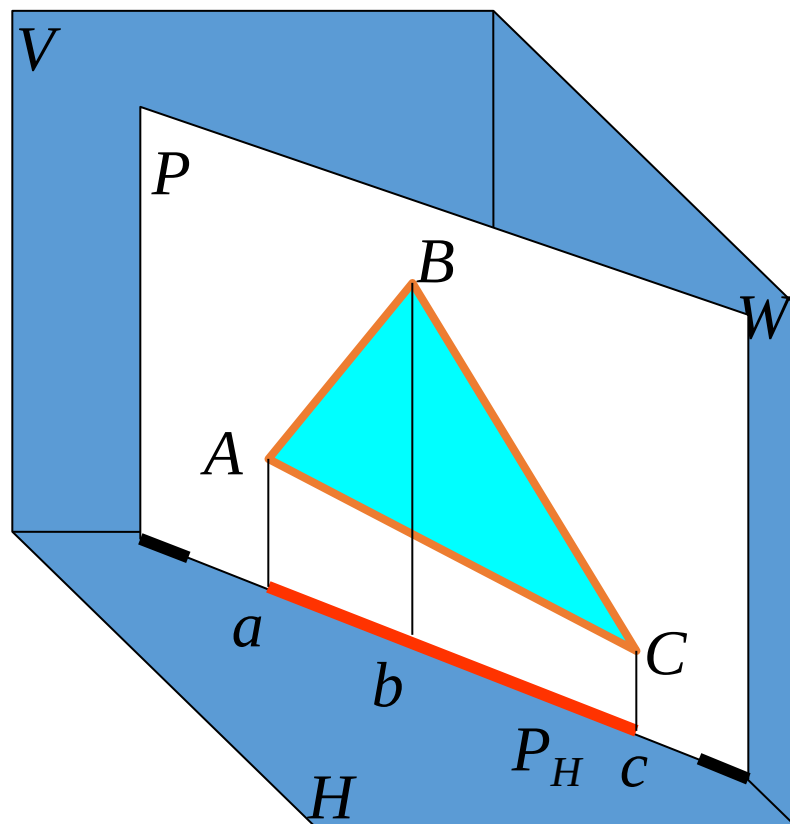


垂直

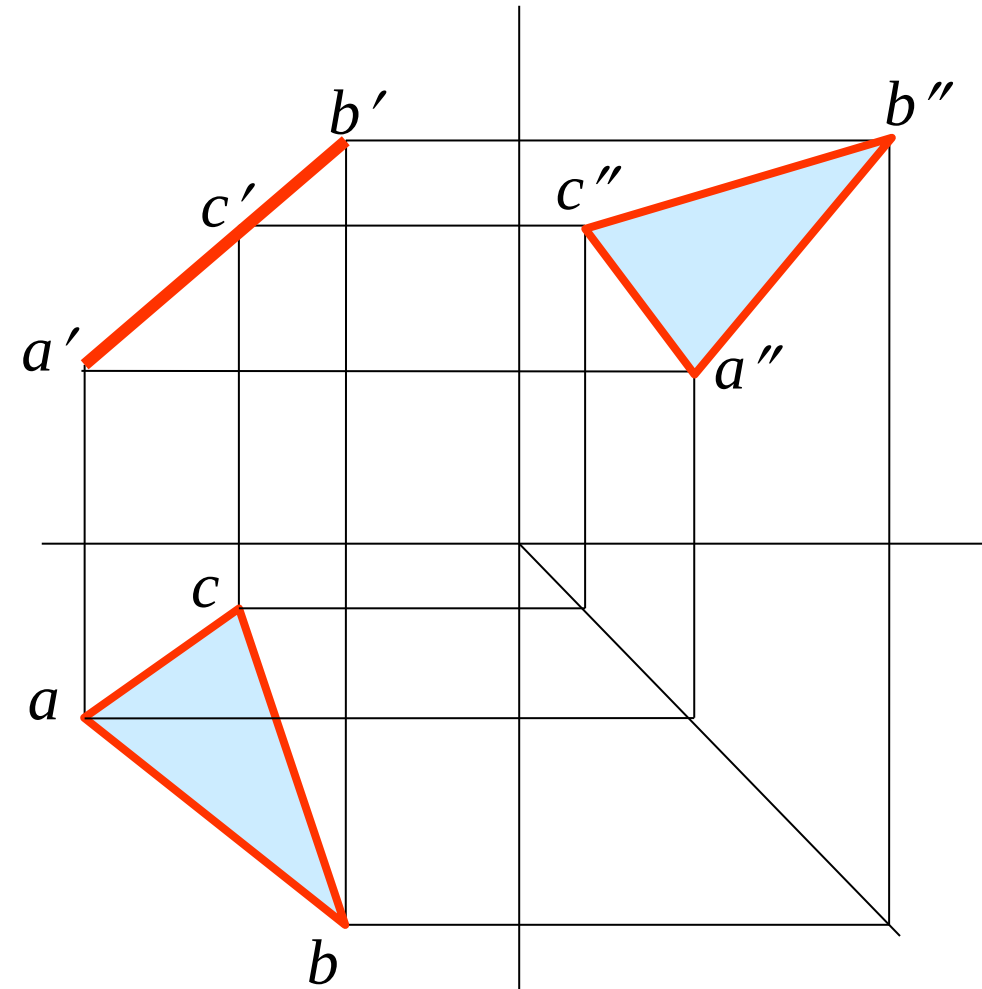
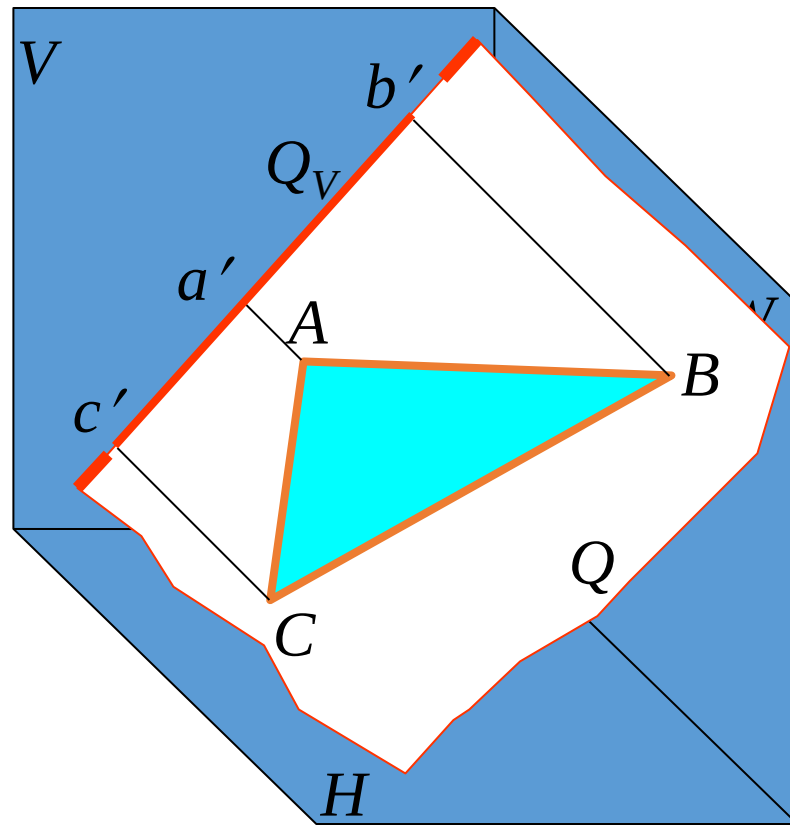


倾斜

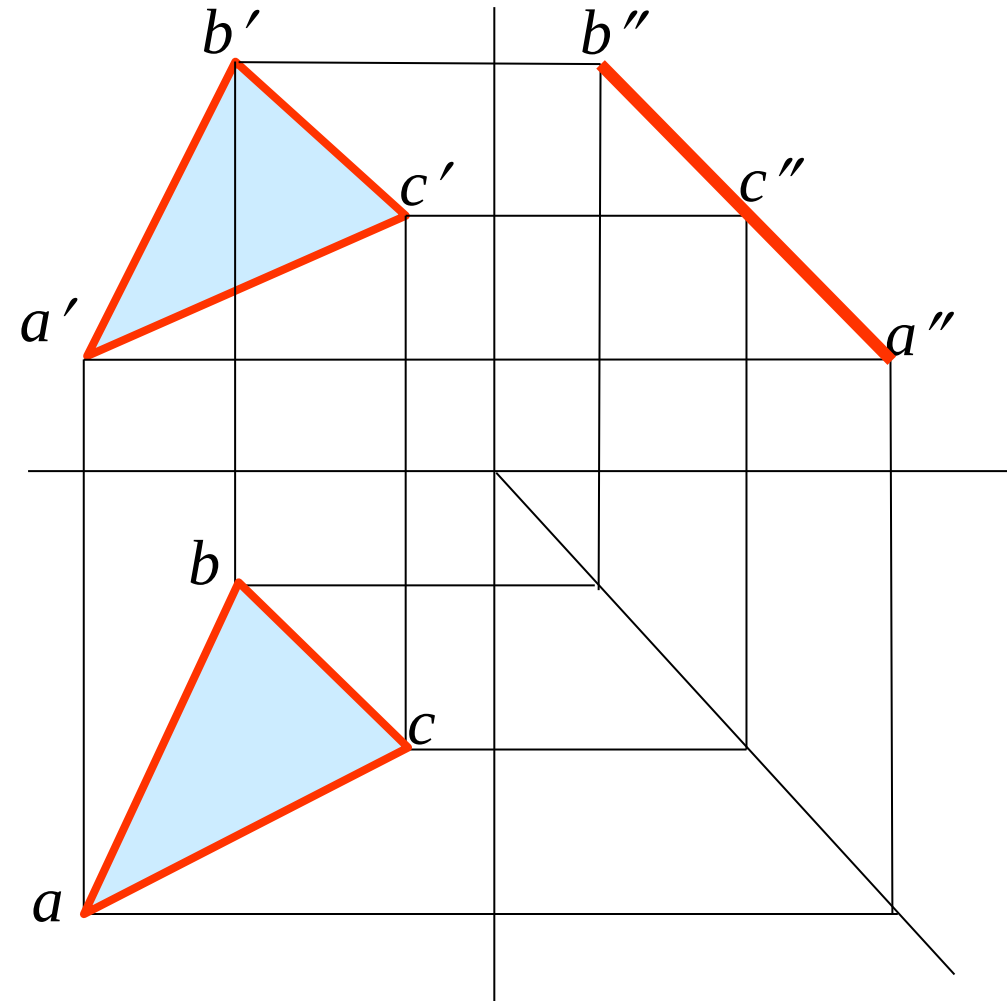
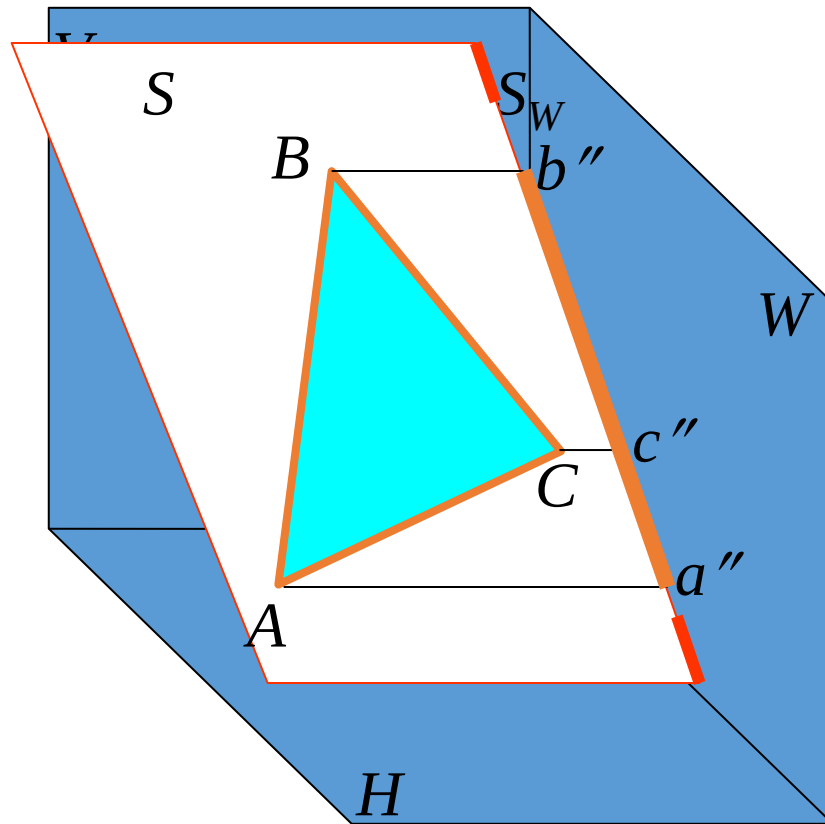
■ 铅垂面

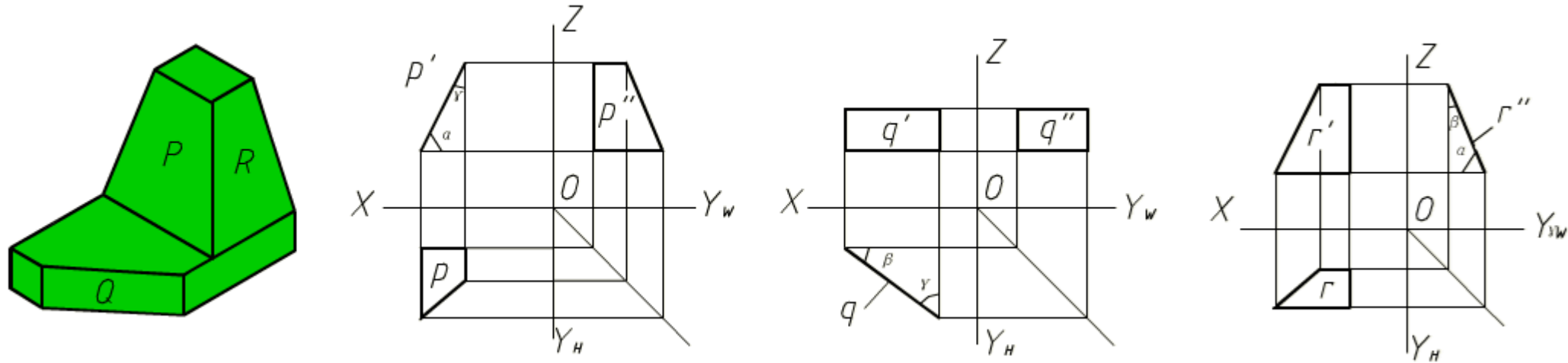


■ 正垂面



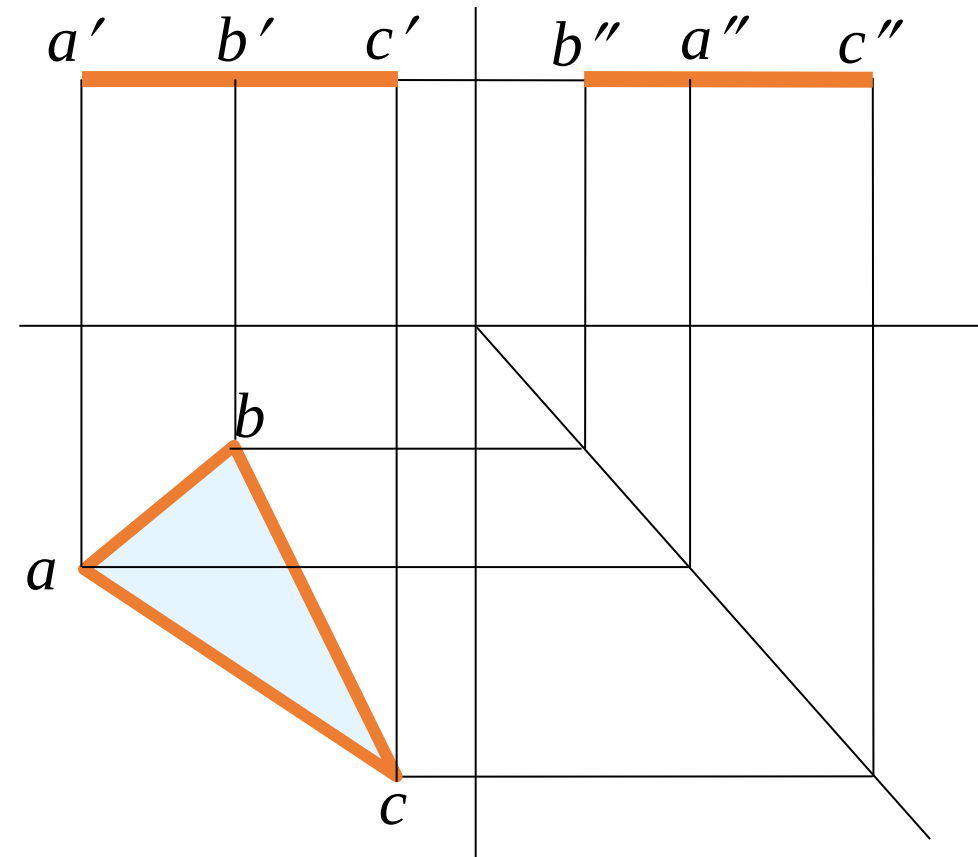
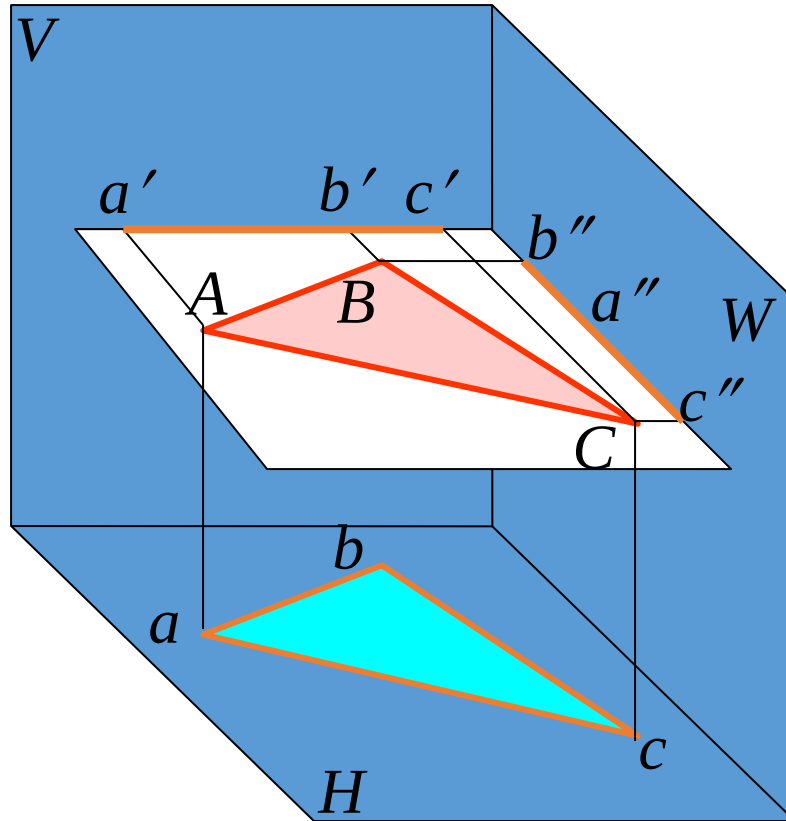
■ 侧垂面



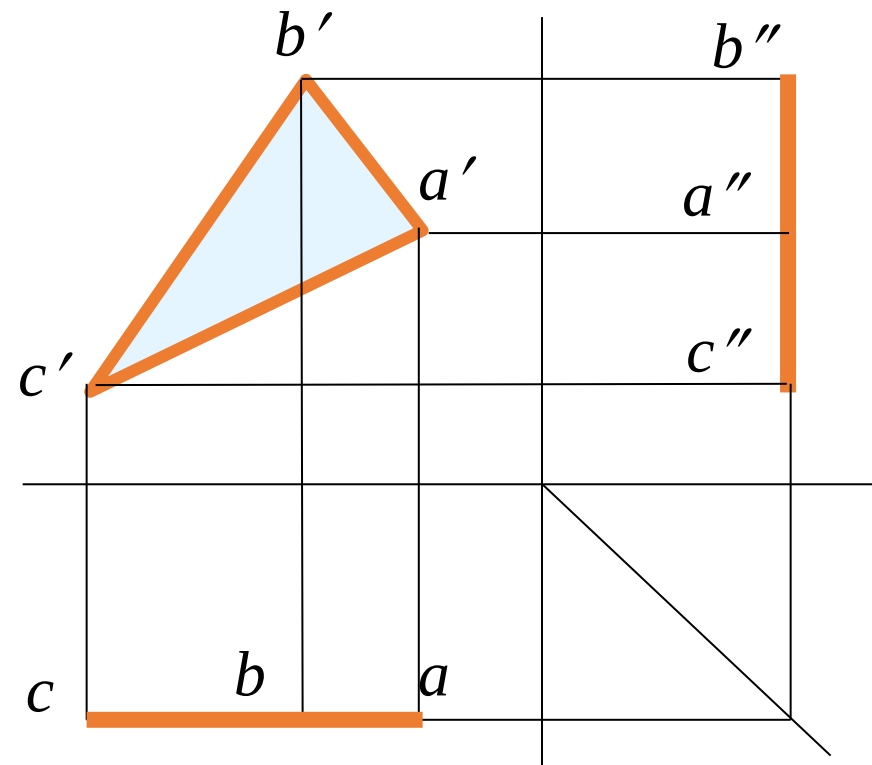
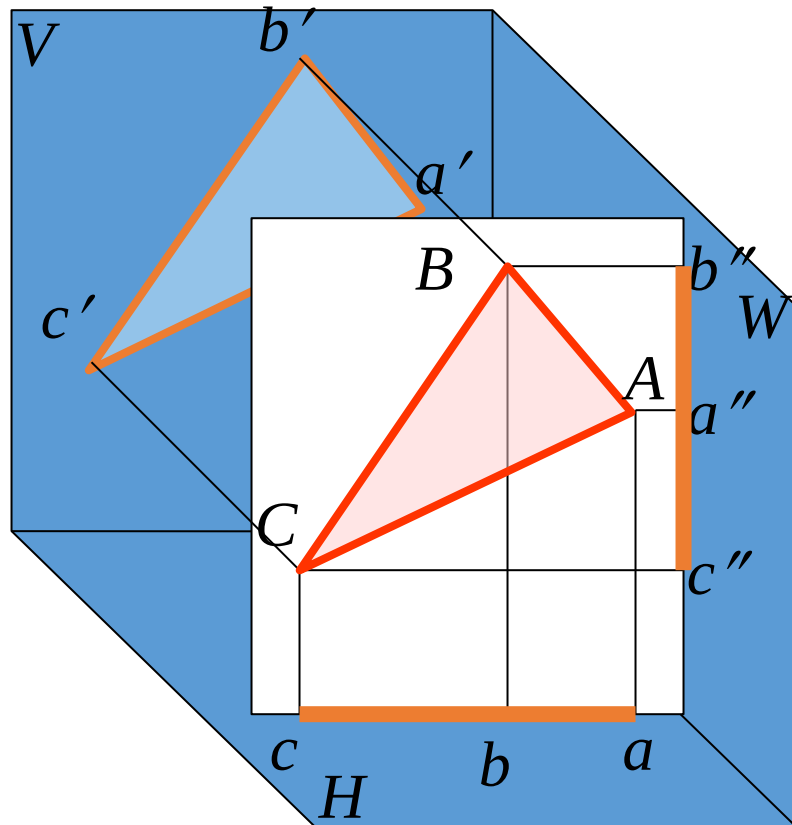


- ✓ 在与平面垂直的投影面上，其投影积聚为一倾斜线段。
- ✓ 其余两个投影都是小于实形的类似形。

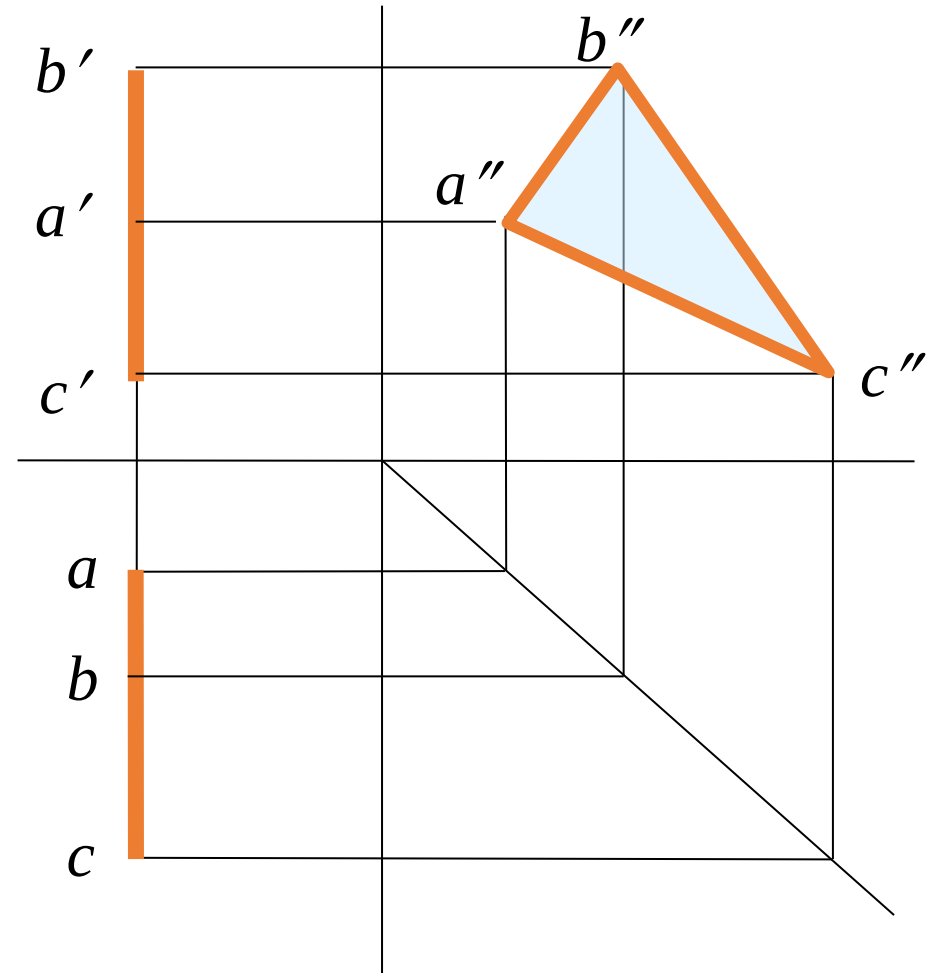
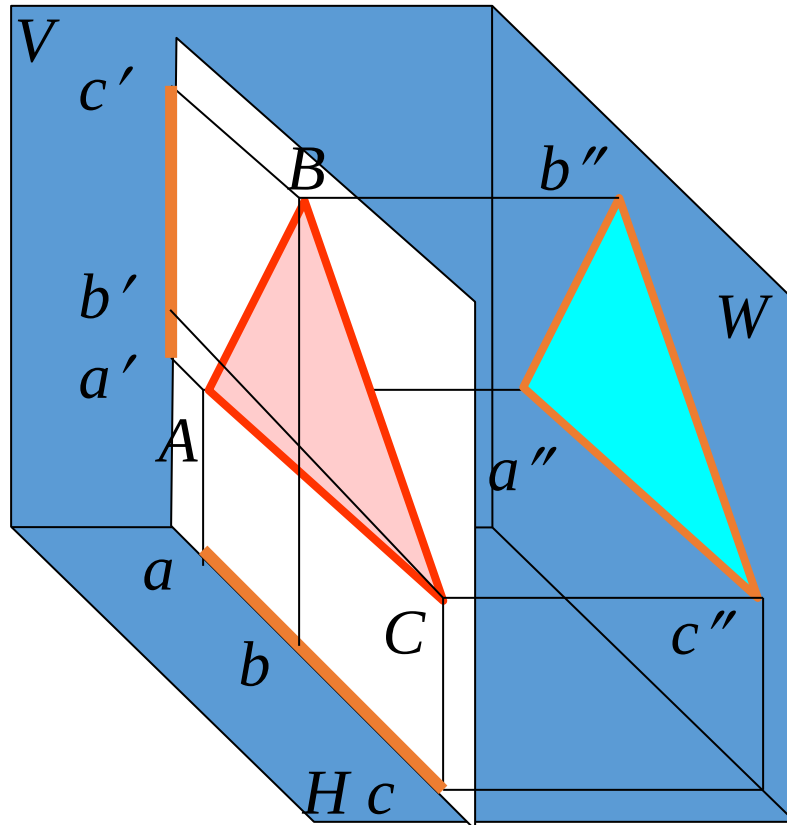
■ 水平面

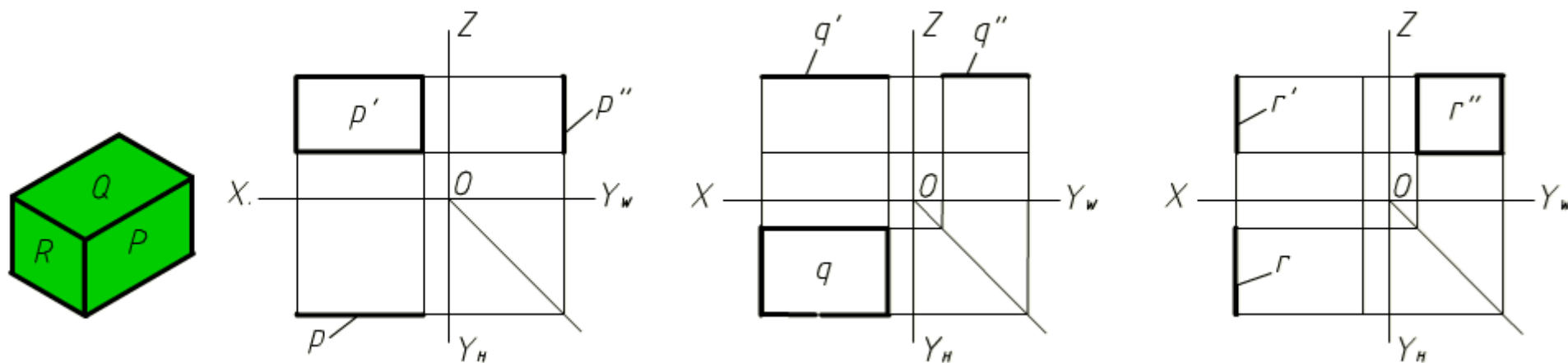


■ 正平面



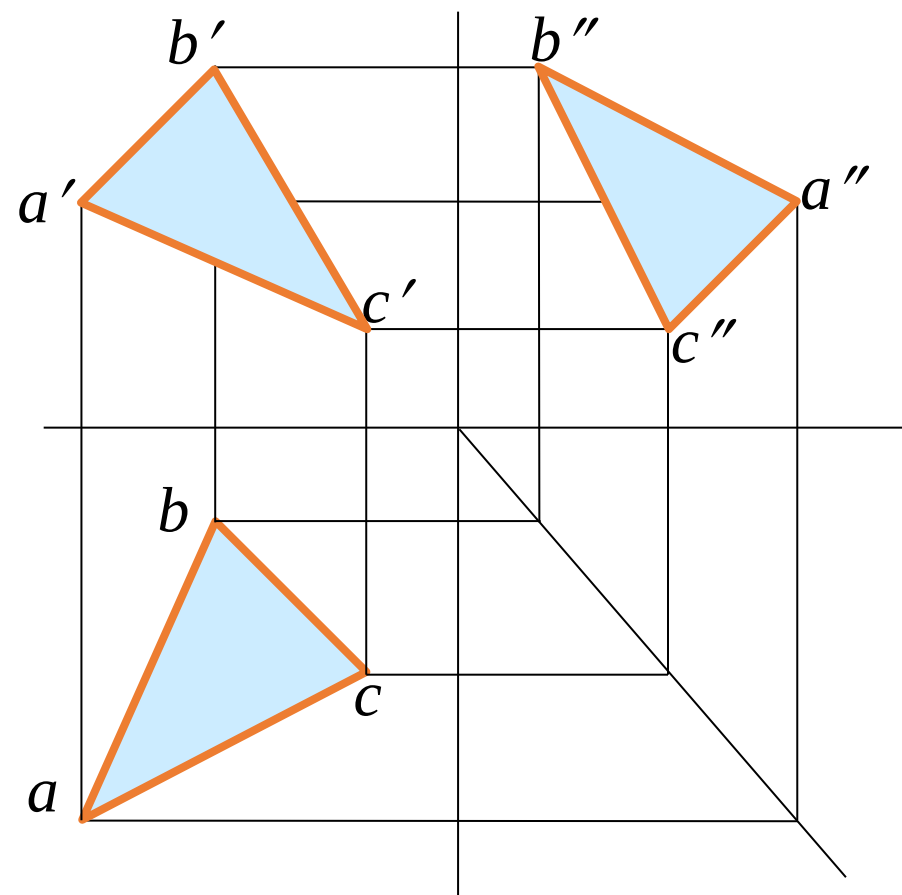
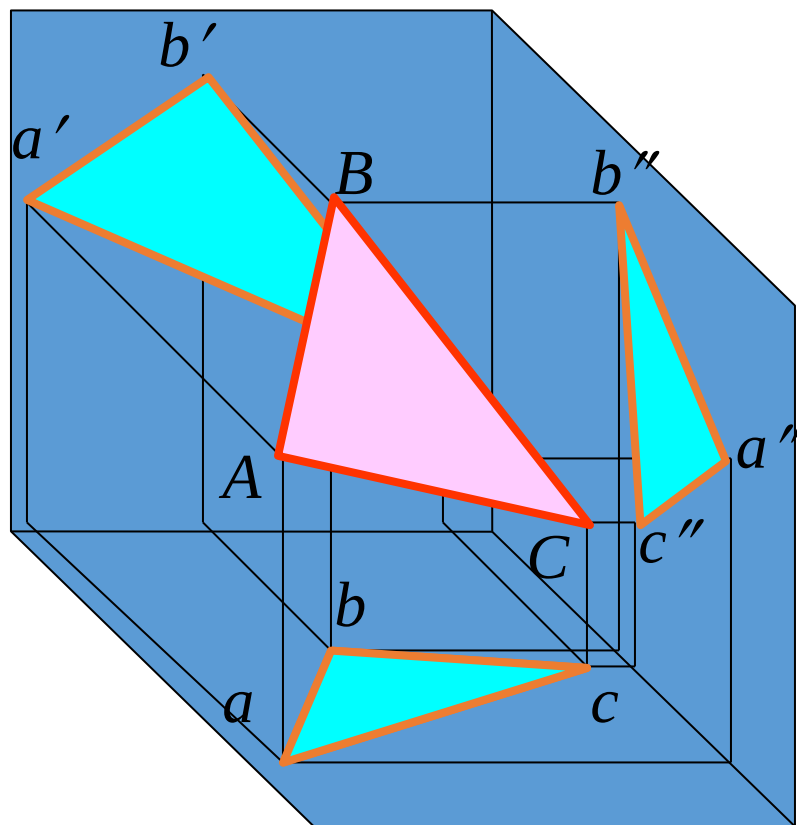
■ 侧平面





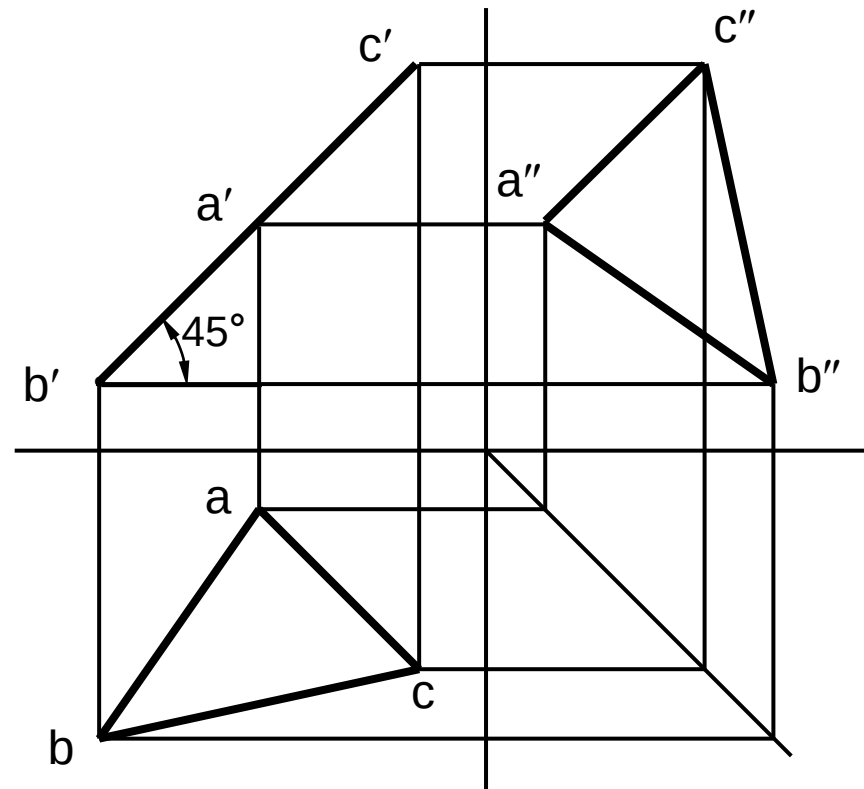
- ✓ 在与平面平行的投影面上，其投影反映实形。
- ✓ 其余两个投影分别为水平线段或垂直线段，且都具有积聚性。

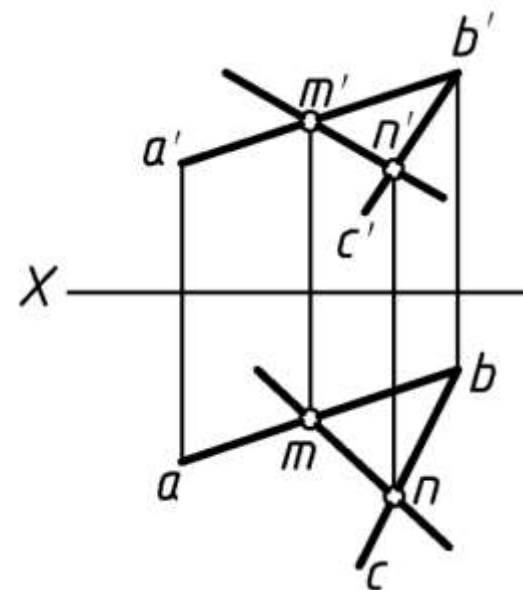
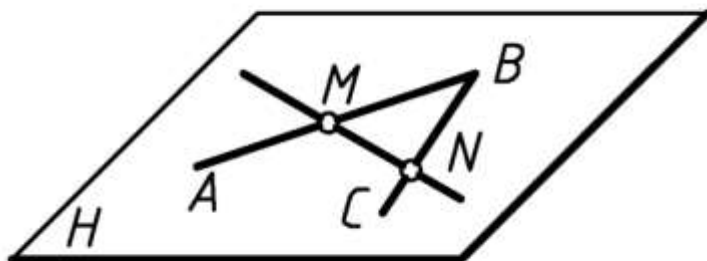
■ 一般位置平面



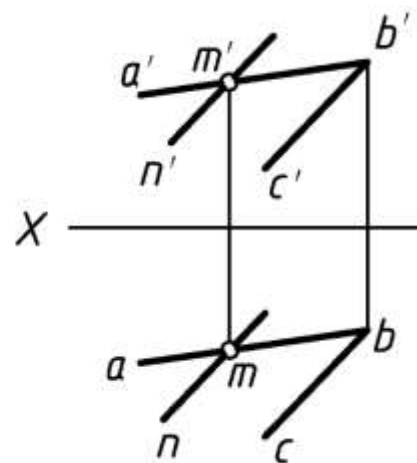
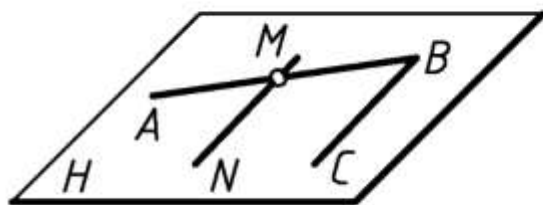
✓ 一般位置平面 S 的三个投影都是小于实形的类似形

正垂面ABC与H面的夹角为 45° ，已知其水平投影及顶点B的正面投影，求 $\triangle ABC$ 的正面投影及侧面投影。

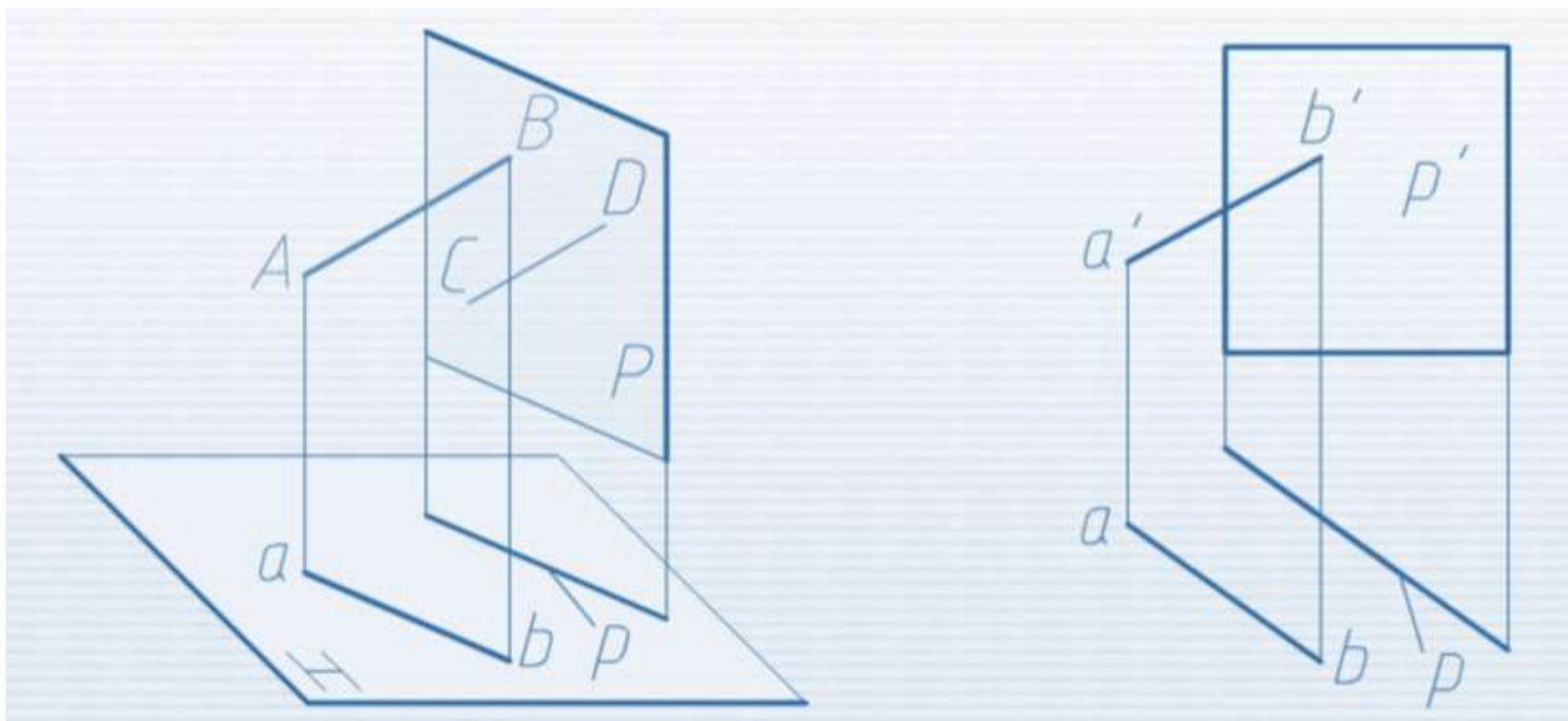




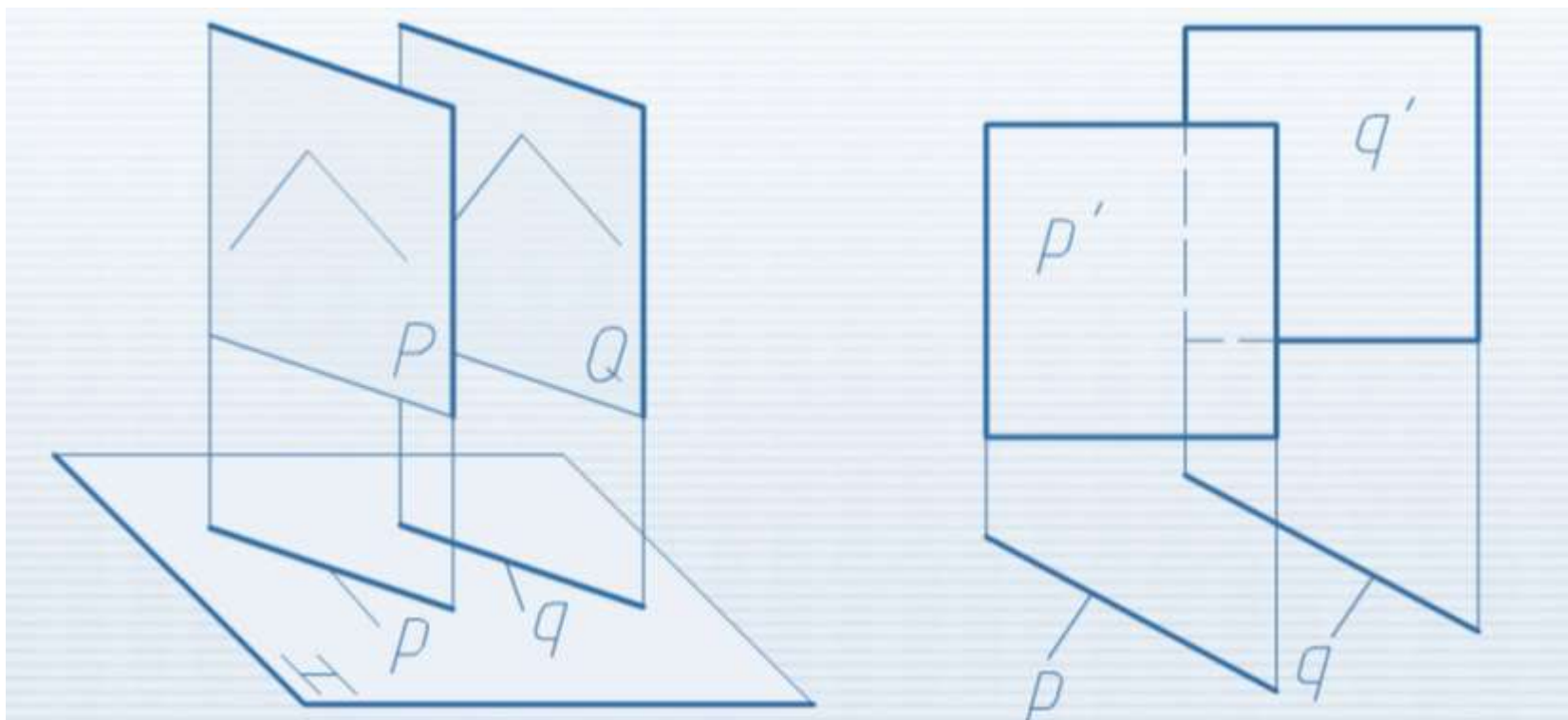
✓ 1. 若一直线过平面上的两点，则此直线必在该平面内。



- ✓ 2. 若一直线过平面上的一点且平行于该平面上的另一直线，则此直线在该平面内。

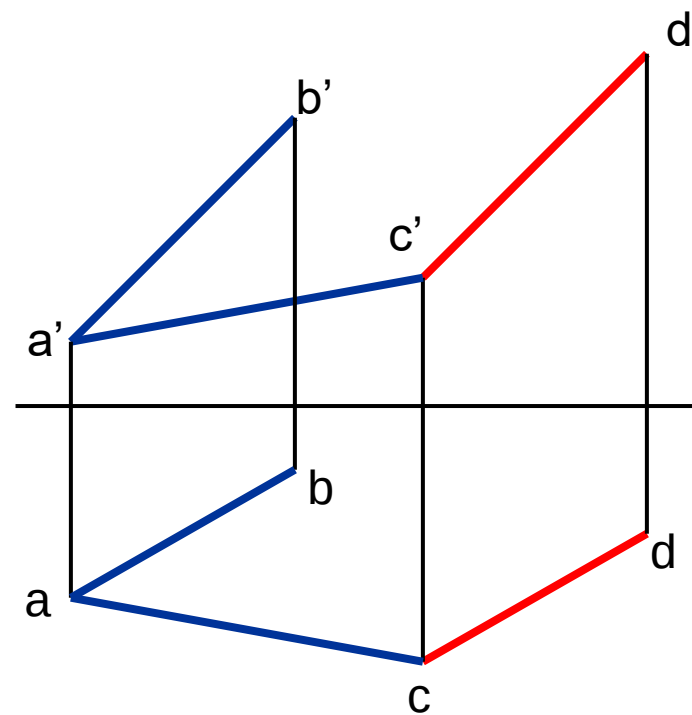
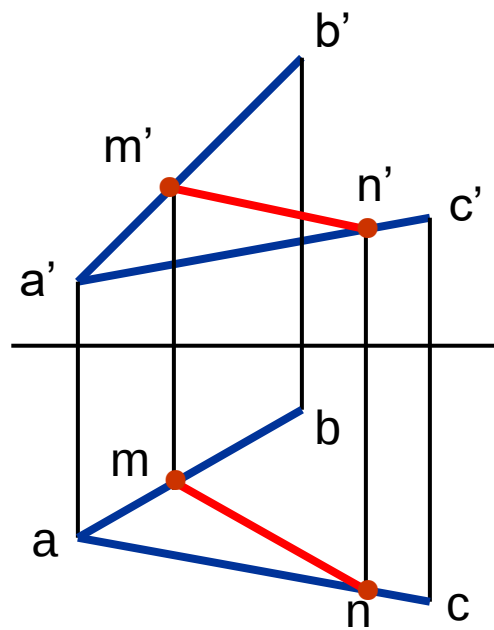


- ✓ 3. 若平面外的一直线与平面内的一直线平行，则此直线与平面相互平行。

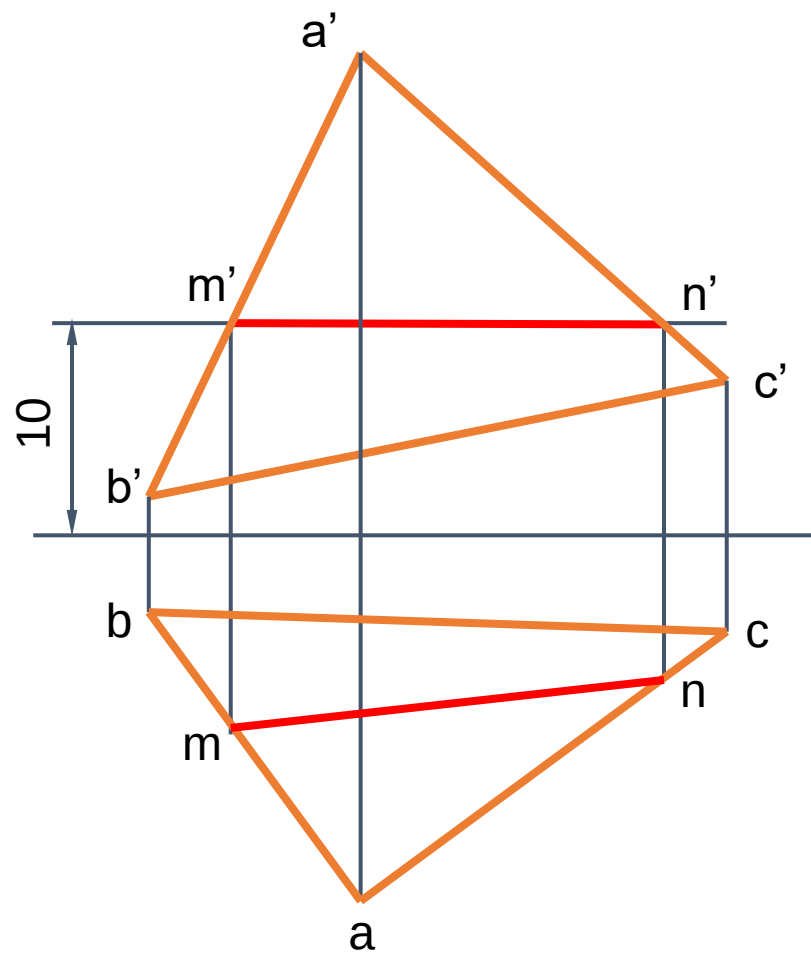


- ✓ 4. 若一平面内的两相交直线分别平行与另一平面内的两相交直线，则这两平面相互平行。

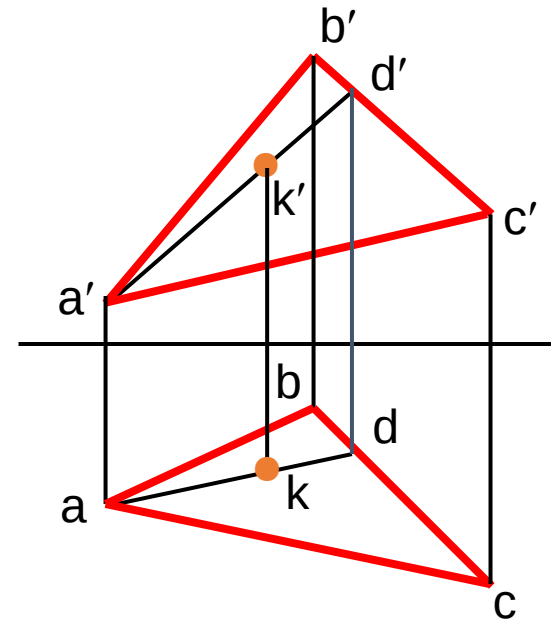
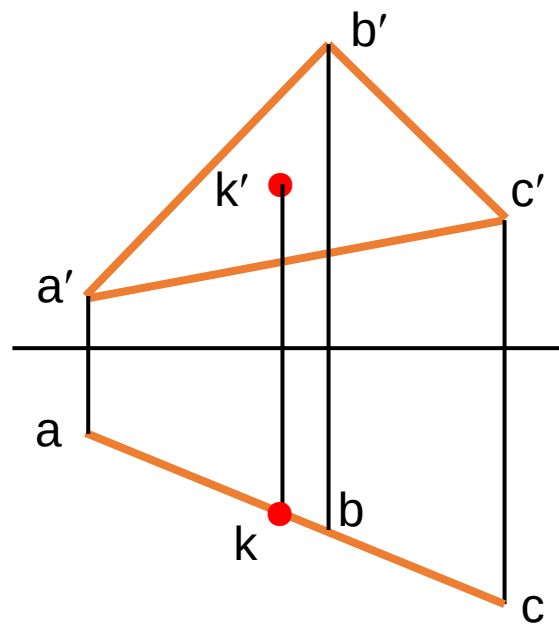
已知平面由直线AB、AC所确定，试在平面内任作一条直线。



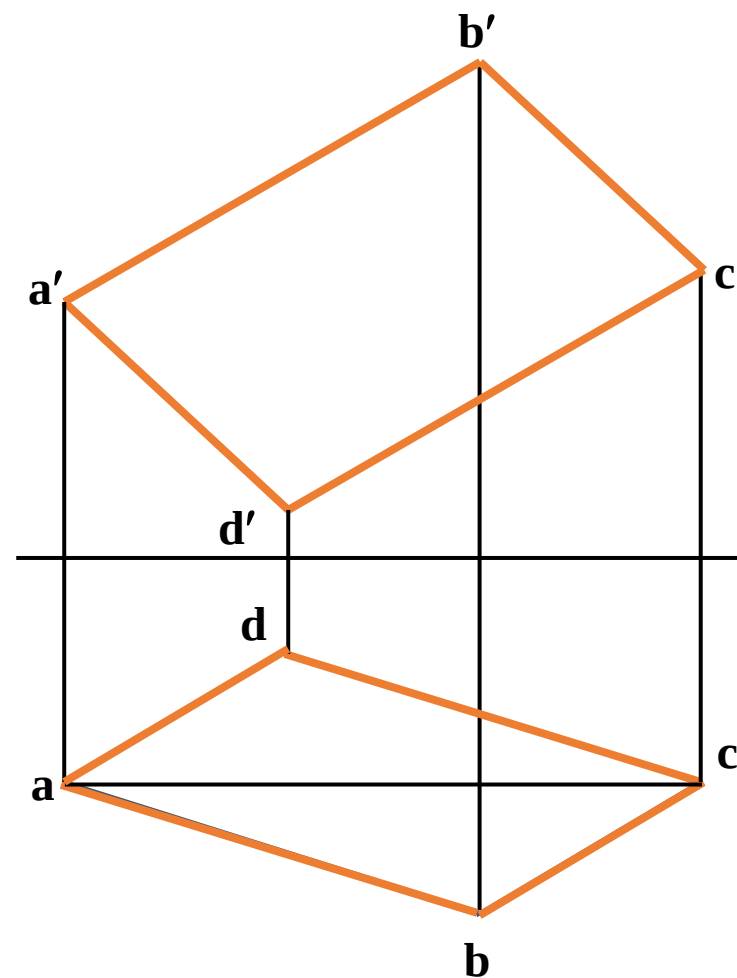
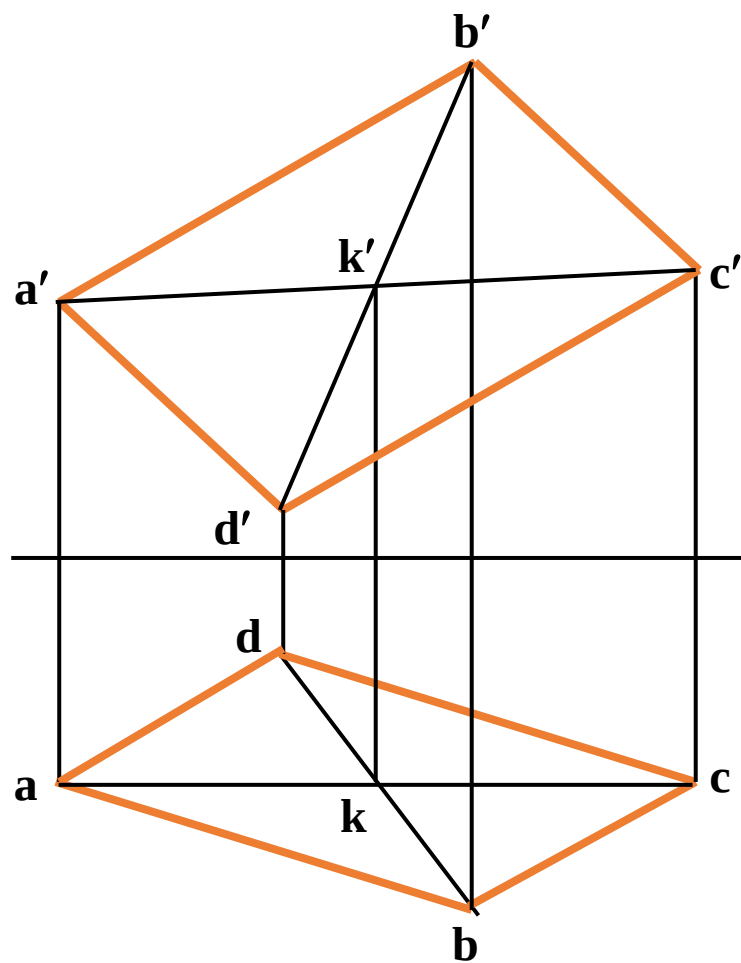
在平面ABC内作一条水平线，使其到 H 面的距离为10mm。



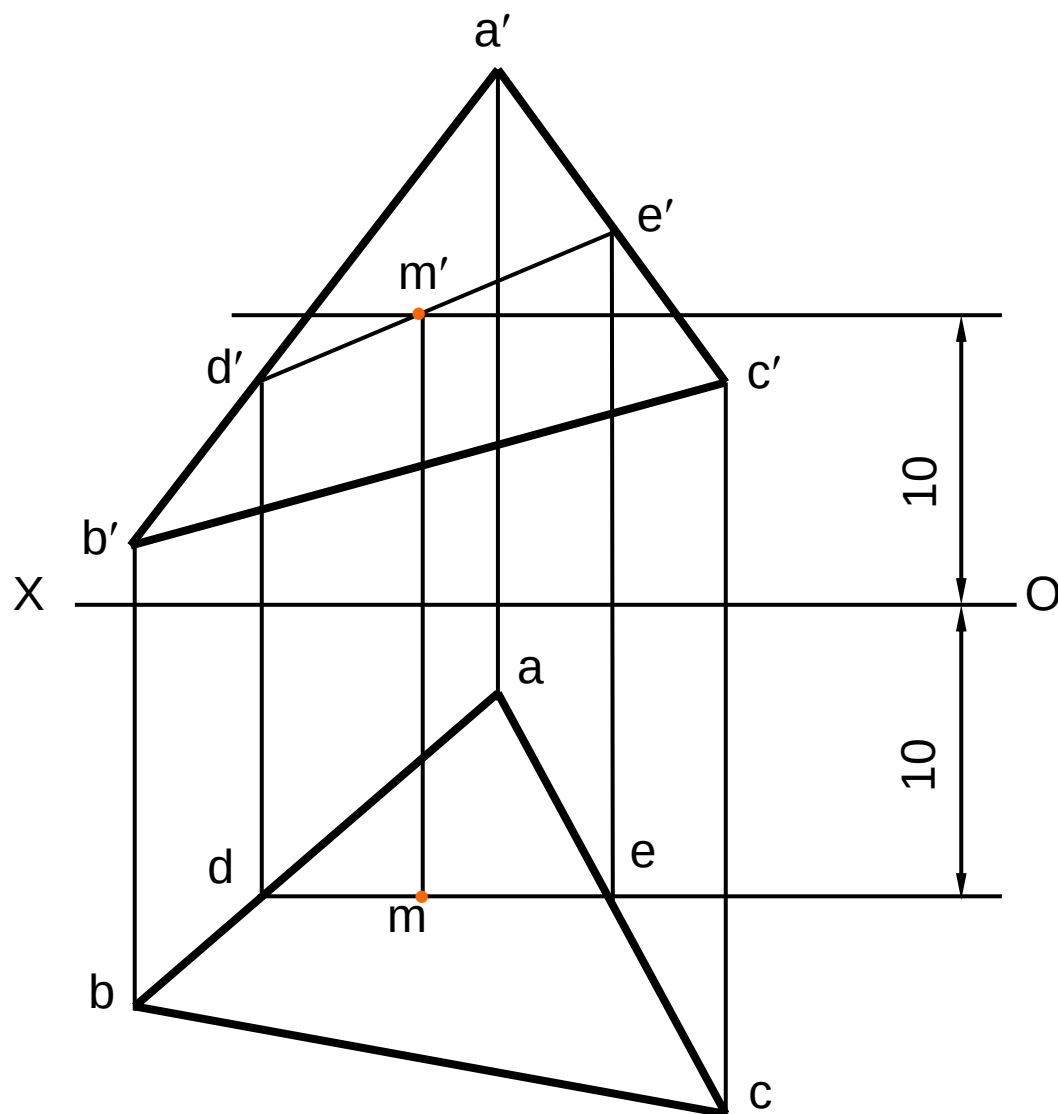
已知K点在平面ABC上，求K点的水平投影。



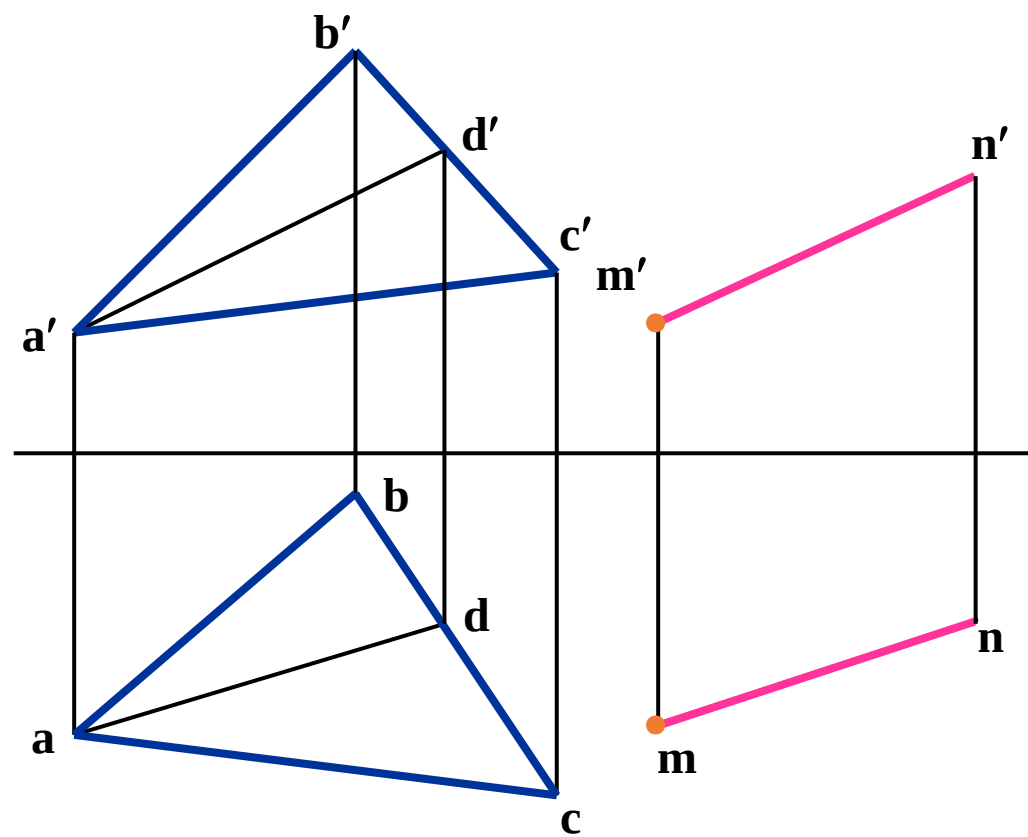
已知AC为正平线，补全平行四边形ABCD的水平投影。



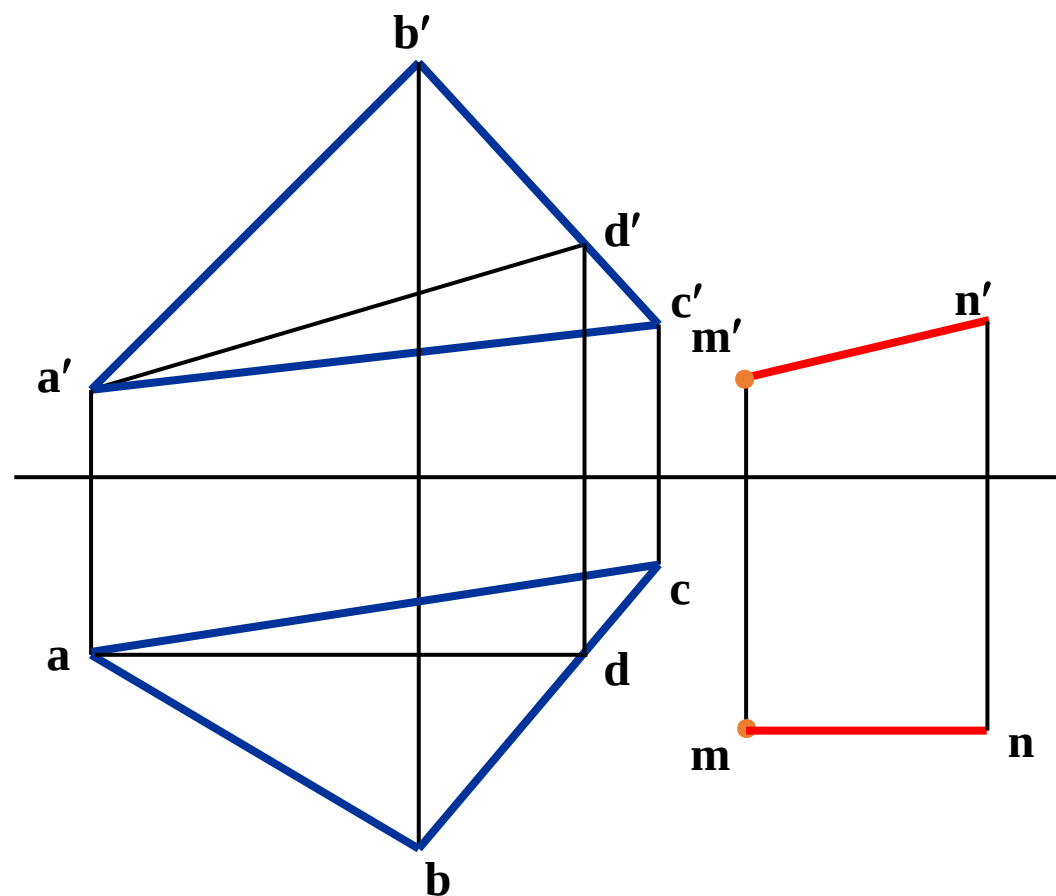
在 $\triangle ABC$ 内取一点 M ，并使其到 H 面 V 面的距离均为10mm。



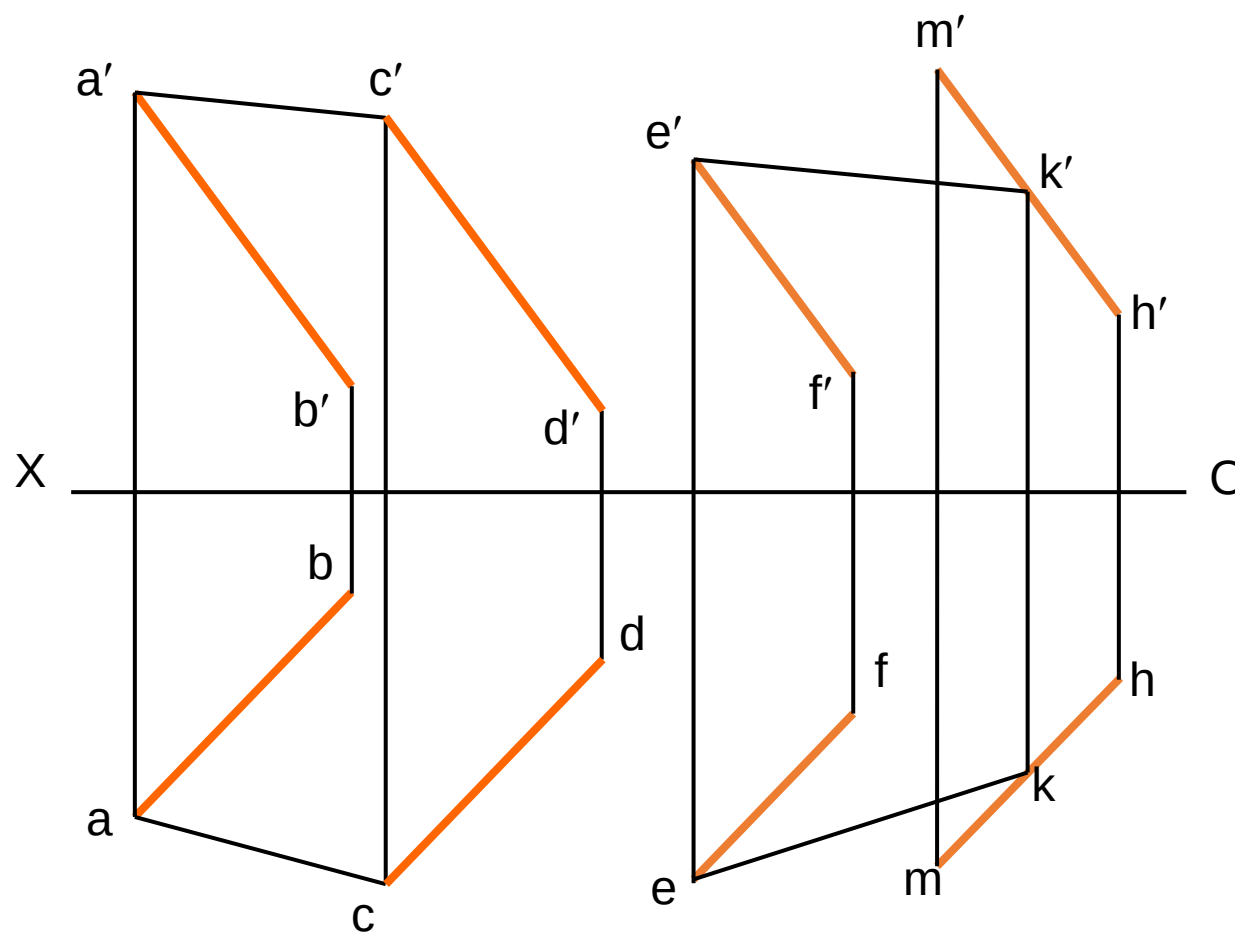
过M点作直线MN平行于平面ABC。



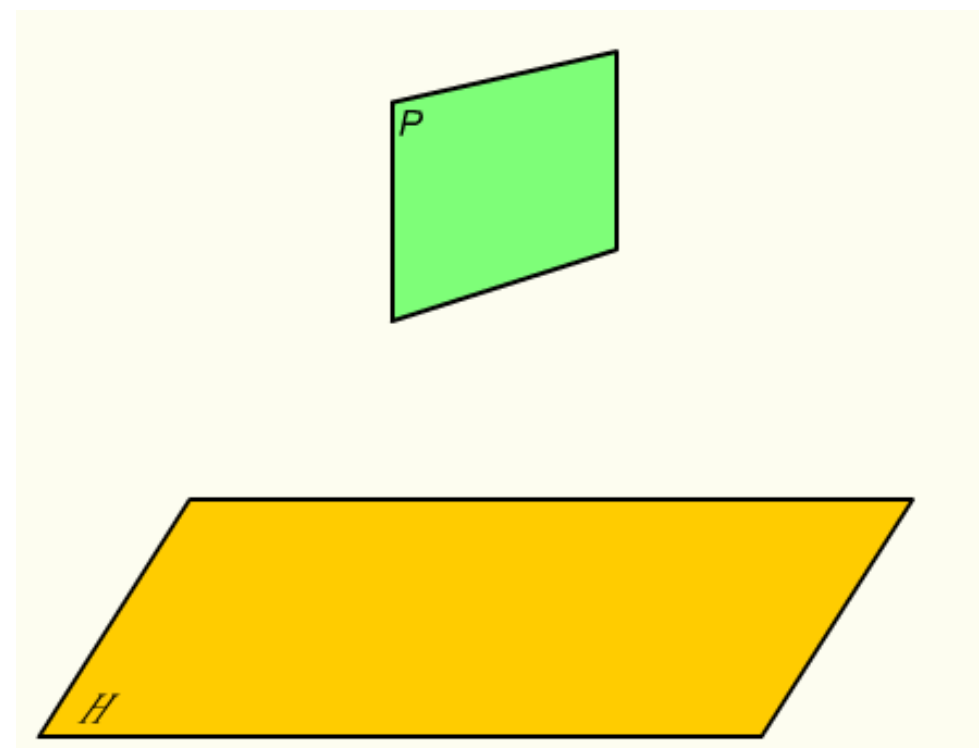
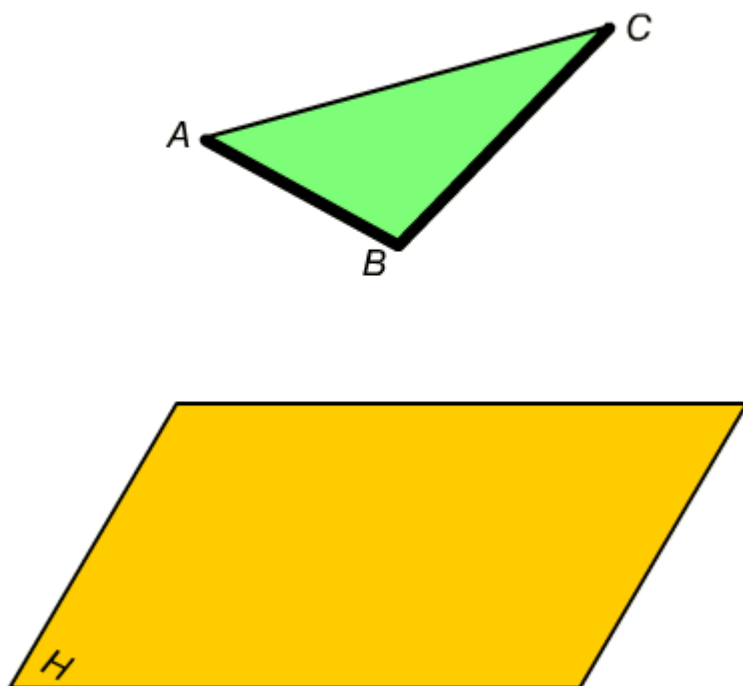
过M点作直线MN平行于V面和平面ABC。



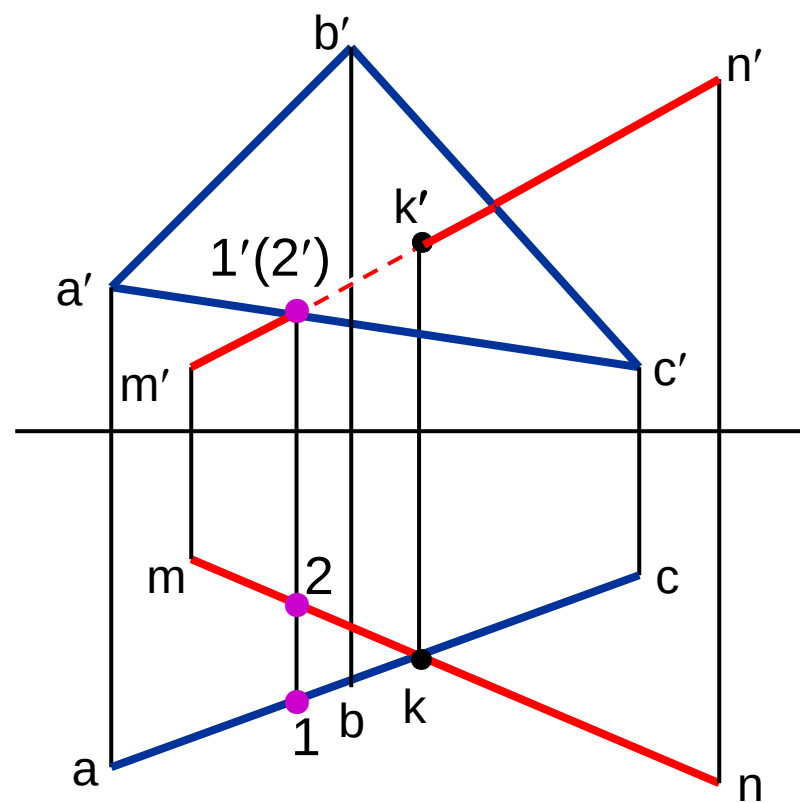
判断平面ABDC与平面EFHM是否平行，已知 $AB \parallel CD \parallel EF \parallel MH$ 。



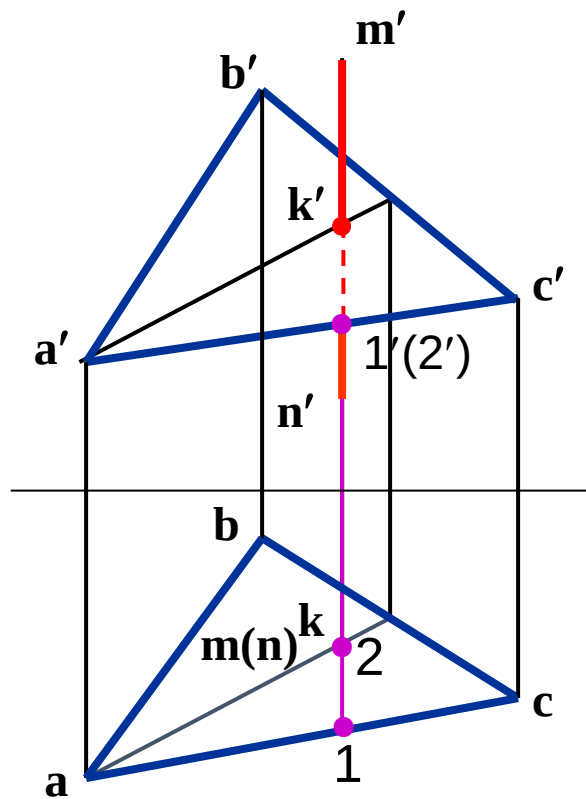
直线与平面相交



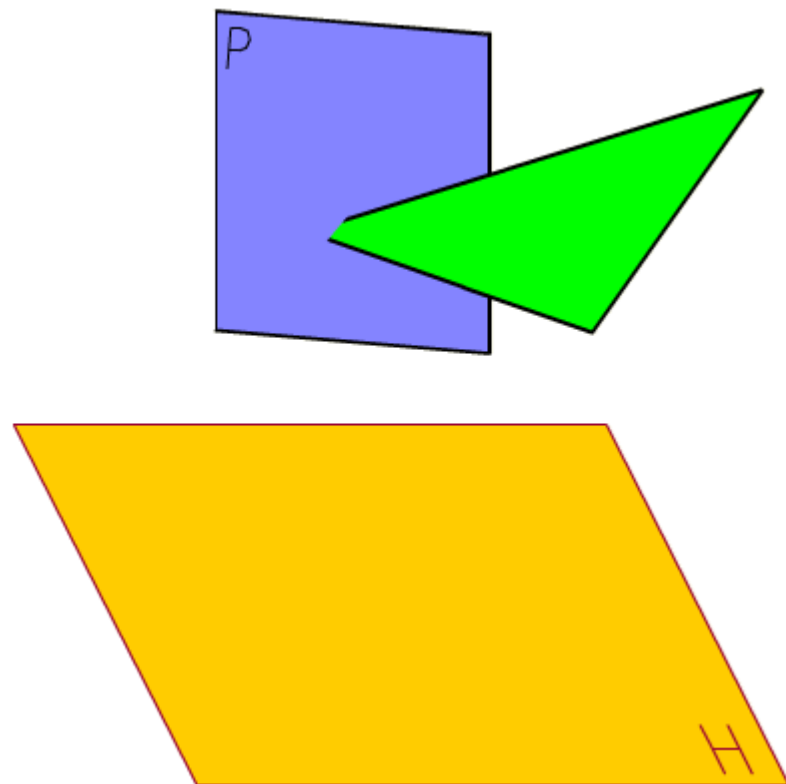
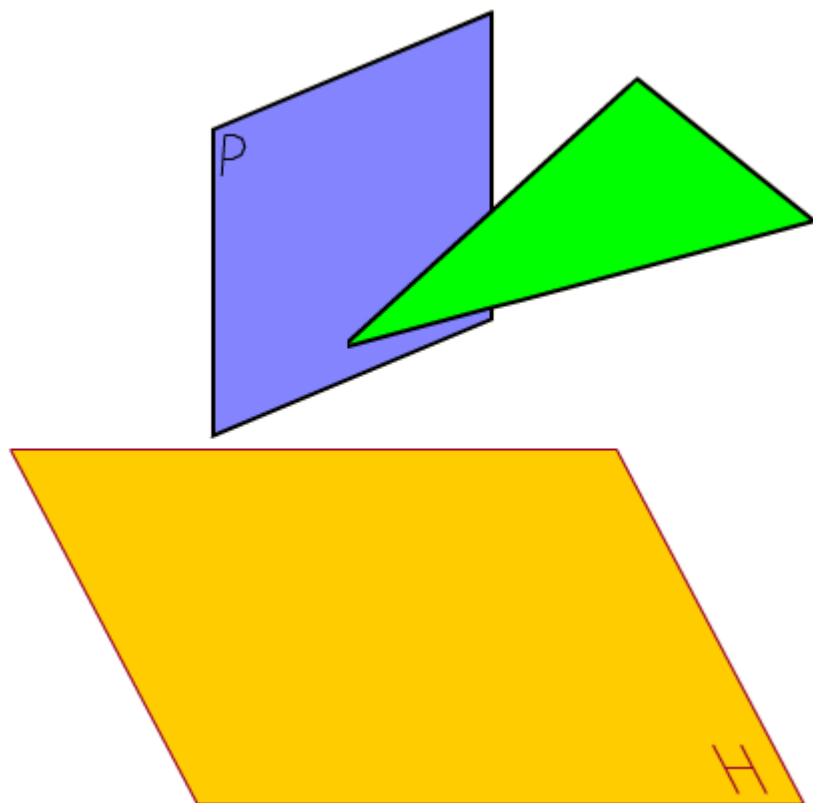
求直线MN与平面ABC的交点K并判别可见性。



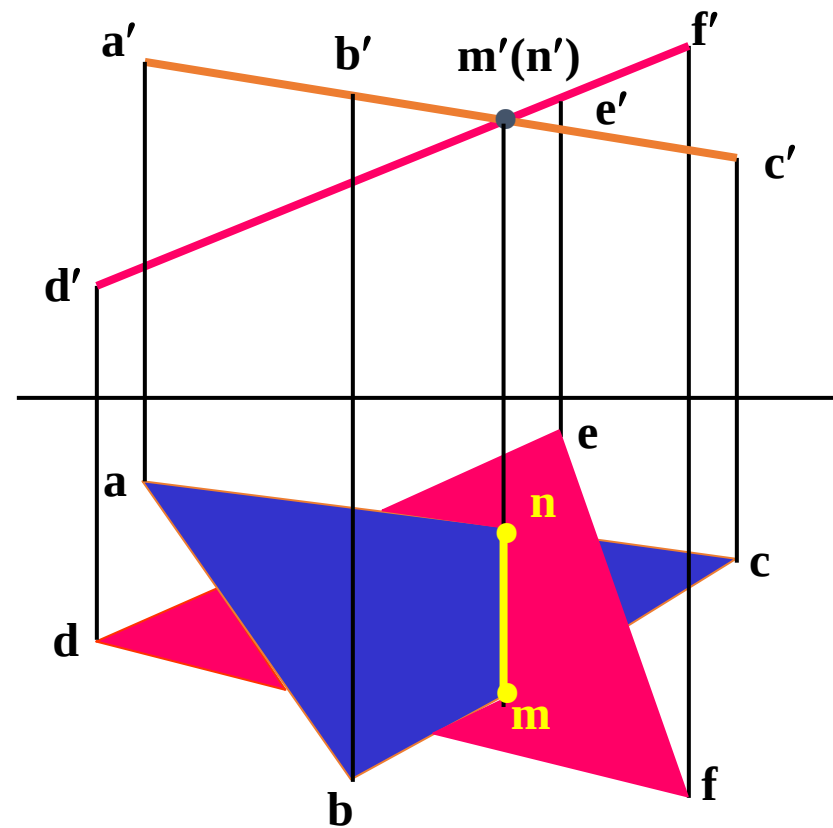
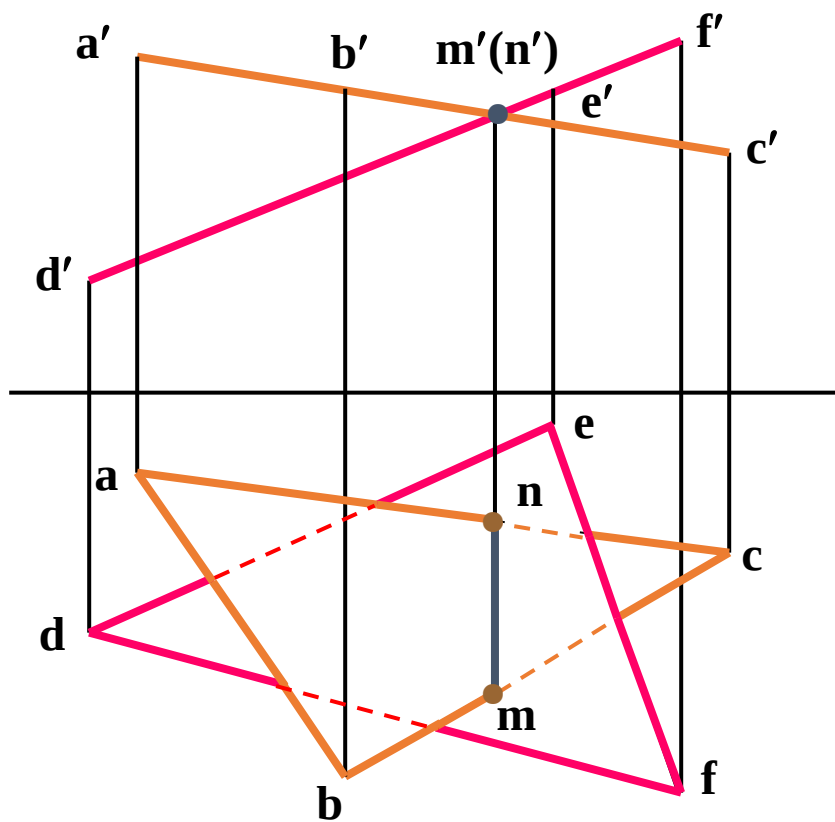
求直线MN与平面ABC的交点K并判别可见性。



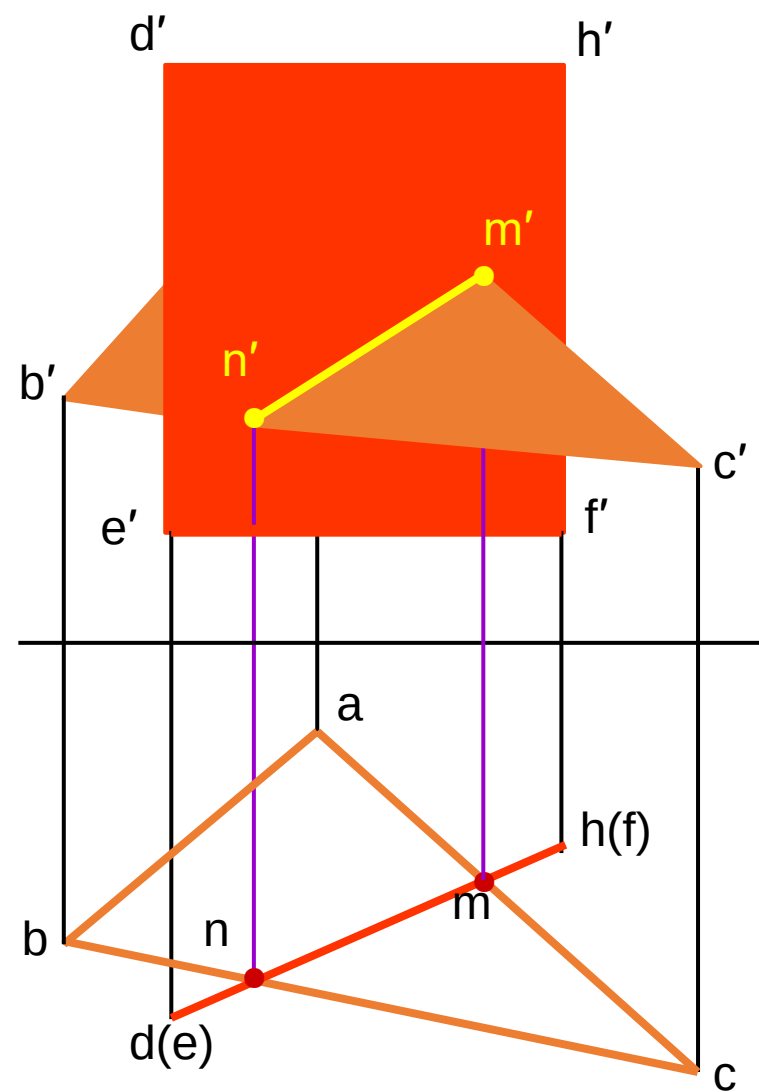
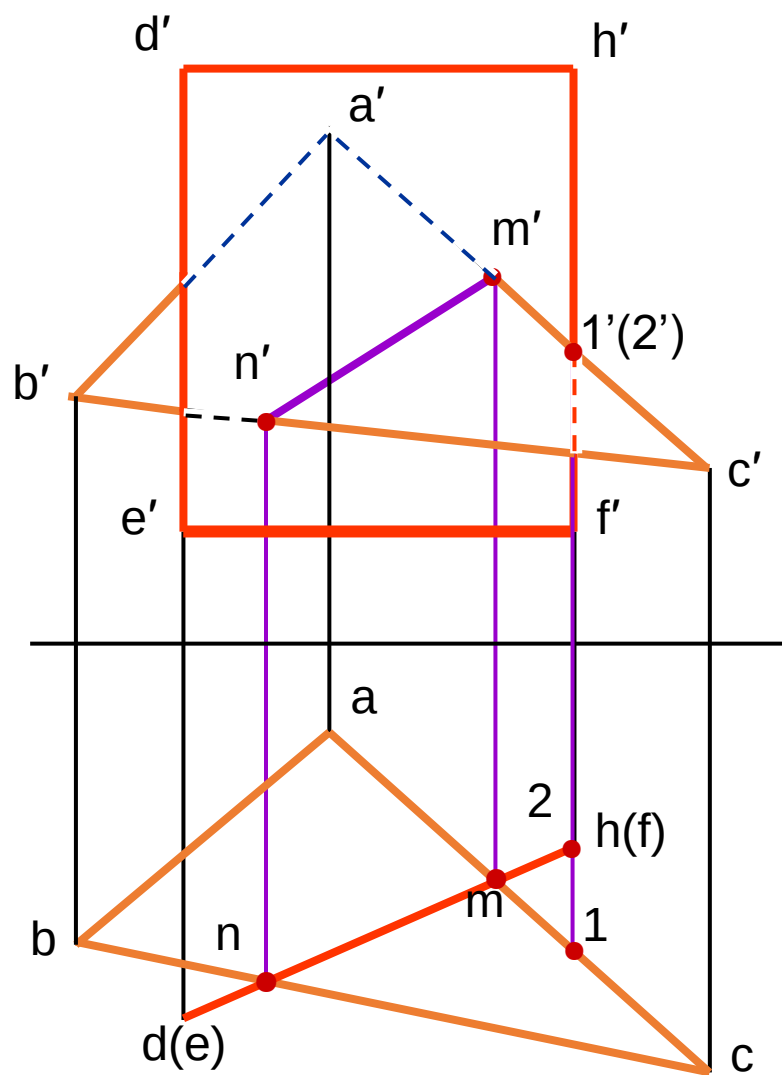
平面与平面相交



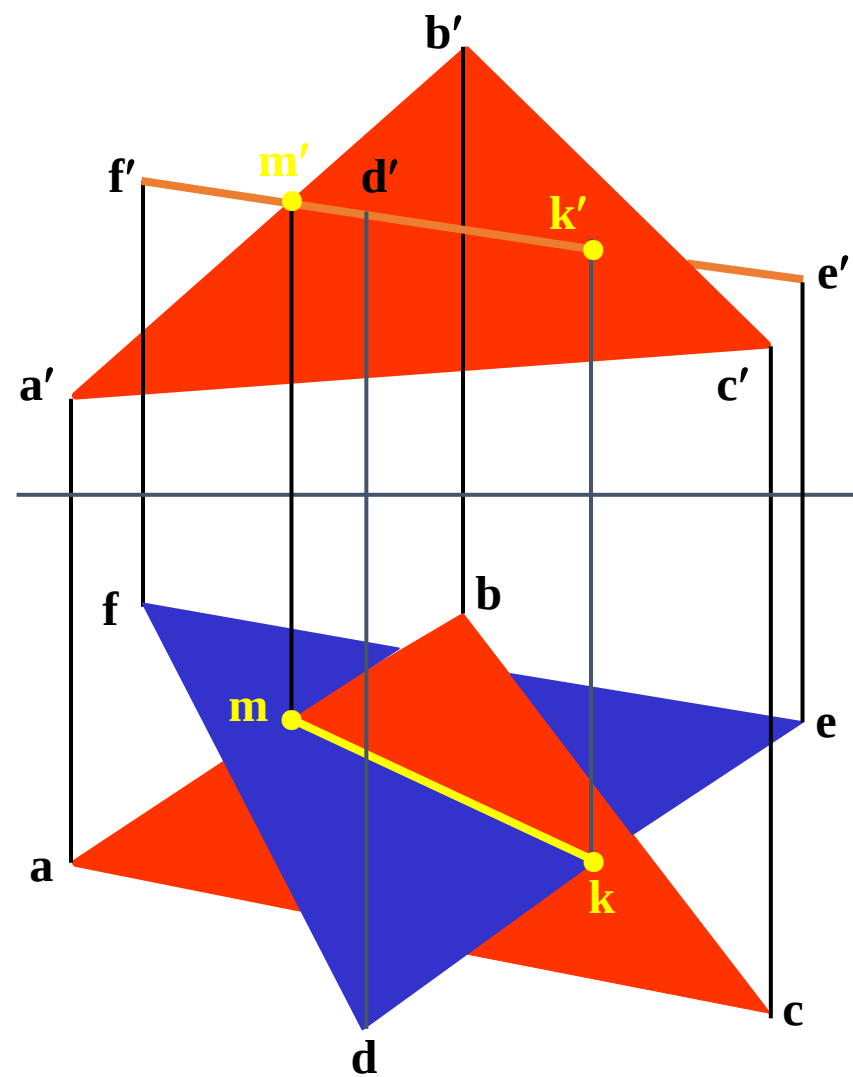
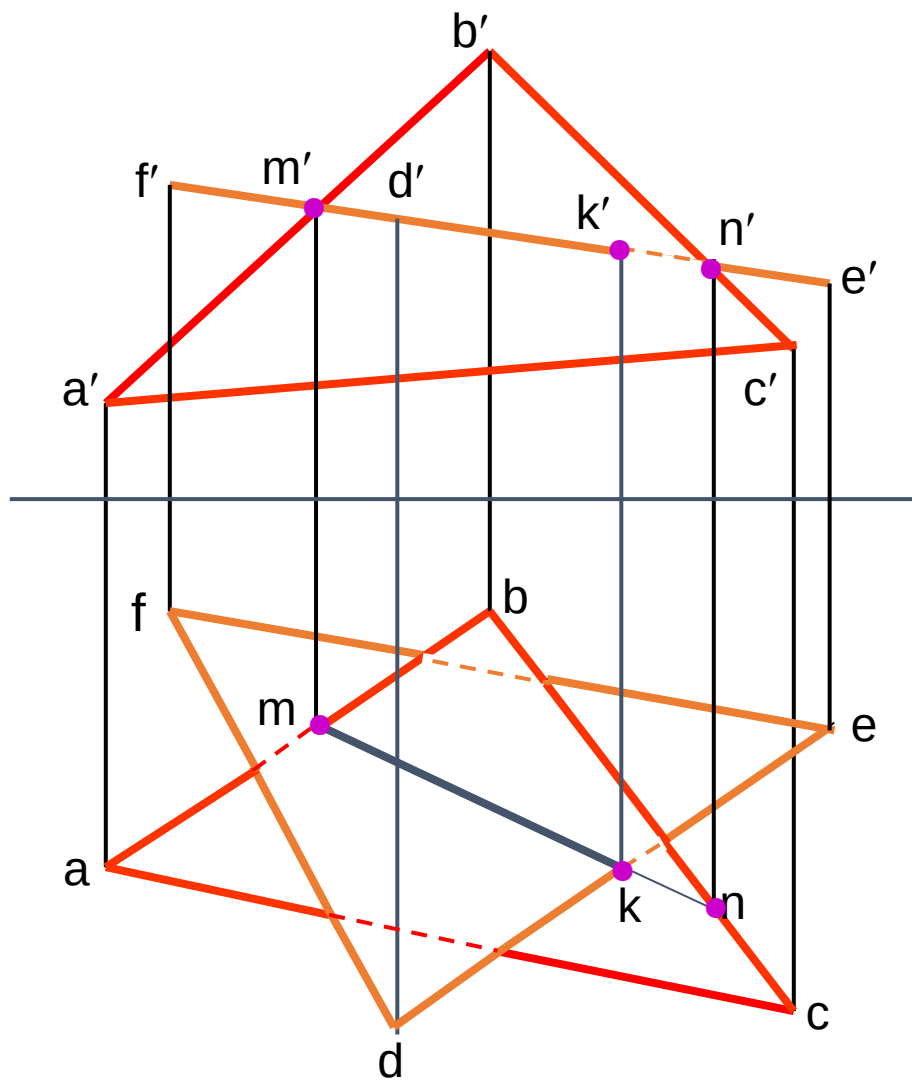
求两平面的交线MN并判别可见性。

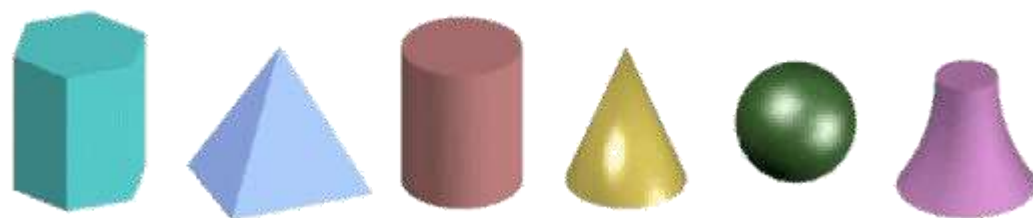


求两平面的交线MN并判别可见性。

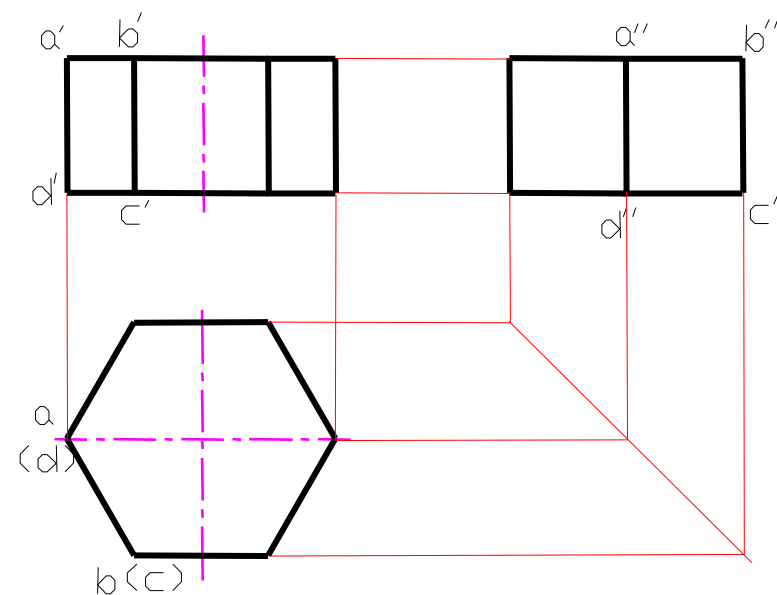
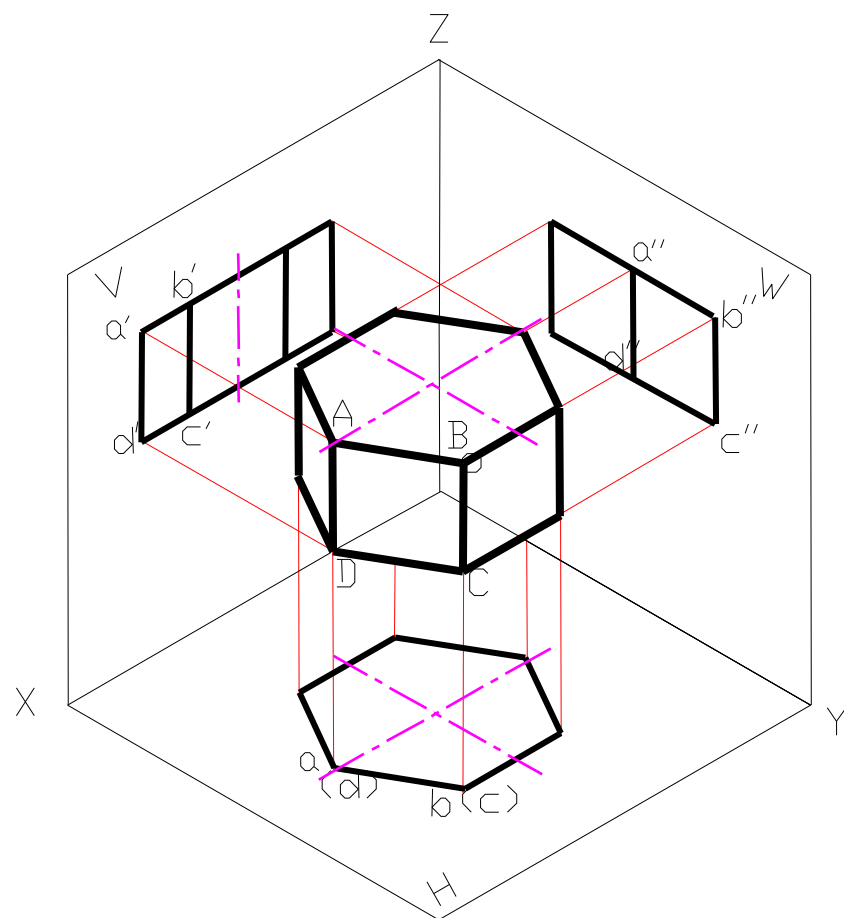
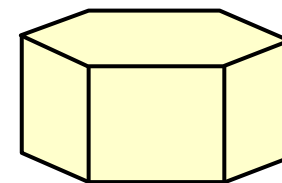


求两平面的交线MN并判别可见性。

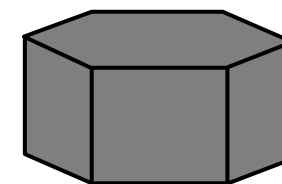
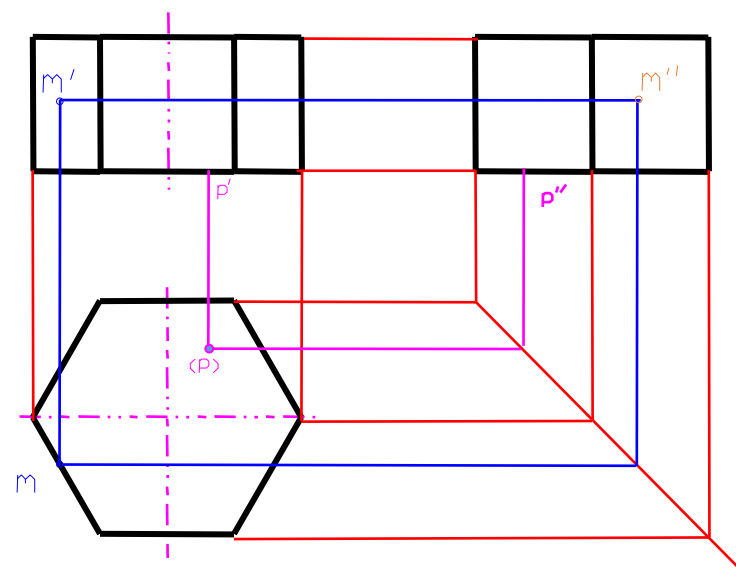
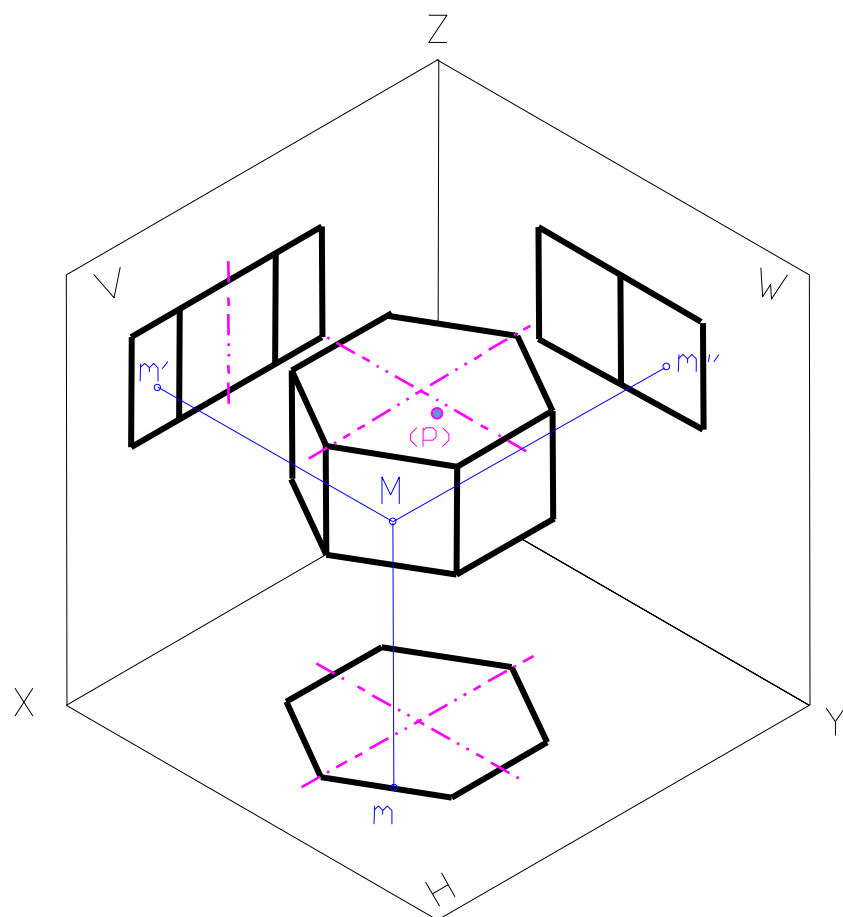




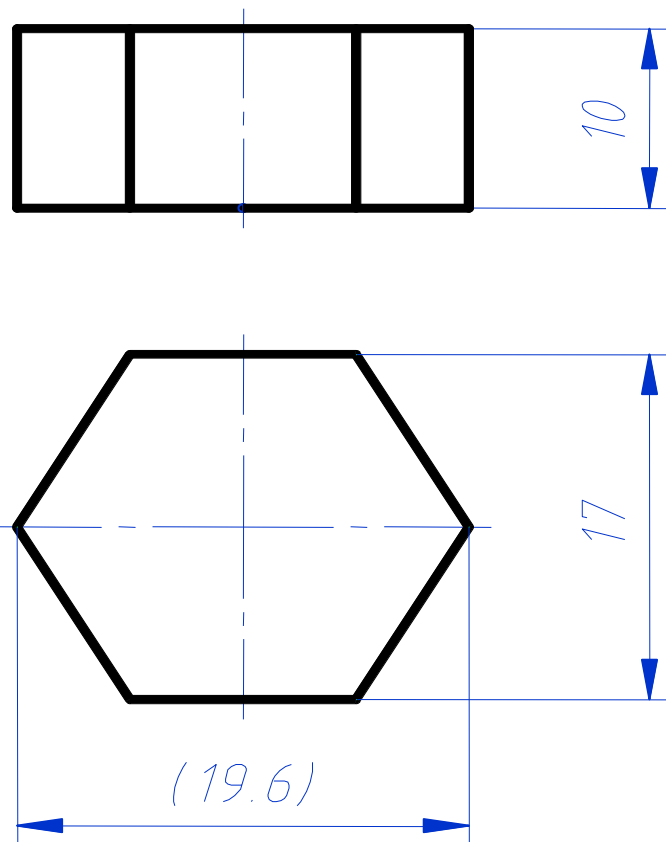
■ 正六棱柱



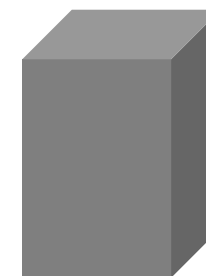
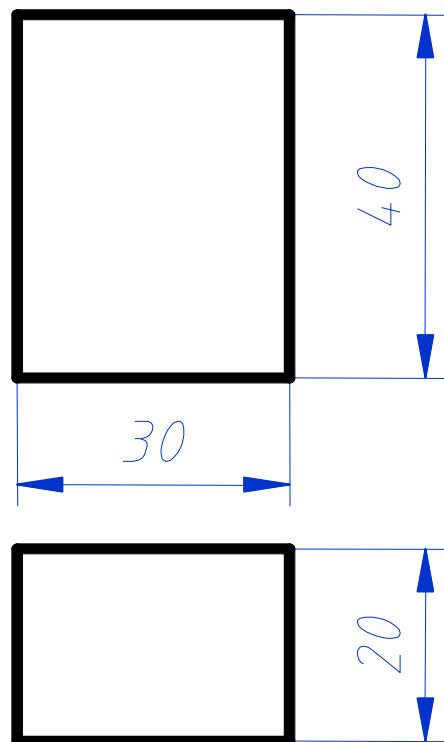
■ 正六棱柱

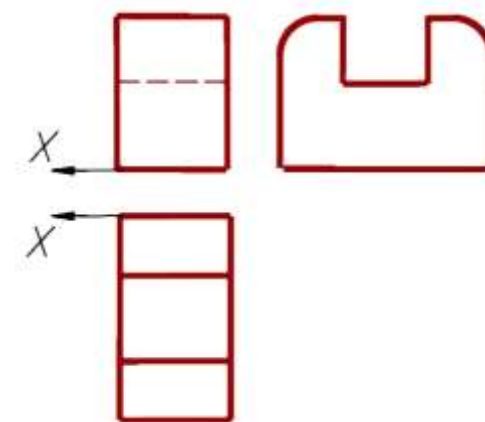
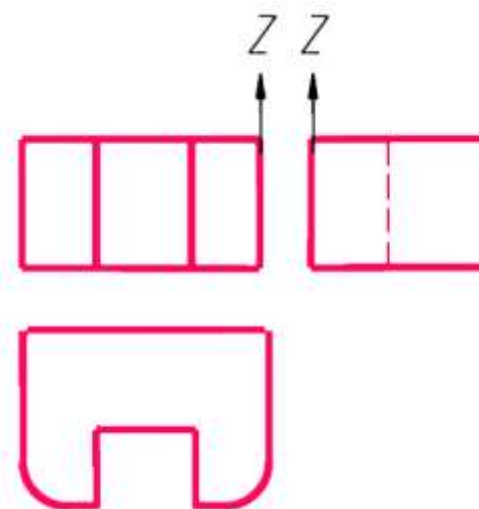
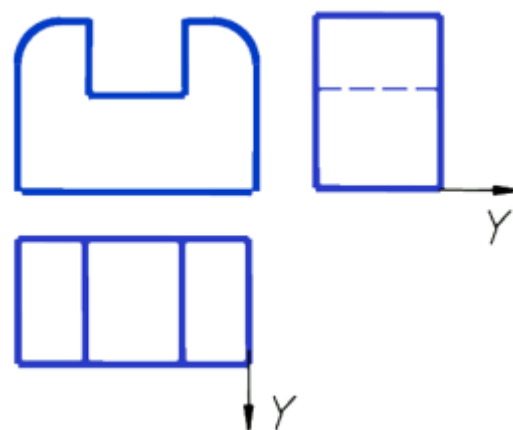
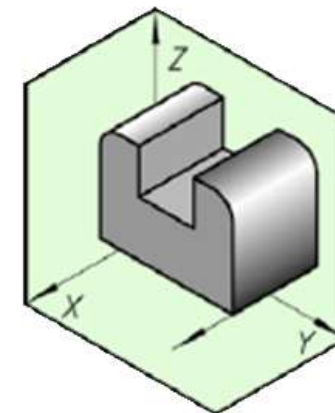
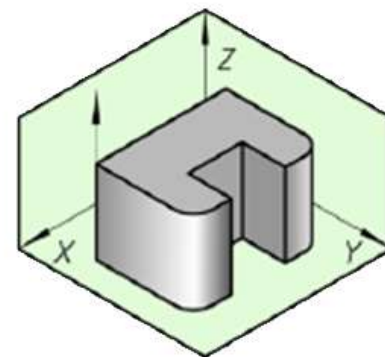
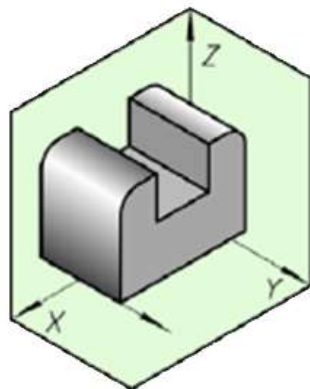
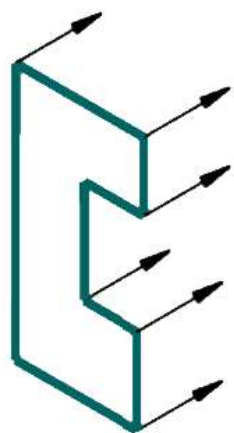


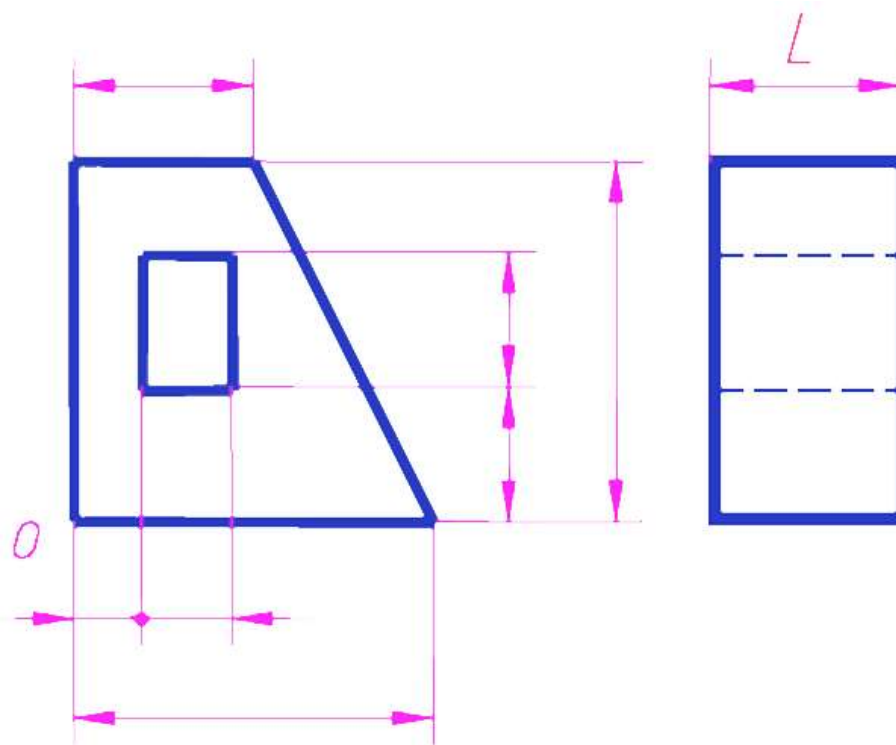
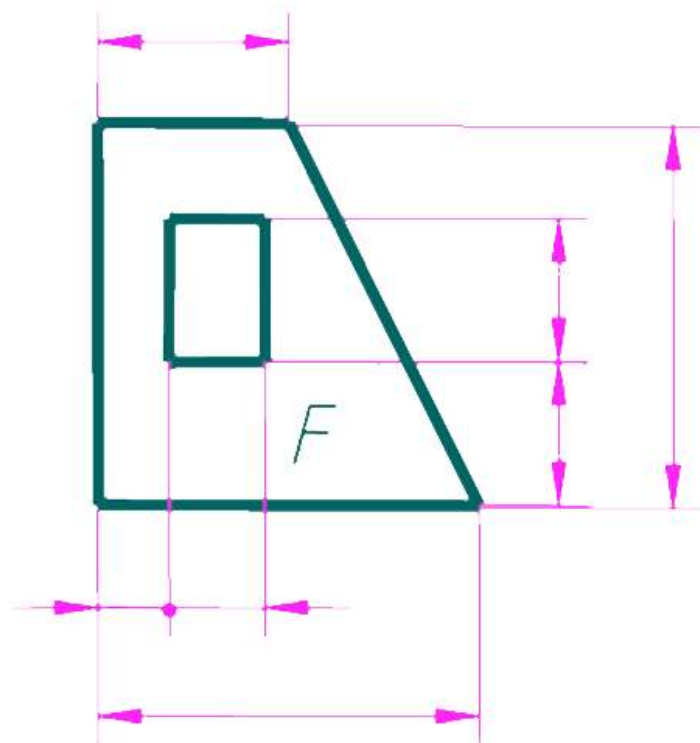
■ 正六棱柱



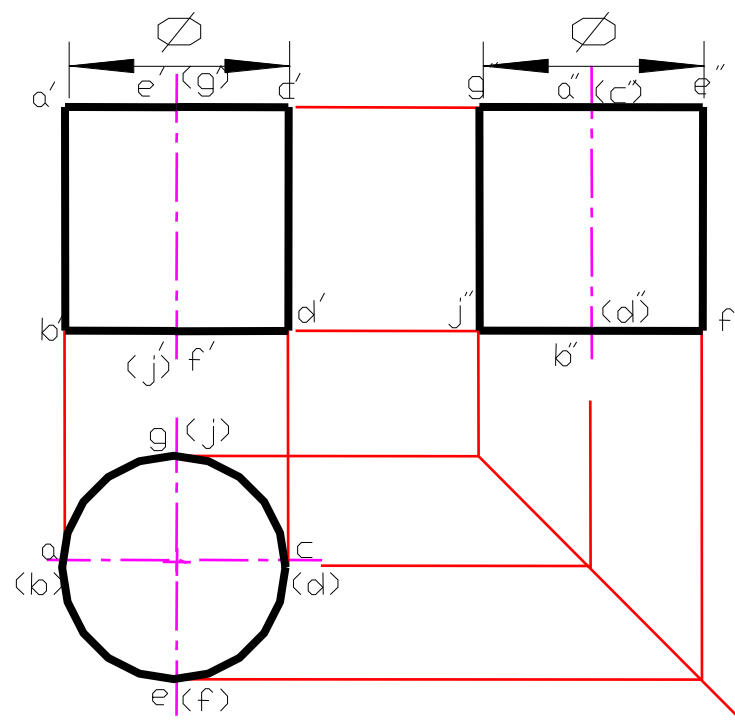
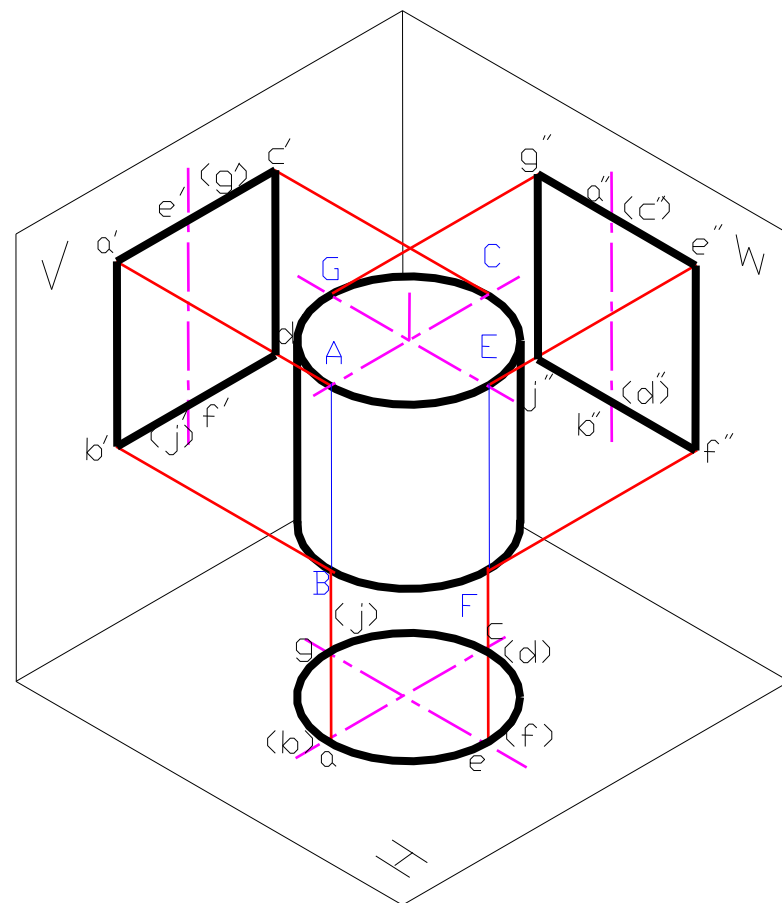
■ 长方体



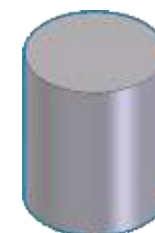
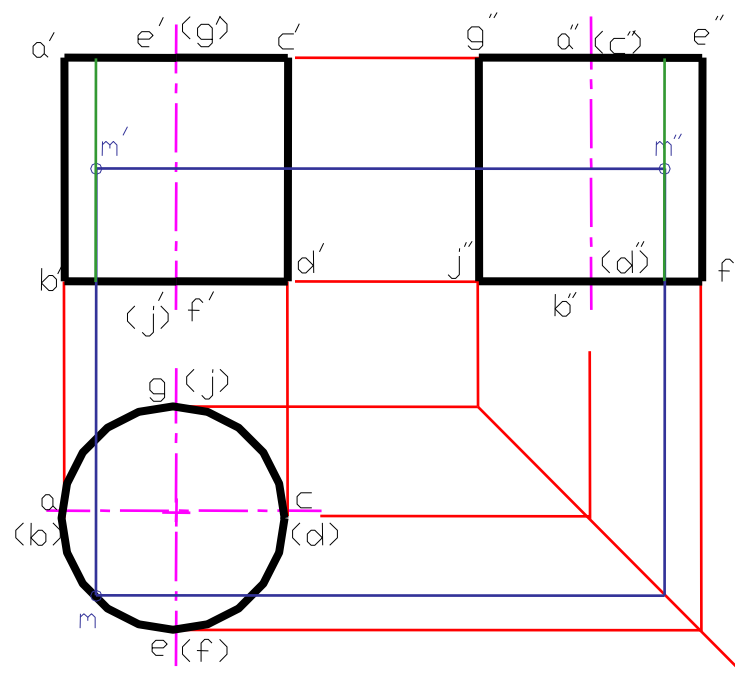
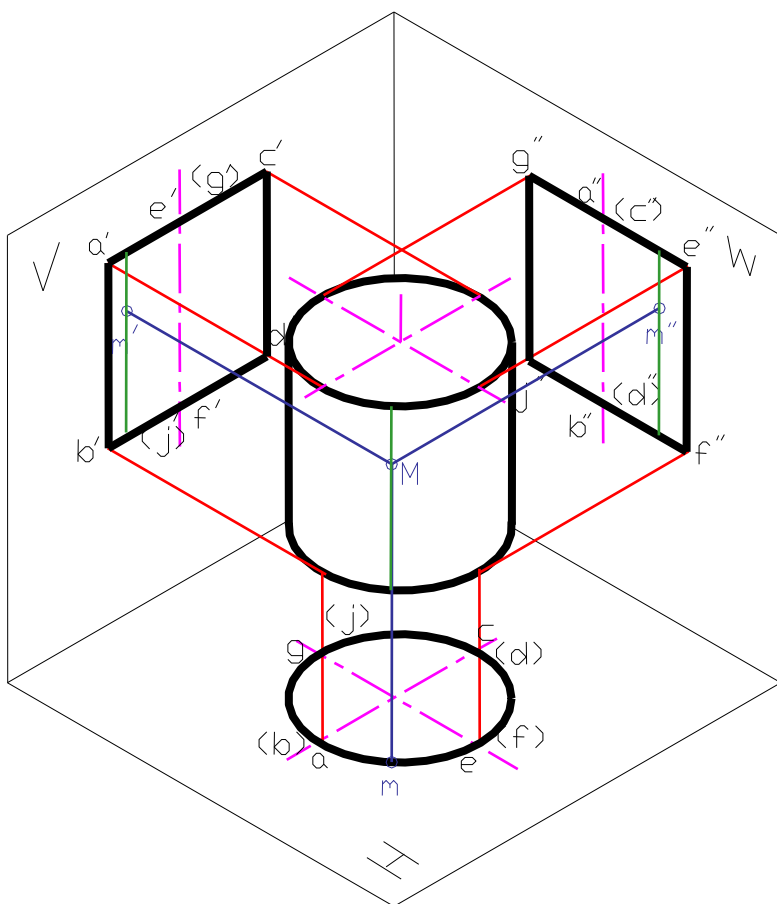




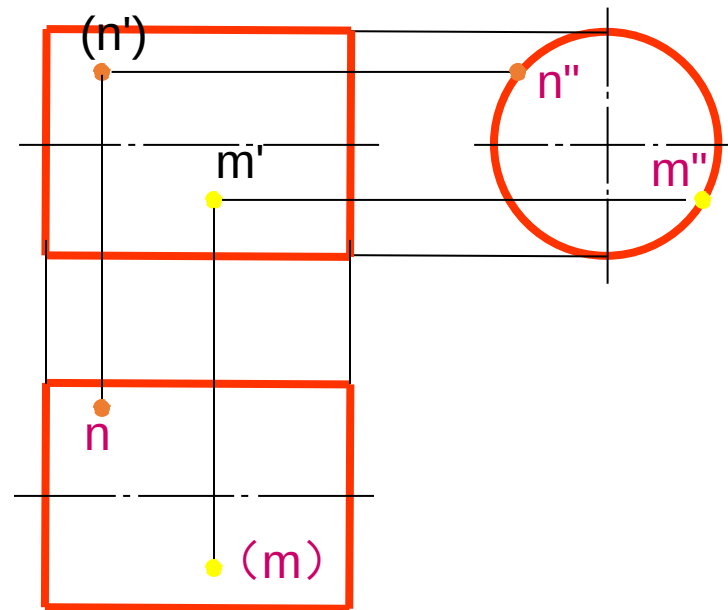
■ 圆柱



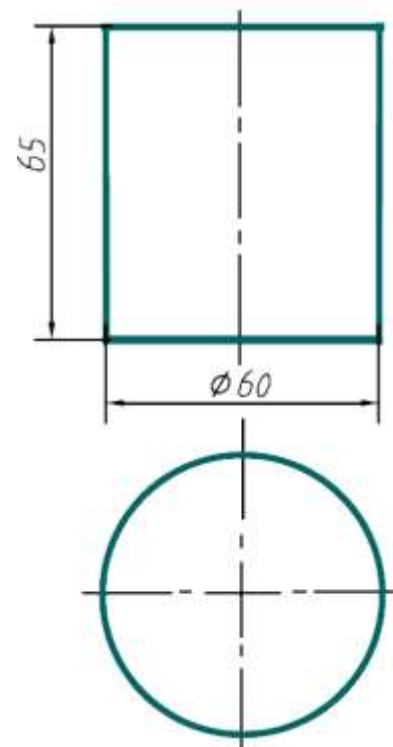
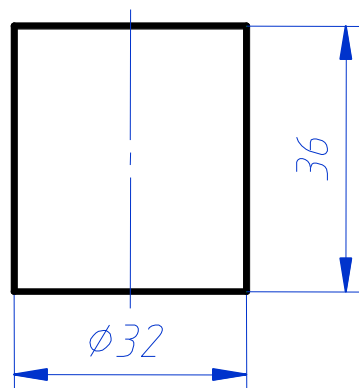
■ 圆柱



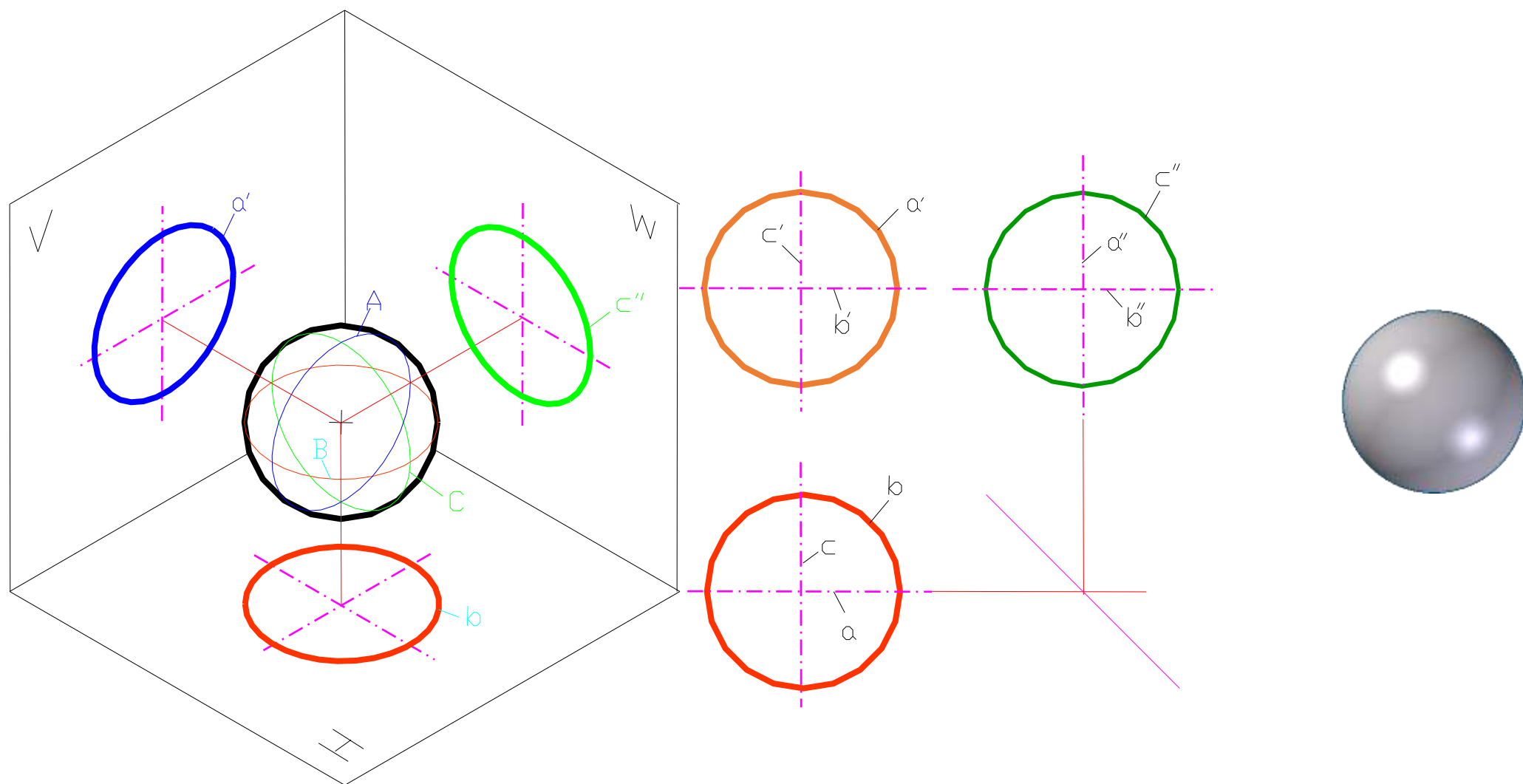
已知圆柱体表面上M、N两点的正面投影 m' 、 (n') ，求其它两面投影。



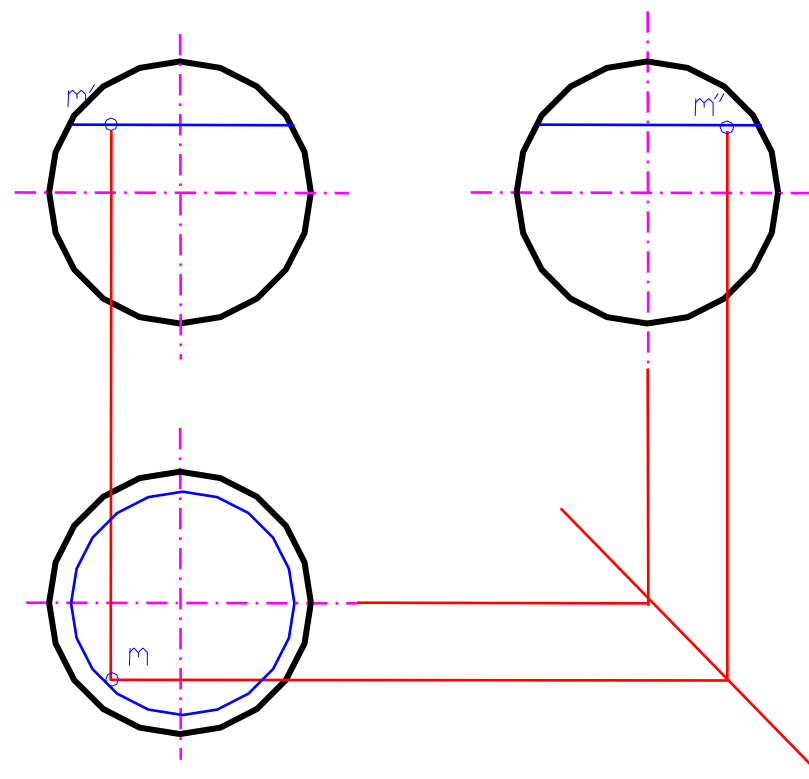
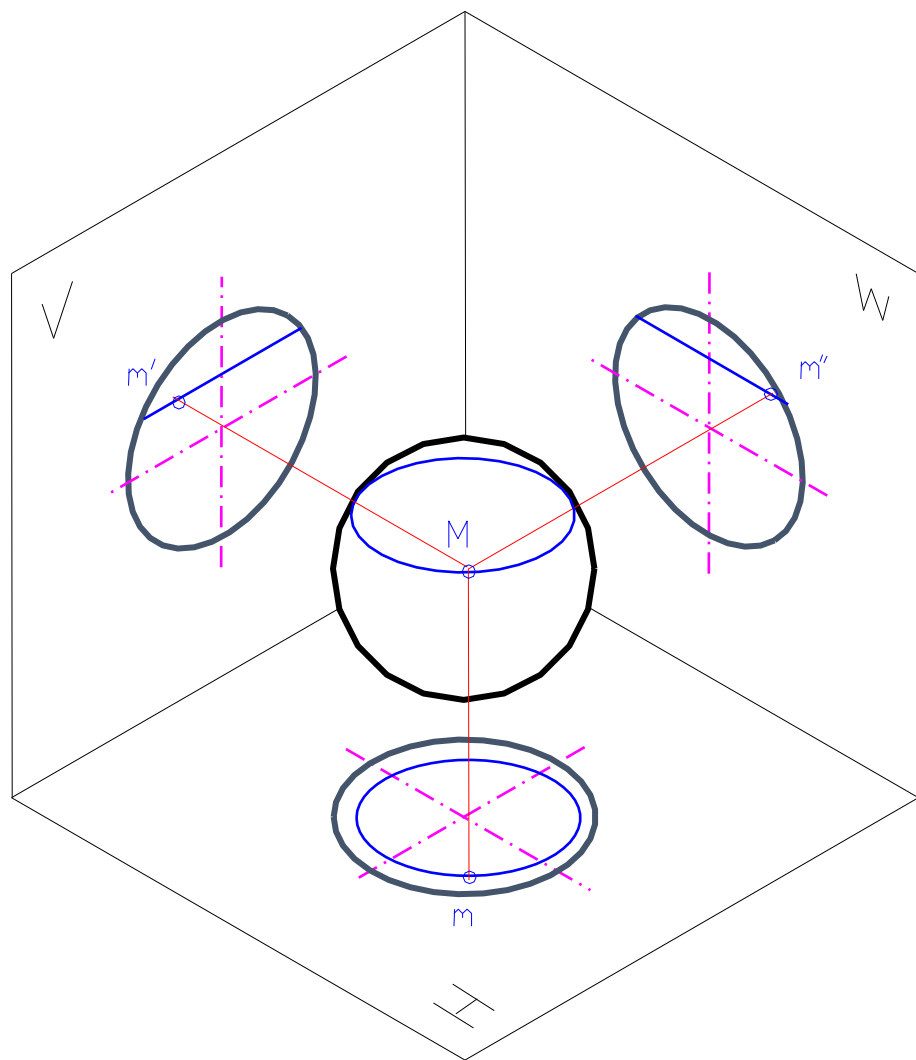
■ 圆柱



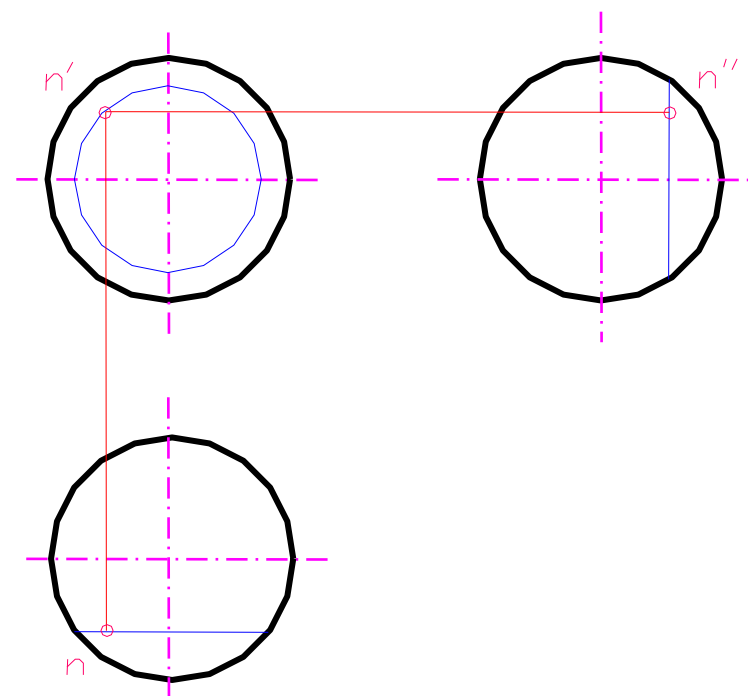
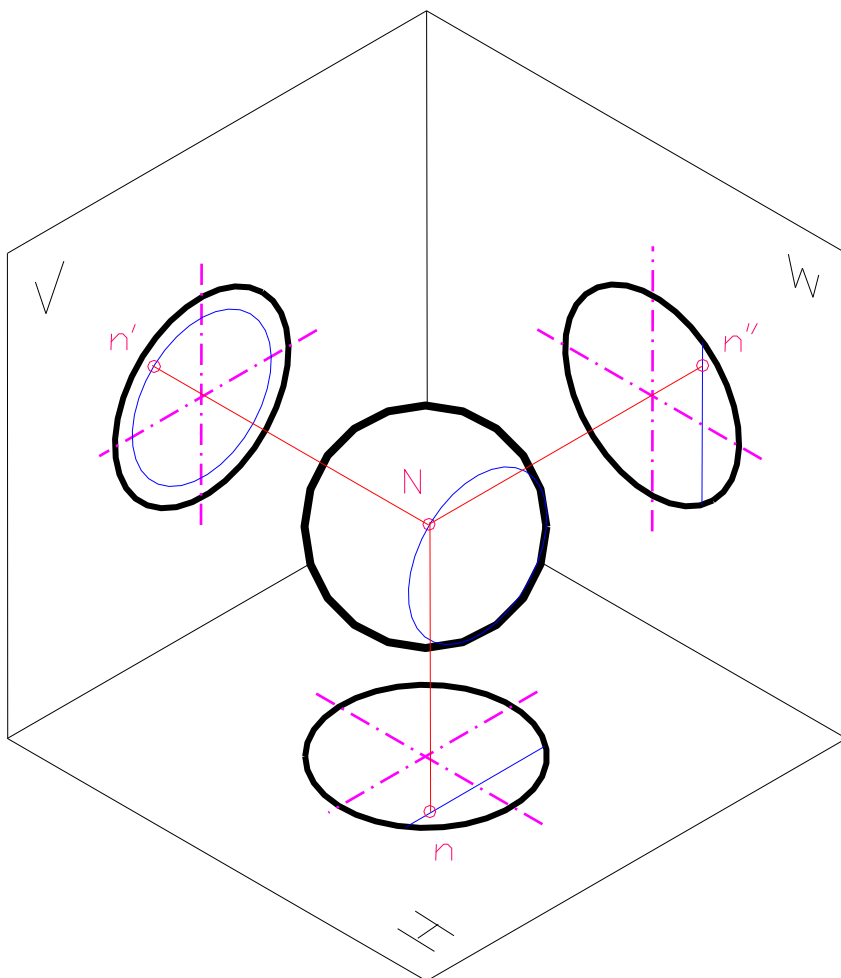
- 圆球



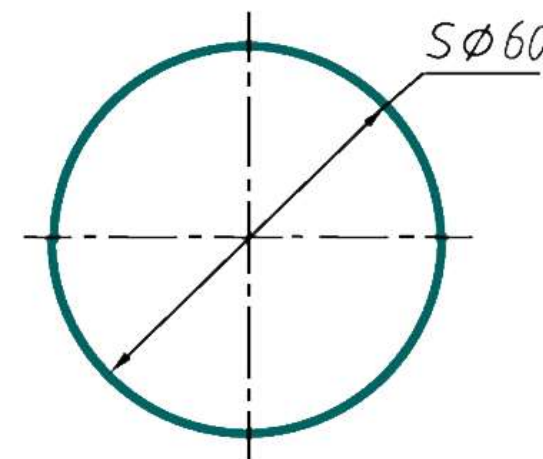
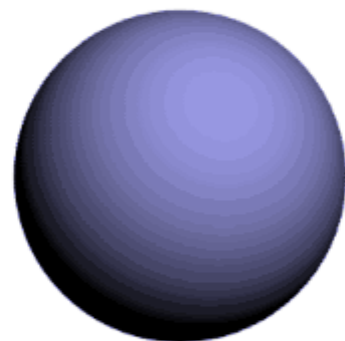
■ 圆球



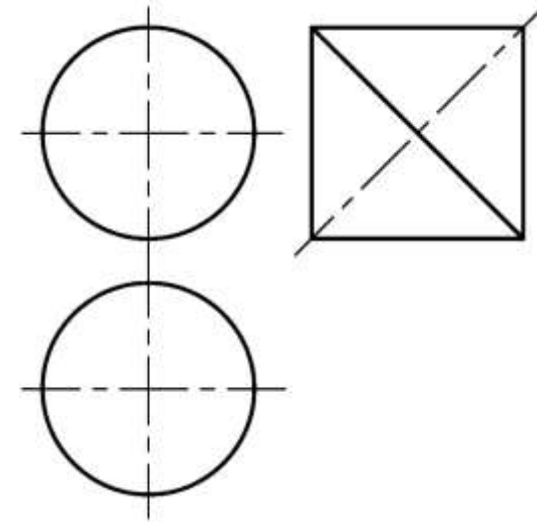
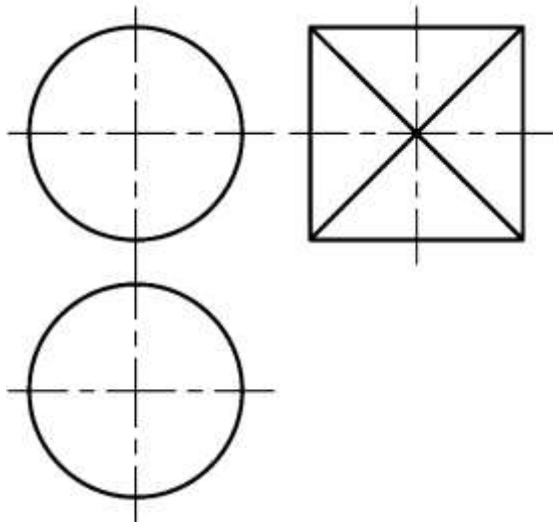
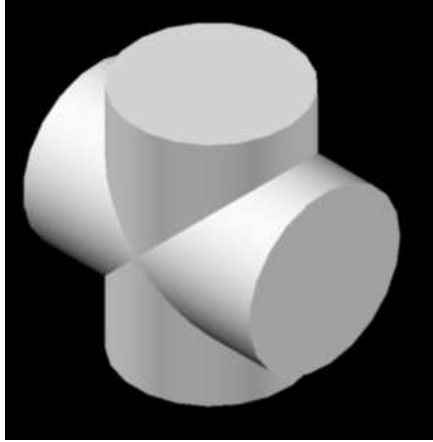
■ 圆球



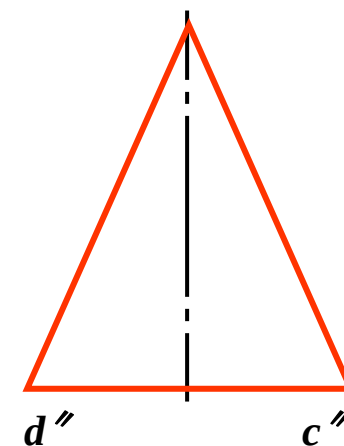
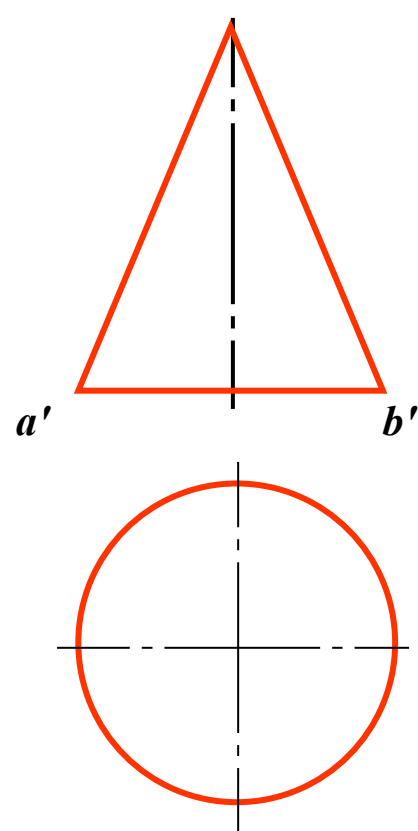
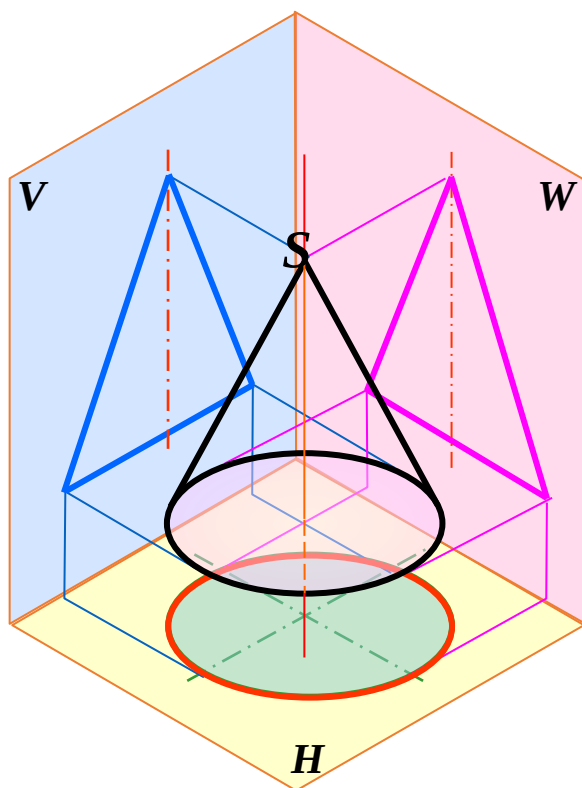
■ 圆球



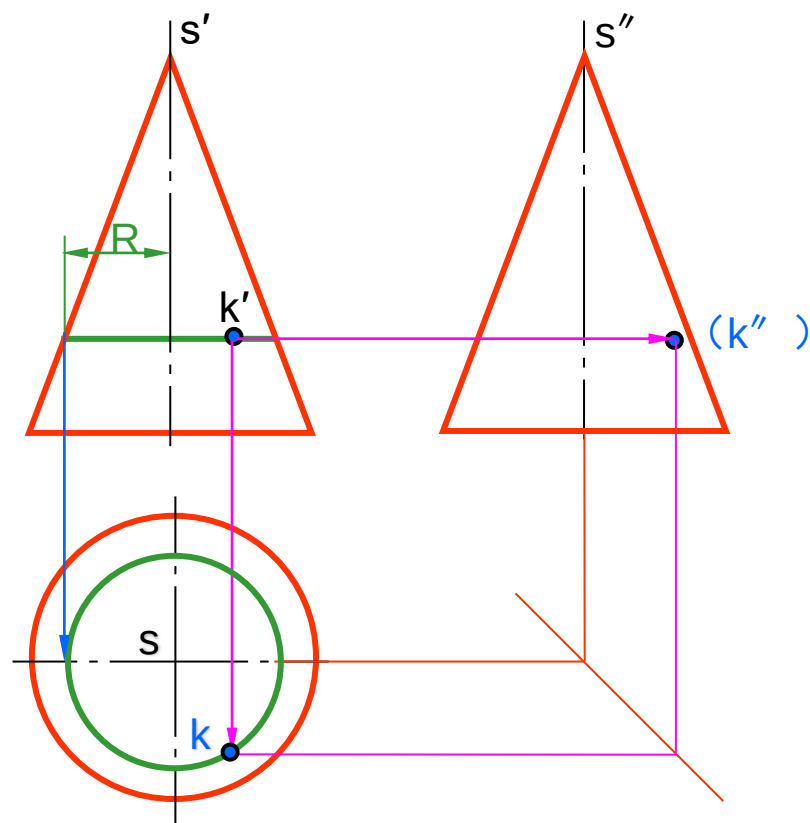
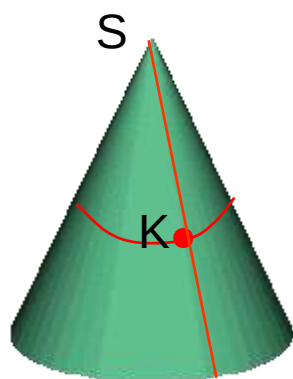
■ 圆球



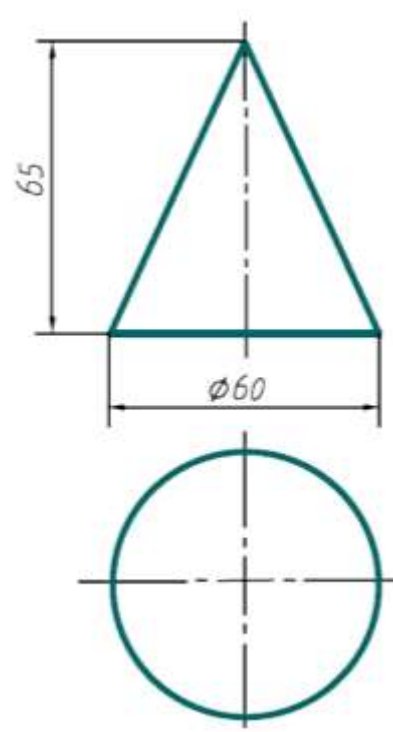
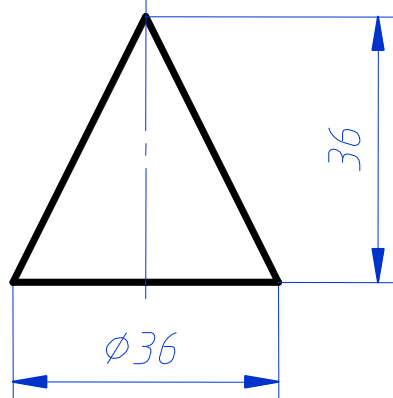
■ 圆锥

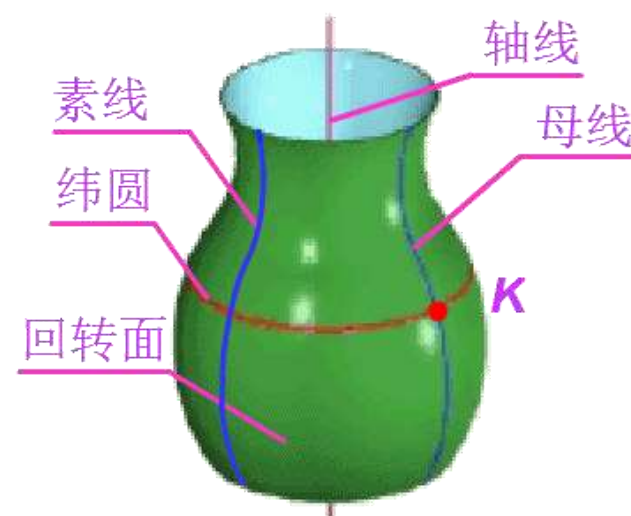


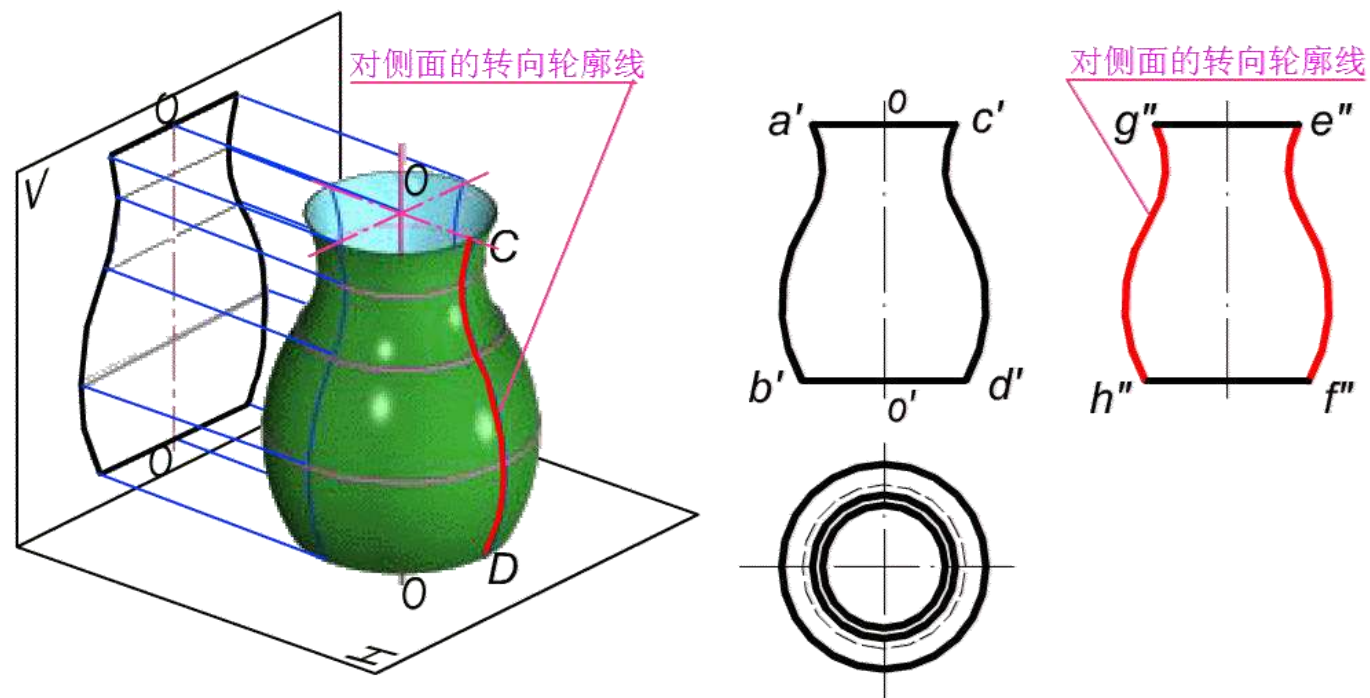
■ 圆锥



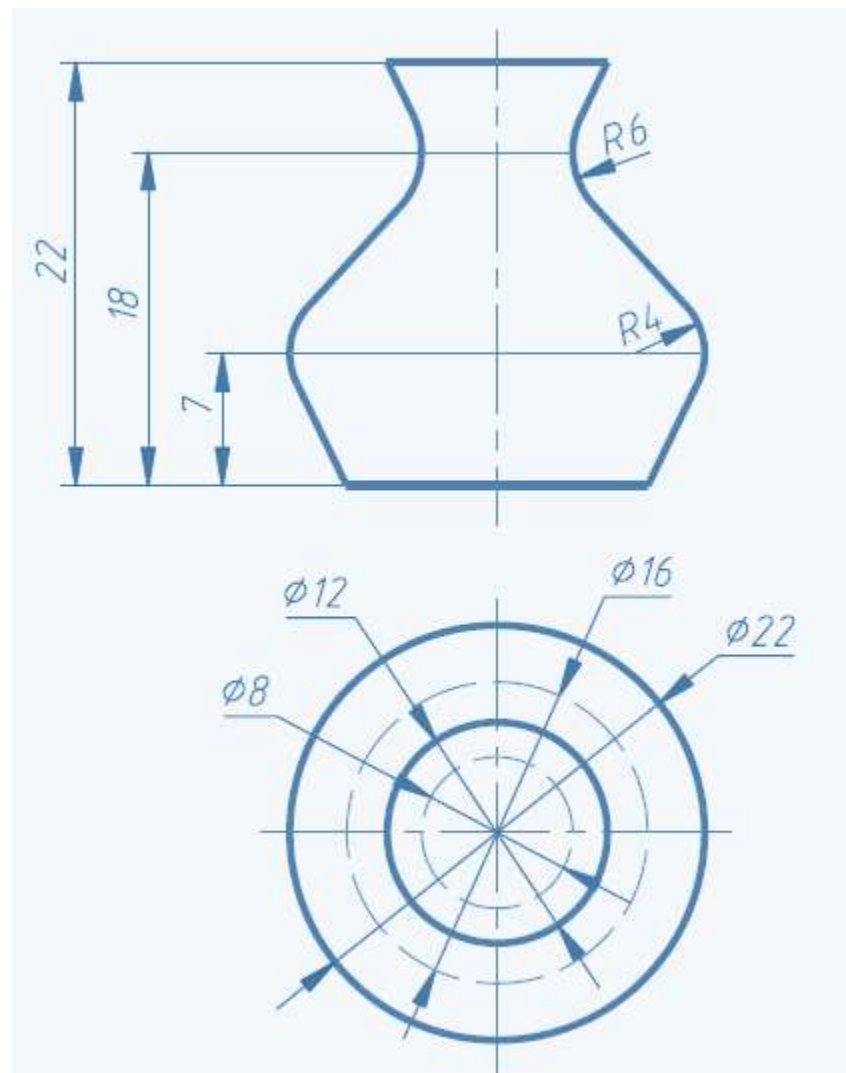
■ 圆锥

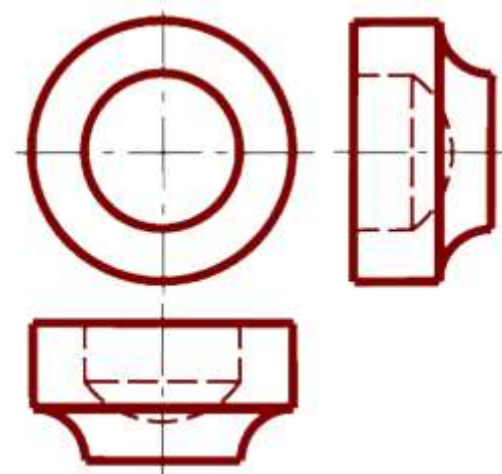
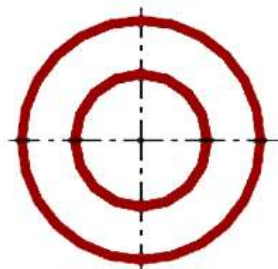
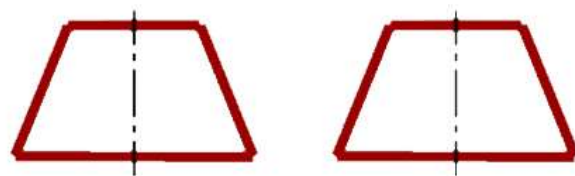
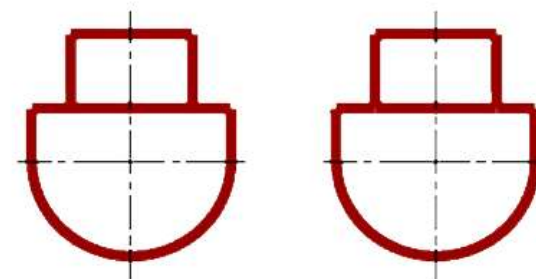
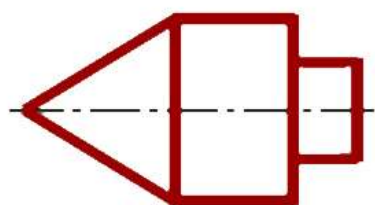
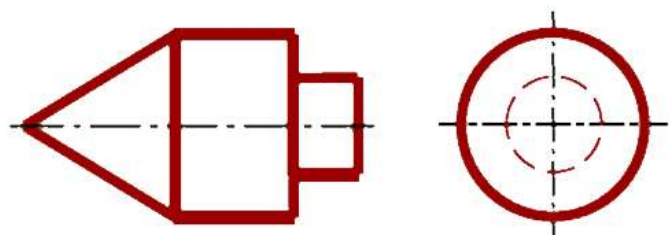


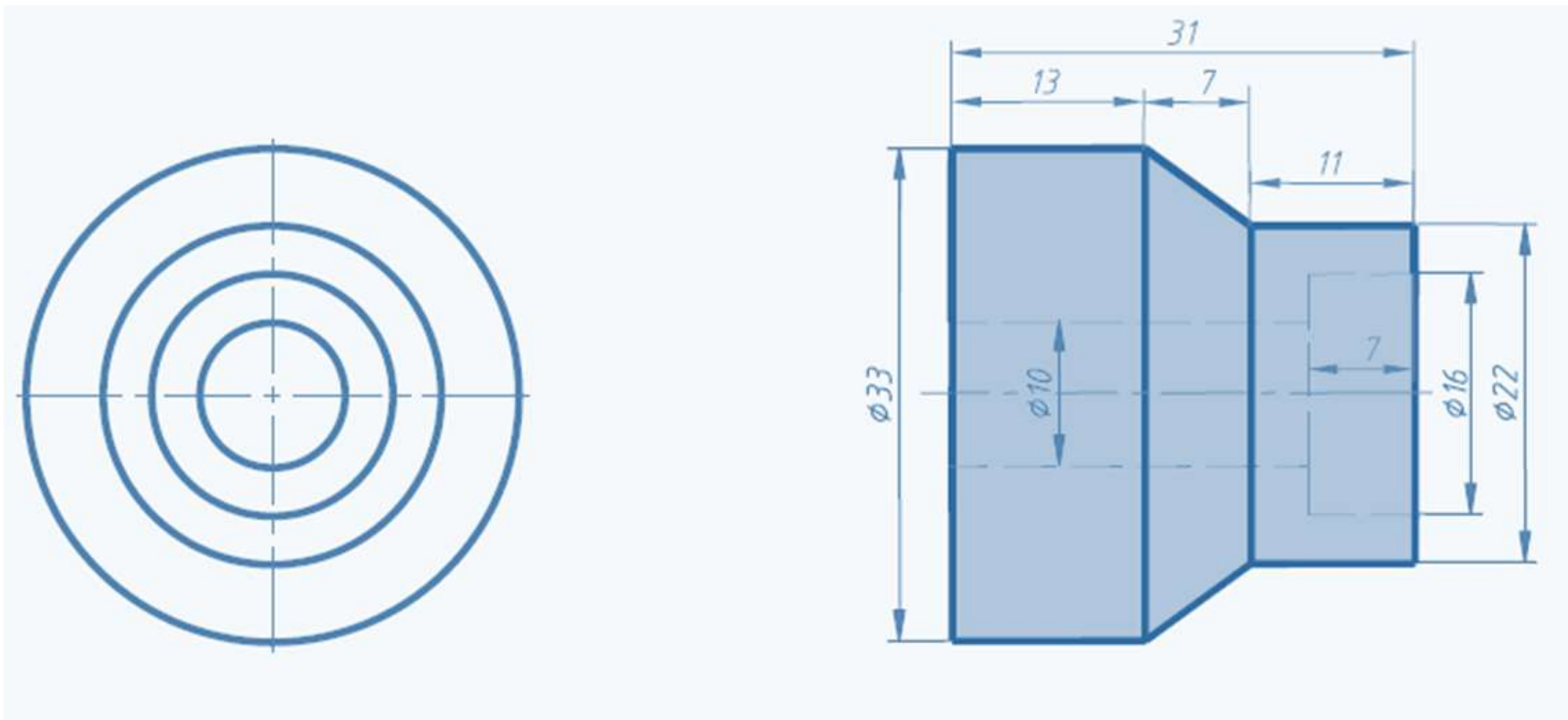




- ✓ 回转面的轮廓线是回转面上对某投射方向可见部分与不可见部分的分界线。
- ✓ 在垂直于轴线的投影面上的投影是圆或同心圆。
- ✓ 其余两投影是大小完全相同的轴对称图形。

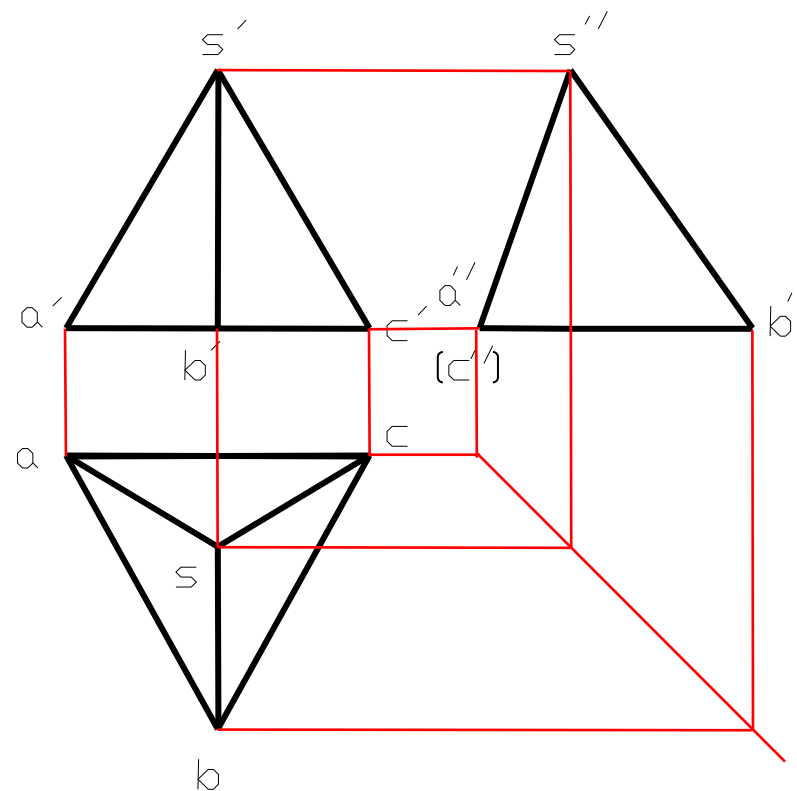
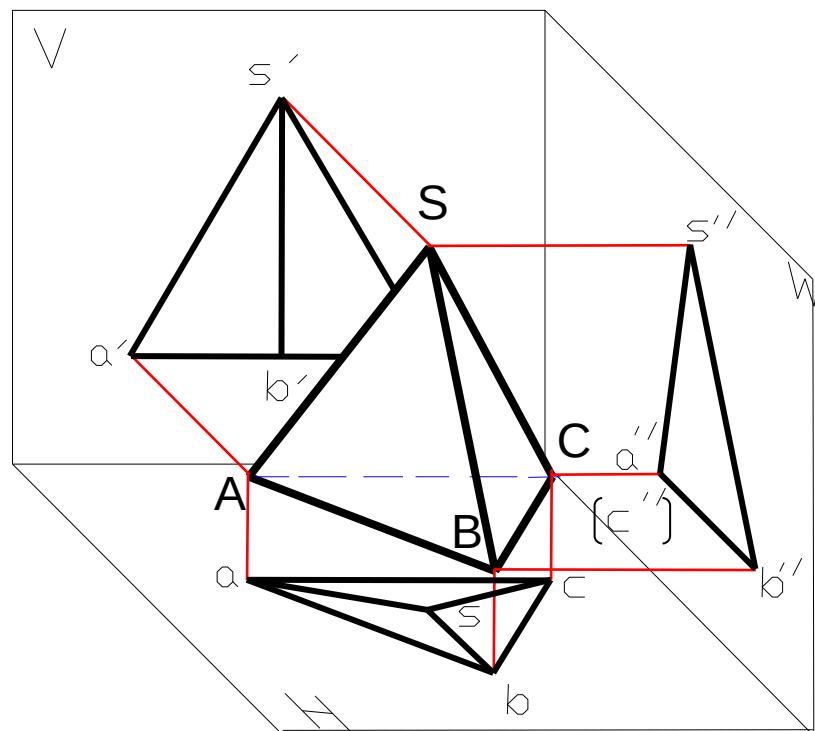
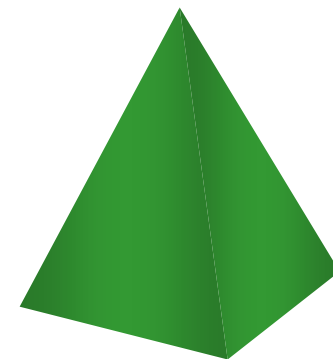




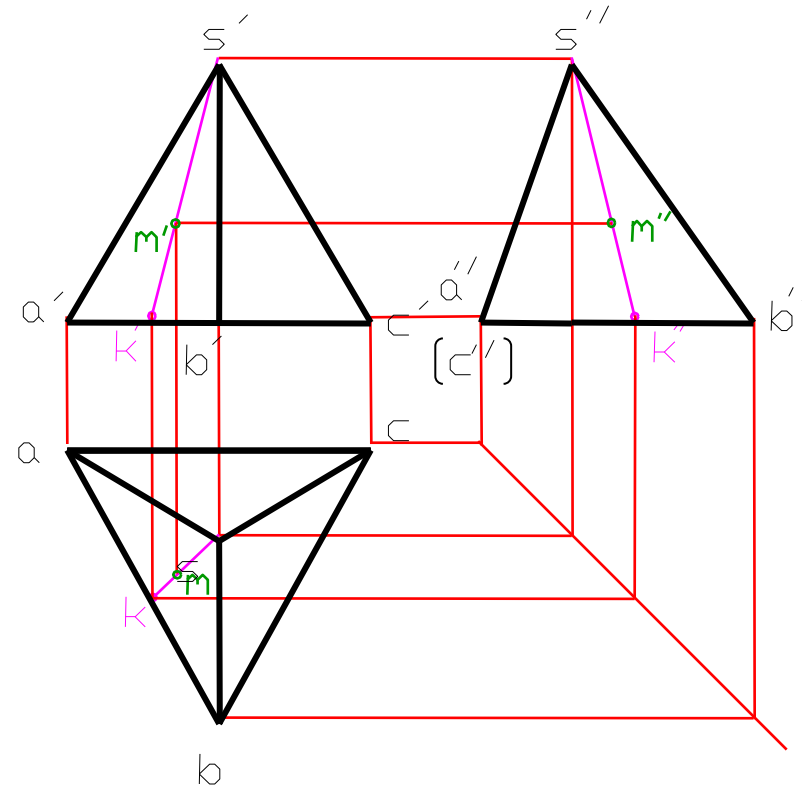
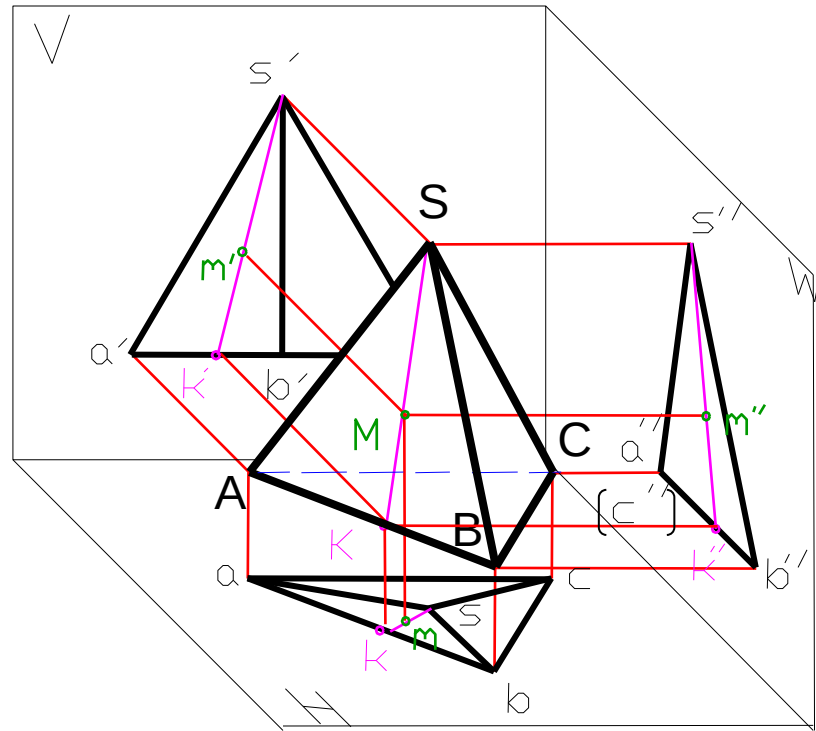
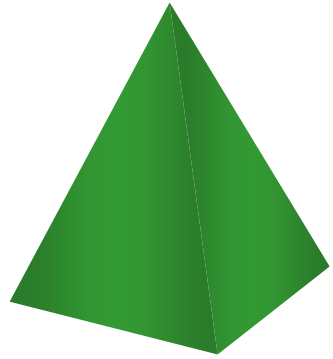


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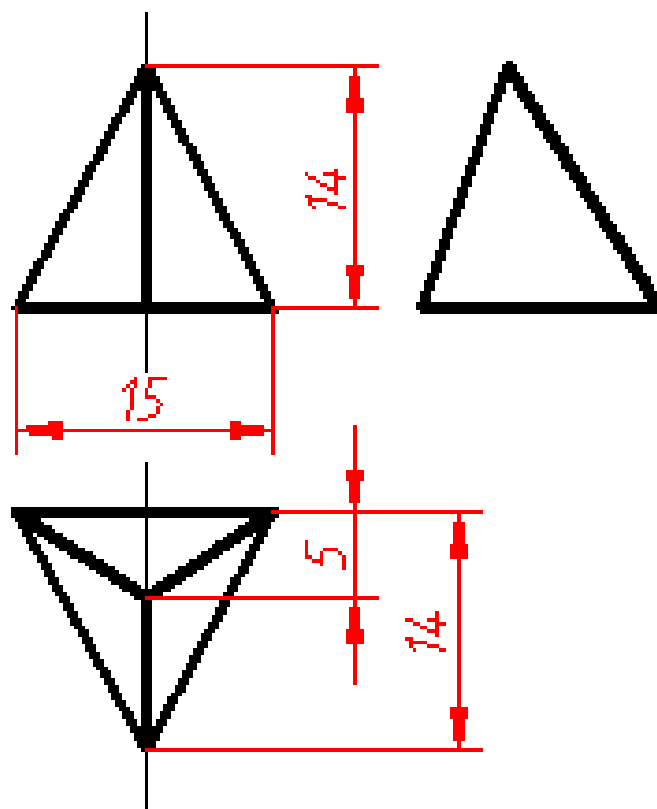
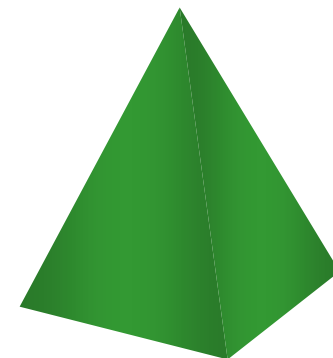
■ 三棱锥



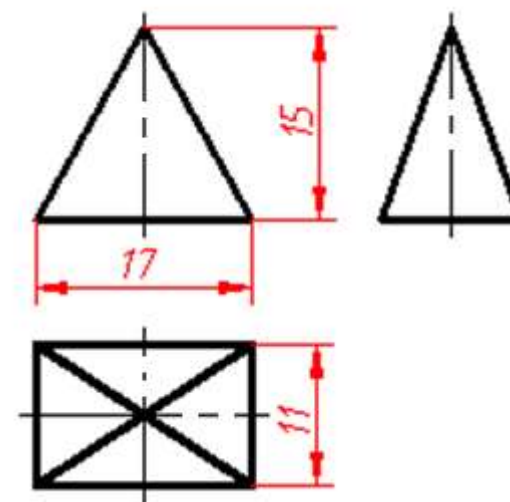
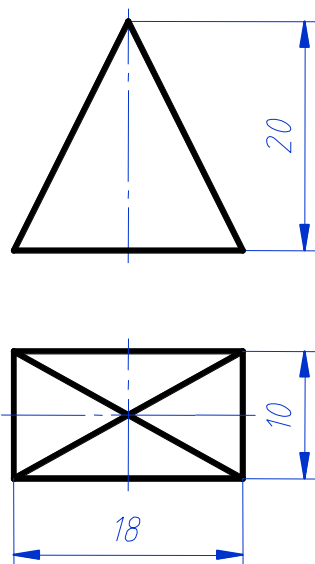
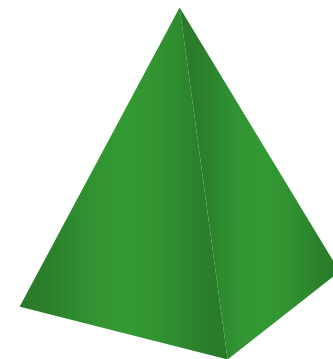
■ 三棱锥



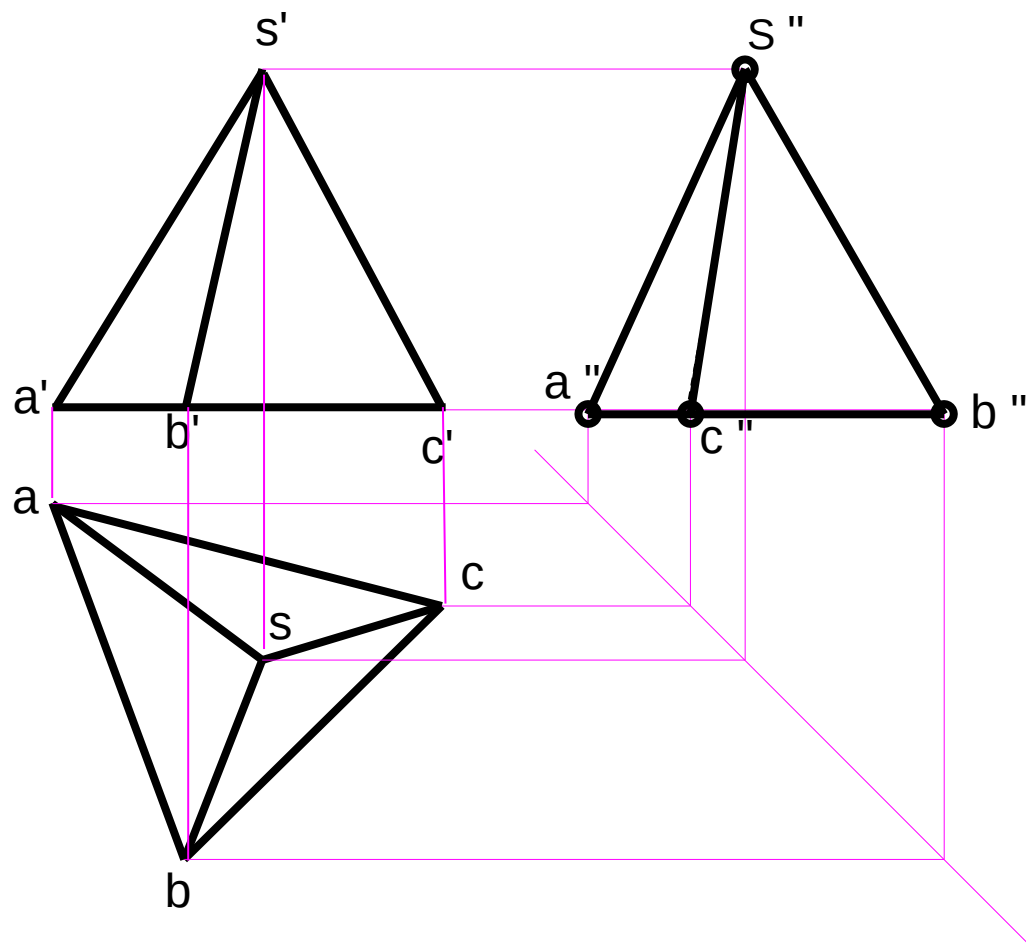
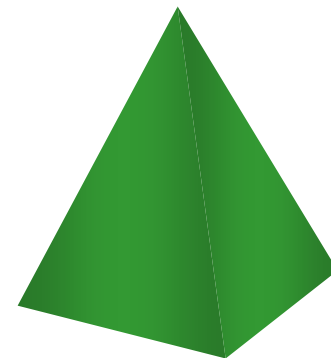
■ 三棱锥



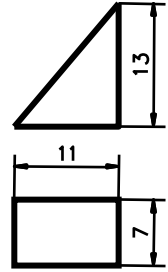
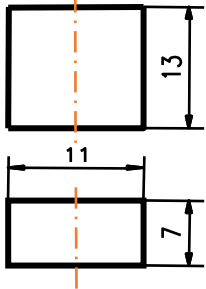
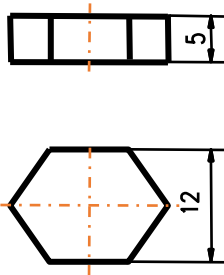
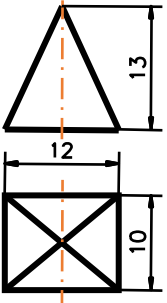
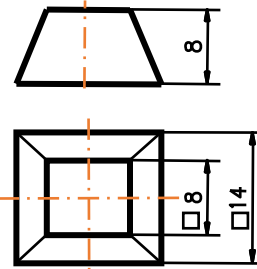
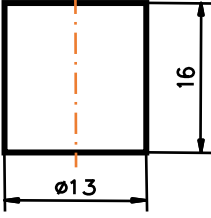
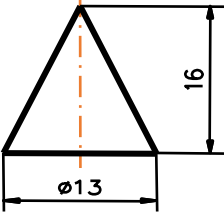
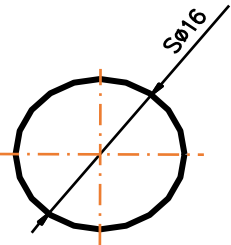
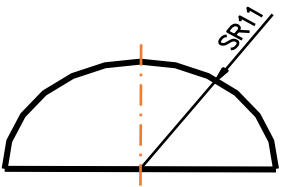
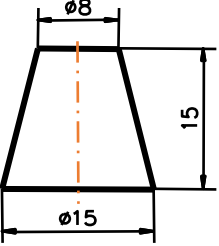
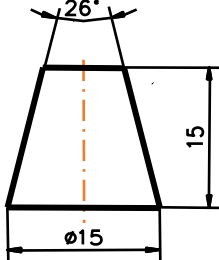
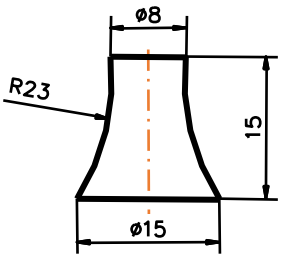
■ 四棱锥



求三棱锥的第三面投影。



基本立体的尺寸标注

三棱柱	四棱柱	六棱柱	四棱锥
			
四棱台	圆柱	圆锥	圆球
			
半球	圆台		内环
			

■ 轴测图

